

A Strategic Community of Practice and Observatory for the Agrifood P4S

D5.1 Training design and Delivery methodology	
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1. Foreword: context and case for change

Deliverable 5.1 (Training Design and Delivery Methodology) sets out how the AGRIFOODSKILLS project will **design, deliver, assess, evaluate and credential** training for the agrifood and veterinary sectors. It follows directly from the **skills intelligence** produced in Work Package 3, which mapped the current and emerging skills needs of the sector against the ESCO framework, and which defined twelve new occupational profiles and the subsequent eight that were selected as the training that this report is built to serve. The present deliverable takes those profiles as its starting point and **works backwards** from them, in the manner of the **New World Kirkpatrick** model: it begins from the results the sector needs, identifies the behaviours and competences that produce those results, and only then specifies the learning that develops them. It builds on the descriptive survey of training methodologies and the review of supporting pedagogical literature provided in I-RESTART D4.1 (2023), which it deliberately does not reproduce; readers seeking the underpinning literature for each methodology are referred to that deliverable. What the present report adds is the comparative judgement of each methodology's suitability for the agrifood and veterinary workforce, and the credential and data architecture that has matured since D4.1 was written.

This task produces a methodology that the consortium will use to design the micro-credential learning of Task 5.2, the work-based learning scheme of Task 5.3 and the training content of Task 5.4, and that will be presented at the train-the-trainer session of Task 6.1. It is, in short, the methodological spine of Work Package 5. Its purpose is to ensure that the eight occupational profiles within scope are translated into training in a consistent, quality-assured and recognised form, rather than being designed eight separate ways by eight separate partners. **Quality assurance** is central to that purpose, which is why the European Quality Assurance in Vocational Education and Training framework (**EQAVET**) is adopted as the overarching reference for the whole methodology. EQAVET supplies the continuous quality cycle (plan, implement, evaluate, review) within which every design, delivery, assessment and credentialing decision in these deliverables sits, although much of the evaluation activity will come with the pilots in later packages. Without an agreed quality framework, a federated set of **micro-credentials** issued by many providers across many Member States would have no shared basis for trust. EQAVET provides that basis for vocational provision, as the Standards and Guidelines for Quality Assurance in the European Higher Education Area provide it for higher education.

The agrifood sector faces a persistent and well-documented challenge: despite significant investment at EU level, workers are not upskilling and reskilling at the rate needed to support workforce mobility and sectoral resilience. Validation through an EU-level focus group (as part of T3.4) confirmed that the barriers are practical and structural: for example, workers are rarely aware that training can be completed in flexible, modular, remote formats that do not require a full day away from the job. Addressing this within the scope of curriculum design and delivery means that **how training is communicated is as important as how it is designed**: without awareness, even well-structured, accessible content will go unused.

The methodology is organised around important contemporaneous developments that bear on every design decision and that signpost the structure of the rest of the report. The first of these is the workforce demand documented in WP3 and WP4: a **time-poor, geographically dispersed** and **often classroom-averse** adult workforce whose training need is acute and whose tolerance for traditional formats is low. The second is the arrival of **generative artificial intelligence** in the workplaces of that workforce. The third is the maturation of the **European credential infrastructure**, in particular the **European Learning Model**, the European Digital Identity Wallet under the revised eIDAS Regulation, the EQAVET framework, and the ESCO classification of skills and occupations, which together make portable, verifiable, stackable micro-credentials a near-term reality for vocational education and training.

This deliverable is written to meet the requirements of Task 5.1, but its scope reaches beyond them. The questions it engages, namely **how adults learn when AI is embedded in their work**, how **competence** rather than course **completion** becomes the unit of **recognition**, and how **credentials** travel with the learner across providers and borders, are questions for adult and vocational education generally, not only

for the agrifood and veterinary sectors. The deliverable therefore reads in places as a position on the future of training in a **post-AI-informed era**. That era is dated here, with deliberate imprecision, from November 2022 and the public release of large-language-model chat interfaces, perhaps not the best thing to happen to education, but undoubtedly the most disruptive. This was the moment at which these tools became ubiquitous in the working lives of ordinary learners, and the moment from which the design of training could no longer proceed as though they were not there.

1.1 Terminology: Micro-credential and Verifiable Learning Credential

The report uses a small set of **credential-related terms** that are interrelated but require distinct definitions. The **micro-credential** as defined in the 2022 Council Recommendation is treated here as one form of a wider class of **verifiable learning records** issued under recognised European and international standards. The wider class is the verifiable credential, as set out in the W3C Verifiable Credentials Data Model, which is the technical foundation of the **European Learning Model** and of **European Digital Credentials for Learning**.

The **verifiable credential** model is issuer-neutral: it does not assume that the issuer is a higher education institution, a vocational provider, a sector body, an employer or a regulator. It is also **composition-neutral**, a verifiable credential can present a single attestation or a composition of several. There is at present no settled European term that captures this wider class in a way that is both VET-inclusive and operationally precise.

The **MICROBOL framework**, which informed the 2022 Council Recommendation, was developed within the Bologna Process and is consequently oriented toward higher education. National implementation of the Recommendation has reinforced that orientation, and vocational providers in the agrifood and veterinary sectors report a sense of exclusion from the micro-credential agenda as it has developed in practice. To avoid that exclusion in the working vocabulary of AGRIFOODSKILLS, this report uses the placeholder term **Verifiable Learning Credential (VLC)** for the wider class.

- A **micro-credential** as defined in the 2022 Council Recommendation as a specific kind of VLC.
- An attendance attestation, an assessment record and competency attestation reflect **atomic units** of learning or assessment.
- Traditional competency **badges** issued widely in the IT sector are also VLCs, although often delivered outside either higher education or the EQAVET framework
- Attendance certificates, clinical specialization certificate should also be considered as learning credentials, although often lack the verifiable certainty provided by VLCs

The placeholder of VLC is offered for adoption or replacement by a settled European term once one emerges; the use of Verifiable Learning Credential within this report is not intended to pre-empt that settlement. The distinction is developed in Sections 6 and 7. Its principal practical consequence is that the question of who is permitted to issue a micro-credential, which has occupied much of the post-MICROBOL discussion, is reframed: many providers may issue assertions, and recognising authorities compose VLCs from them. The result is a federation of issuers rather than a small set of gatekeepers, and a corresponding reduction in the institutional bunker around micro-credential authority. The interaction between these credentials is summarised in Figure1.

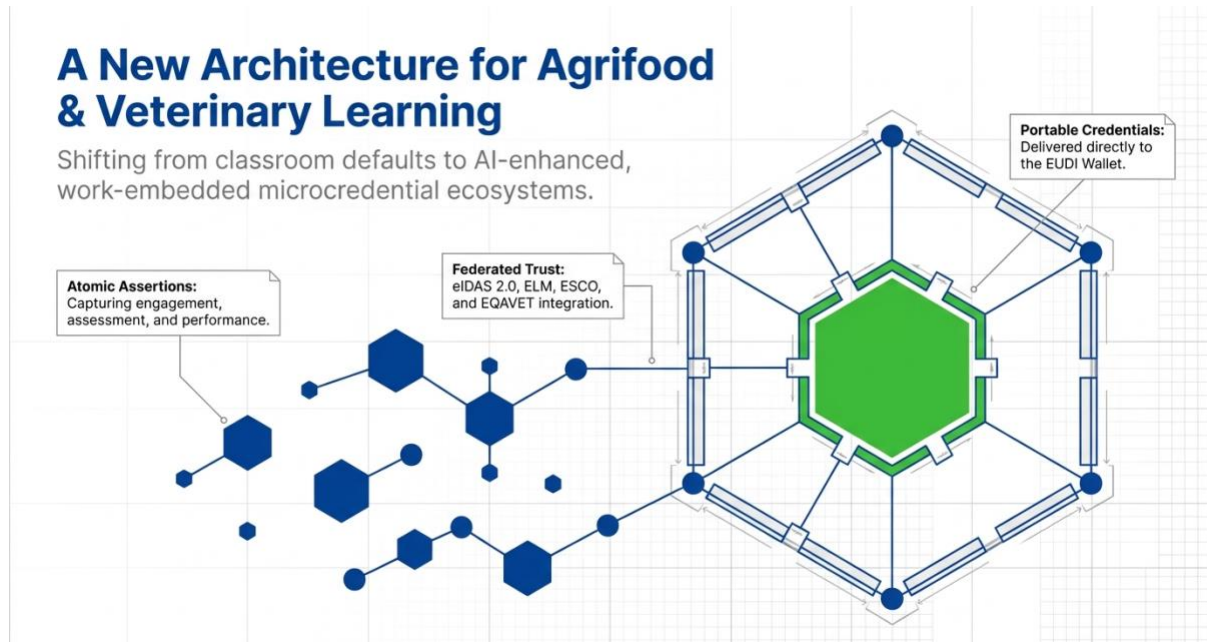


Figure 1: Graphic recommendation of how atomic learning assertions can be brought together to form verifiable learning credentials, under the principles of the Micro-credential system.

1.2 Fundamental positions of this paper:

A fundamental position of this methodology is the transition from **grant-dependent, time-limited training cycles** to a **permanent infrastructure for skills intelligence**. To reach a workforce currently facing a **40% qualification mismatch**, the methodology moves away from the **multi-day classroom format**, which removes the learner's control over pace and location. Instead, it adopts a **modular, assertion-based model** where training is triggered at the moment of workplace need and results in **verifiable learning credentials** that are portable across the EU via the **European Digital Identity Wallet**. This document sets out a number of guiding principles informed by evidence in earlier work packages, previous projects (FIELDS and I-RESTART) and emerging understanding of modern learning theory. They are stated here but can be seen across the document.

1. **Reduce reliance on the multi-day classroom format.** Available completion and transfer-of-learning data for the agrifood and veterinary workforce indicate that multi-day, single-venue training underperforms shorter and work-embedded alternatives. The AGRIFOODSKILLS pilot programme should therefore adopt multi-day classroom formats only where physical co-presence is necessary.
2. **Design at the competency level rather than the syllabus level.** Each micro-credential should publish its prerequisite competencies and its outgoing competencies, both referenced to the ESCO classification, so that credentials are compositionally tractable.
3. **Distinguish assertions from credentials.** An assertion is treated in this report as an atomic verifiable claim concerning a learner (attendance, assessment outcome, performance attestation). A credential, including a micro-credential, is treated as a composition of assertions issued under a recognised authority. The distinction supports stackability and recomposition across providers and reframes the question of credential ownership. Where post-MICROBOL practice has concentrated on which institutions are permitted to issue micro-credentials, the architecture proposed here permits many providers to contribute assertions and a smaller number of recognising authorities to compose Verifiable Learning Credentials from them.
4. **Design credentials in alignment with the European Learning Model**, under EQAVET quality assurance, signed under qualified eIDAS trust services, and presentable through the EUDI

Wallet. The constituent specifications and regulatory instruments are now in place, and the agrifood/veterinary sector is well positioned to adopt them.

5. **Treat artificial intelligence as a transformation of the learning ecosystem** rather than as a single additional methodology. The deployment of generative and predictive AI in workplaces and learning environments alters the set of competencies required of learners, the role of the trainer, and the design space for learning experiences.

What this means for you, today

- If you are designing a micro-credential: read Sections 6 and 7 first, write down its prerequisite and outgoing competencies in ESCO terms before you write a single learning outcome.
- If you are a trainer planning a pilot: read Section 10 and pick the methodology that fits the learner's calendar, not yours.
- If you sit on the consortium or the Pact for Skills board: read Section 9, the Observatory-hosted LRS concept, and decide whether to put it on the M18 agenda.
- If you commission training: read Section 12 (the directional recommendations) and ask any provider you fund to align with them.

1.3 Relationship to the rest of D5.1

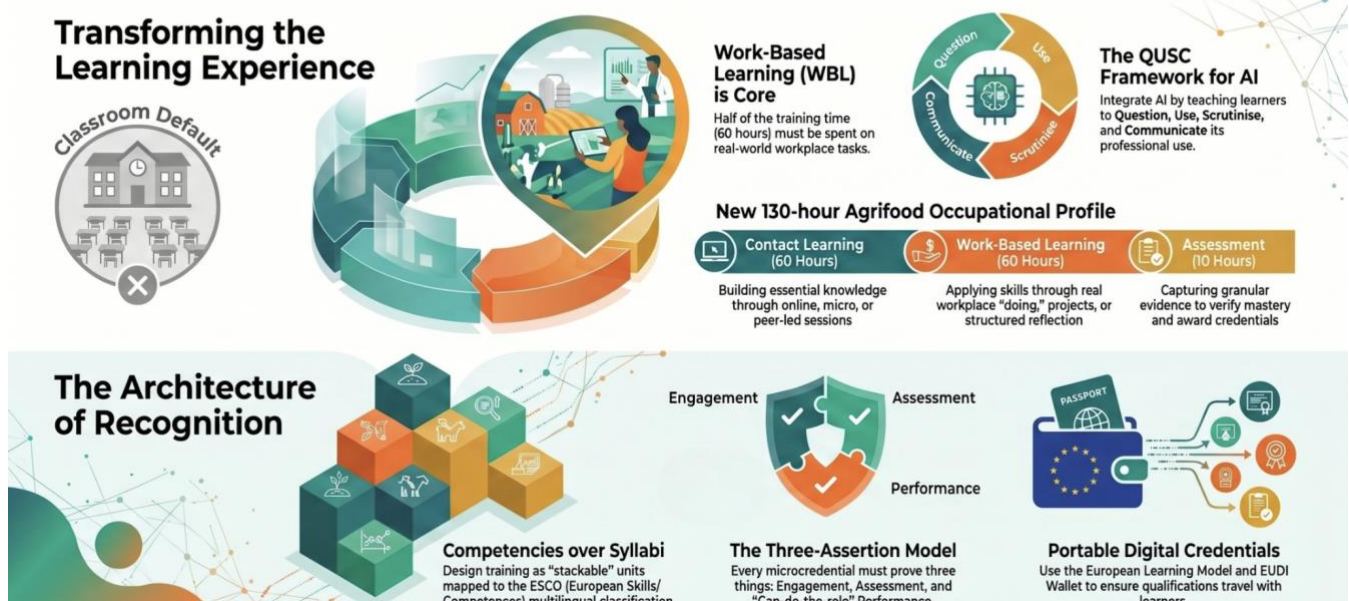
D5.1 as a whole adopts ADDIE as the design-process backbone, ABC Learning Design (Laurillard, 2012) for combining formal training with work-based learning (contributed by POLITO), and Universal Design for Learning (CAST, 2018) as the engagement and inclusivity frame. EQAVET and the ESG guidelines provide the quality references. This report does not repeat that framing. It focuses on (i) how adult learning is changing in an AI-informed era, (ii) the engagement methodologies that follow, (iii) the EQF differentiation across them, (iv) the competency, assertion and credential architecture they all rely on, and (v) the conceptual case for an Observatory-hosted Learning Record System.

SUMMARY INFOGRAPHIC

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Reimagining Training: The AGRIFOODSKILLS Methodology

A shift to a modern, **competency-based**, AI-informed training ecosystem for the agrifood and veterinary sectors.



2. Foundations of adult learning in an AI-informed era

Section 2 sets out the body of **adult learning theory** on which the design choices in subsequent sections rest. It opens with the **foundational paradigms** of learning, develops the practical distinction between **pedagogy** and **andragogy** in mixed learner populations, presents Knowles' andragogy as the principal tradition guiding AGRIFOODSKILLS design, and considers how the AI-informed era changes the conditions under which adult learning happens. The **QUSC** framework that AGRIFOODSKILLS adopts for AI competency is introduced within the AI-era subsection.

2.1 Foundational paradigms of learning

Understanding how individuals acquire knowledge and skills is fundamental to designing effective training programs. Learning theories provide frameworks for interpreting the learning process, guiding instructional design, and optimizing educational outcomes. While numerous theories exist, the dominant paradigms that inform contemporary educational practices are behaviourism, cognitivism, and constructivism. Each offers a unique lens through which to view and facilitate learning, particularly in diverse and dynamic environments. The evolution of learning theories provides a comprehensive understanding of human learning, from observable behaviours to complex cognitive processes and social interactions.

2.1.1 Behaviourism: Learning Through Association and Reinforcement

This theory, popular in the early 20th century, posits that learning is a change in observable behaviours, acquired through interactions with the environment. It emphasises the role of external stimuli and reinforcement in shaping responses. Learners are often seen as passive recipients whose behaviour is conditioned by consequences. Key concepts include classical conditioning (associating stimuli) and operant conditioning (modifying behaviour through rewards and punishments). In a training context, this translates to clear instructions, drill and practice, and immediate feedback to reinforce desired actions.

2.1.2 Cognitivism: The Mind as an Information Processor

Emerging as a response to behaviourism, cognitivism views the learner as an active information processor. It focuses on internal mental processes such as memory, perception, problem-solving, and metacognition. Learning is understood as a change in a learner's internal mental structures or "schemata," rather than just observable behaviour. Effective learning strategies under this paradigm often involve organising information logically, using analogies, and encouraging active engagement to build robust mental models.

2.1.3 Constructivism: Building Personal Understanding

Constructivism proposes that learners actively construct their own understanding and knowledge of the world through experiencing things and reflecting on those experiences. Knowledge is not passively received but is actively built upon existing knowledge and personal experiences, often influenced by social and cultural contexts.

This theory champions learner autonomy, encouraging individuals to explore, question, and formulate their own hypotheses. Training methods derived from constructivism often involve collaborative projects, problem-based learning, and real-world applications where learners can connect new information with their prior knowledge and build personal meaning.

While distinct, these theories often complement each other in practical application. For instance, basic skill acquisition (e.g., memorising facts or procedures) can benefit from behaviourist principles, while developing deeper understanding and critical thinking often requires cognitive strategies. Constructivism becomes paramount for stimulating complex problem-solving abilities and transferring knowledge to novel situations, particularly in adult learning environments where prior experience is rich. The optimal approach frequently involves integrating elements from each theory to cater to different learning objectives and learner profiles.

2.2 Pedagogy and andragogy for mixed learner populations

Understanding how people learn is the cornerstone of effective training design. While we often associate "pedagogy" with teaching children and "andragogy" with teaching adults, these aren't rigid age categories but rather two influential frameworks that guide how we approach learning and instruction. For programmes which cater to a diverse group of primarily adult learners, including farmers, foresters, farm advisors, and SME professionals, as well as university students wishing to work in the sector, it's vital to recognise that learning needs can vary even within an adult population. Grasping the distinctions between these approaches allows us to combine their strengths, ensuring that course design maximises knowledge transfer for everyone involved.

2.2.1 Pedagogy: Structured Learning for Foundational Knowledge

Traditionally, pedagogy refers to the art and science of teaching children, where the instructor often plays a more direct, guiding role. However, elements of pedagogical approaches are incredibly valuable for *any* learner, including adults, especially when:

Introducing New or Complex Concepts: When learners, even adults, are encountering a completely new subject area or highly complex foundational theories, a more structured, teacher-guided approach can be highly effective in building a solid base of understanding. University students, for example, might be new to certain agricultural concepts and benefit from clear, sequential instruction.

Ensuring Foundational Knowledge: A structured approach ensures that essential prerequisites and core concepts are thoroughly covered before learners move on to more self-directed application. This can prevent gaps in understanding that might hinder later, more independent learning.

Facilitating Guided Practice: Pedagogical elements can include clear assignments, guided exercises, and direct feedback from an instructor, which helps reinforce learning and correct misunderstandings in a supportive learning environment.

Managing Learning Pace: For certain modules or learners, a more defined pace and curriculum can help manage the learning process effectively, ensuring all critical areas are addressed.

2.2.2 Andragogy: Self empowered learning for adults

Andragogy is treated here as a tradition of empirical findings concerning the **learning behaviour of adults**, established over six decades of research, and is applied below as a set of design heuristics for the AGRIFOODSKILLS pilot programme. Andragogy is built on the following principles

Self-Directed: Adults generally prefer to take ownership of their learning. They want to be actively involved in deciding what and how they learn, often seeking a personalised path that fits their needs.

Experience-Based: Adults bring a wealth of life and work experience. This prior knowledge is a valuable resource and should be integrated into the learning process, allowing them to connect new information to what they already know. This is particularly relevant for experienced farmers or SME owners.

Problem-Oriented and Relevant: Adults are typically motivated to learn when they see the immediate practical value and how it can solve real-world problems or enhance their current skills. They need to understand the 'why' behind the learning.

Internally Motivated: While external factors can play a role, adult learners are often driven by internal desires for personal growth, career advancement, or increased competence.

Active and Participatory: Andragogical approaches thrive on active learning, encouraging participation through discussions, group projects, problem-solving, and hands-on activities, which leads to deeper understanding and application.

Principle	What it says	Design implication for AGRIFOODSKILLS
Need to know	Adults engage when they understand why a piece of learning matters before they invest time.	Open every microcredential with a sector-specific problem statement; no boilerplate objectives slides.
Self-concept (self-direction)	Adults are responsible for their own learning decisions.	Default to choice in pace, location and order. Avoid lockstep cohort delivery unless genuinely required.
Role of experience	Adults bring substantial prior experience and want it acknowledged.	Surface existing practice in every session. Use peer discussion and case interrogation as core, not as warm-up.
Readiness to learn	Adults are ready to learn what they need to cope with real-life situations.	Trigger learning at the moment of workplace need; bring training to the situation rather than the reverse.
Orientation (problem-centred)	Adults learn through tasks, not subjects.	Organise content around things the learner must do, not around academic disciplines.
Motivation (intrinsic)	Adults are driven by mastery, identity, autonomy, not extrinsic rewards.	Tie credentials to professional identity ('qualified to lead biosecurity audits') rather than to grades.

The **classroom multi-day model fails on five of Knowles' six principles at once**: it removes the learner's control over pace and location, it discounts prior experience, it organises content by subject rather than task, it disconnects from the moment of workplace need, and it substitutes extrinsic completion incentives for the intrinsic motivation that drives transfer. The reason this format persists is institutional inertia, not

educational efficacy. AGRIFOODSKILLS should not contribute to that inertia. Section 10's catalogue of methodologies is built specifically to give trainers alternatives that work better.

Literature in brief: andragogy and its successors

Foundational: Knowles, Holton & Swanson (2020) on andragogy. Connecting strands relevant to AGRIFOODSKILLS: Mezirow on transformative learning and disorienting dilemmas in adult professional development; Lave & Wenger on communities of practice and situated learning (directly relevant to peer demonstration and workplace cohorts); Brookfield on critically reflective practice in adult education. Field synthesis: EPAL policy briefs on adult learning principles in European VET.

Design summary: andragogy checklist for every microcredential

- Need to know: does the opening 30 seconds of the module state the operational problem the learner brought with them?
- Self-direction: can the learner set pace, location and order without losing assessment credibility?
- Experience: where in the module is the learner's prior practice surfaced and put to work?
- Readiness: is the module triggerable at the workplace moment of need, or only at calendar moments?
- Problem-centred: is content organised by tasks the learner must perform, not by academic subject?
- Intrinsic motivation: does the credential connect to a recognisable professional identity ('qualified to lead X'), not just a grade?

2.3 How adult learning is changing in an AI-informed era

Generative and predictive **artificial intelligence** has been adopted rapidly within the working environments of the AGRIFOODSKILLS target learner population. AI-enabled applications are now embedded in farm management systems, processing-line supervisory tools, veterinary diagnostic workstations and field-level decision-support applications. The presence of these tools in the workplace has two principal implications for training. First, the set of competencies required of learners in order to perform their roles effectively now includes the **critical and ethical use of AI tools** alongside the established domain competencies. The Skills Intelligence work in WP3 has identified these AI-related competencies as emerging requirements across all four AGRIFOODSKILLS focus areas (plant production, animal production, food industry and processing, and veterinary activities), and the AGRIFOODSKILLS occupational profiles (T5.2) should reflect that finding. Second, the same generation of **AI tools enables learning designs** that were not feasible at scale prior to 2023. This section identifies the theoretical foundations that should guide pedagogical responses to these developments, summarises the design patterns enabled by current AI capability, and considers their application within the agrifood and veterinary sectors.

2.3.1 Adult learning theory meets AI: four anchors

The andragogical use of artificial intelligence is best considered within the established traditions of adult education research. Anchoring the discussion in those traditions assists in avoiding two analytical errors that recur in the wider literature. The first is the **substitution thesis**, which holds that AI replaces the human educator. The second is the **throughput thesis**, which holds that AI is principally a means of producing more learning content at lower marginal cost. Four theoretical frames are referenced in this report and are summarised below.

Theoretical anchor	Core claim about adult learners	How AI intensifies it
Andragogy (Knowles)	Adults are self-directed, experience-rich, problem-centred and intrinsically motivated.	AI makes genuine self-pacing, contextualisation to the learner's own operation, and problem-centred dialogue feasible at scale. Adaptive pathways respond to the learner's own data.
Transformative learning (Mezirow)	Adults change practice through disorienting dilemmas, critical reflection, dialogue and integration of new perspectives.	AI can generate structured dilemmas, surface counter-arguments, sustain reflective dialogue over time, and prompt the moves through Mezirow's phases. This approach suits professional continuing professional development (CPD).
Self-directed and lifelong learning	Adults plan, monitor and reflect on their own learning across a career.	AI as a meta-learning partner: it scaffolds planning ('what do I need to know to move into this role?'), monitoring (against a competency framework) and reflection (over months of journal entries).
Communities of practice (Lave & Wenger)	Adults learn through participation in communities of practitioners, with peripheral participation evolving into central contribution.	AI as discussion-weaver in online and hybrid communities: summarising threads, surfacing unresolved tensions, translating contributions across languages, prompting quieter members. The infrastructure of practice becomes more inclusive.

Each of the four frames identifies a category of design in which current AI capability supplies an affordance that previous technologies did not. Adaptive contextualisation, the structured staging of transformative learning sequences, meta-learning support over extended periods, and the weaving of multilingual practice communities are each documented in the recent research literature, and each remains located within an established theoretical tradition.

2.3.2 From content delivery to intelligent learning ecosystems

The change in **adult learning architecture** may be described as a transition between two organising models. The **legacy model** organises learning around content units, such as a curriculum, a course, a module or a syllabus, delivered to a cohort by an instructor. The **emerging model** organises learning around a learner pathway through a defined competency landscape; the pathway is shaped iteratively in response to learner performance and reflection, and is supported by a combination of AI-enabled tools and human educators. Under the emerging model the role of the educator extends to the design of learning experiences, the curation of resources, the provision of ethical oversight and the verification of evidence of competence.

Dimension	Content-delivery model (legacy)	Intelligent learning ecosystem (emerging)
Unit of design	Course, module, lesson	Competency, assertion, pathway
Pace and order	Fixed, cohort-paced	Reshaped continuously to the learner; adaptive sequencing is supported only where a Learning Record System (Section 8) captures learner performance and reflection at sufficient granularity
Trainer's primary role	Source of content; assessor of recall	Designer of experience; curator of resources; ethical guardrail; reviewer of can-do-the-role evidence
Content production	Authored once, delivered many	Co-created with AI per learner; reviewed by humans
Assessment	Synchronous, end-of-	Continuous and embedded; captured at the assertion

	module; assessment treated as separable from learning	level (Section 7) in a Learning Record System using Experience API (xAPI) or an equivalent schema, so that assessment evidence simultaneously informs learning design and credential composition
Social layer	Cohort, classroom, occasional forum	AI-woven community spanning languages and time zones
What durable competence looks like	Recall, application of known procedures	Judgement, ethics, situational adaptation, critical AI use

Two implications of the table are central to subsequent sections of this report. First, adaptive sequencing of learning, listed in the second column of the table, is contingent on the presence of a **Learning Record System**; without an LRS, the data required to drive adaptation does not accumulate in an interoperable form. Section 9 sets out the conceptual proposal for an Observatory-hosted LRS that would supply this dependency. Second, the **integration of assessment** with learning, also in the second column, depends on the capture of learner performance at the granularity of the assertion. When assertions are captured (for example using the xAPI specification) the same data may be used to inform **adaptive sequencing** within a module, to compose a Verifiable Learning Credential at module completion, and to support flipped-classroom designs in which scarce synchronous time is reserved for those topics on which learners have demonstrated remaining uncertainty.

2.3.3 Design patterns enabled by current AI capability

The most frequently cited application of generative AI in education is the rapid production of content. That application offers efficiency gains; it is not, however, the principal source of pedagogical or androgical innovation. We describe five design patterns, each feasible at scale only as a consequence of recent AI capability, and each is consistent with the four adult learning frames set out in 2.1. The AGRIFOODSKILLS pilot programme should reference these patterns when designing micro-credentials, in preference to designs that incorporate AI solely as a content-generation aid. These are presented at differing levels of readiness for the AGRIFOODSKILLS pilot phase. Some are immediately deployable using widely-available tools (notably continuous formative feedback and augmented multilingual community moderation). Others, in particular immersive AI-driven simulations, are documented as aspirational opportunities for vocational education and training to develop in subsequent phases of work; delivery of these patterns within the present project window is not assumed.

1. **Adaptive, co-created learning pathways.** The learner sets professional goals; an AI assistant negotiates objectives against a competency framework, proposes a sequence of resources and tasks, and reshapes the pathway as performance and reflection accumulate. Instead of a fixed course there is a continuously rebalanced pathway, with a human educator overseeing patterns rather than every step. Anchored in andragogy and self-directed learning.
2. **Immersive AI-driven simulations and living cases (aspirational).** AI-controlled clients, animals, regulators or colleagues that respond conversationally, retain prior interactions and adapt complexity in real time. Where implemented, this approach can support veterinary clinical reasoning, food-safety incident response, biosecurity decision-making and client communication. AGRIFOODSKILLS notes that simulations of this kind require substantial development, validation and ongoing maintenance effort, and that delivery within the AGRIFOODSKILLS pilot programme is not assumed. The pattern is documented here as an opportunity for vocational education and training to develop in subsequent phases of work, in collaboration with sector partners and with specialist providers of simulation technology, rather than as a deliverable of the current project.
3. **Augmented social learning spaces.** Online communities where AI acts as a discussion-weaver: summarising threads, translating contributions across 31 languages, surfacing unresolved tensions, prompting quieter participants. The community of practice becomes more

inclusive, less English-dominant, and easier for a busy practitioner to dip in and out of. Anchored in communities-of-practice theory.

4. **Multimodal co-creation studios.** Adult learners produce real workplace artefacts, protocols, training videos, decision documents, data visualisations, with AI as a creative collaborator, then critique and refine the outputs against professional standards. The learner's competence is in the craftsmanship of critique, not in passive consumption. Anchored in problem-centred andragogy.
5. **Continuous AI-mediated formative feedback.** AI coaches that review reflective journals, written case discussions or recorded consultations over weeks and months, identifying patterns, surfacing blind spots, and prompting deeper inquiry. Feedback is narrative and developmental rather than right/wrong scoring. Anchored in transformative and self-directed learning.

2.3.4 What this means for agrifood and veterinary education

The patterns described in 2.3 admit concrete instantiation within the sectors covered by AGRIFOODSKILLS. Four illustrative examples follow.

1. **AI-orchestrated case-based learning for veterinarians.** A vet preparing for a clinical specialisation works with an AI that generates living cases tailored to her practice profile, drip-feeds data through a Mezirow-shaped sequence (disorienting dilemma → critical reflection → reframing → integration), and supports a 90-minute synchronous case conference with peers where the human clinician-educator focuses on the move into integration.
2. **Personal practice coach for farm managers and processors.** A dairy farm manager uploads anonymised herd records, milking-system logs and his own reflective notes. An AI coach maps these against an ESCO-tagged competency framework, surfaces blind spots, and proposes a short pathway of micro-learning units plus an on-site peer demonstration. The human educator reviews the patterns across a cohort, not every interaction.
3. **Competency-linked portfolios with AI curation.** A meat-processing supervisor builds a digital portfolio of work artefacts, revised protocols, audit findings, training videos for her own team. AI auto-tags each artefact to ESCO competencies, drafts evidence statements, and proposes a microcredential. The human assessor refines and validates rather than starting from scratch.
4. **Augmented multilingual sector communities.** The Agrifood Pact for Skills Observatory hosts a sector-wide practitioner community in which AI weaves discussion across the project's nine working languages, summarises emerging themes for the steering group, and surfaces practice innovations that would otherwise be invisible to anyone not reading the language they happened in.

Hero example: narrative spine for the report

An EQF 7 veterinarian, working in mixed practice, enrolling in an PACT FOR SKILLS microcredential on antimicrobial stewardship.

Week 1: AI assistant negotiates a personalised pathway against an ESCO-tagged competency map and the vet's own caseload.

Weeks 2-6: a sequence of AI-driven living cases drawn from her actual practice, structured through Mezirow's phases, with reflective journal entries reviewed by an AI coach.

Week 7: a 90-minute synchronous case conference in her own region, scaffolded by a clinician-educator.

Weeks 8-12: a work-based project in her own practice with mentor sign-off.

The credential issued using the European Learning Model (ELM), ESCO-tagged at the competency level, signed under eIDAS 2.0, held in her EUDI Wallet, and, conceptually, anchored as a set of assertions in an Observatory-hosted Learning Record System that can also recognise VET/ Continuing Professional Development (CPD) she undertakes with other providers across her career.

2.3.5 Quality, ethics and governance in the AI era

The design patterns described in 2.3 are associated with substantial governance, ethical and regulatory considerations. The AGRIFOODSKILLS pilot programme should treat each of the following as a design constraint rather than as an external consideration.

1. **The EU AI Act** treats certain educational uses as high-risk, with specific obligations on transparency, human oversight, data quality and post-market monitoring. AGRIFOODSKILLS designs should be compliant by design, not retrofitted.
2. **General Data Protection Regulation (GDPR)** and learner data. Reflective journals, performance logs, recorded interactions and case data are personal data and often special-category data. Anonymisation, purpose limitation and learner control over their own records are non-negotiable.
3. **Hallucination, bias and equity.** AI output is generated, not retrieved; it is sometimes confidently wrong and frequently biased. The competence we want learners to develop is not blind reliance on AI but critical engagement with it.
4. **Digital divide and accessibility.** AI tools that depend on high-bandwidth always-on connectivity exclude many sector workers. AI tools that translate, caption, simplify or provide audio alternatives include many learners who were previously excluded. Both effects are real and have to be designed for.
5. **Trust infrastructure.** Claims about competence made on the back of AI-mediated learning need the same trust architecture as any other claim: qualified issuers, eIDAS trust services, ELM-conformant data, EQAVET quality assurance. AI does not exempt anyone from the credentialing architecture; if anything, it raises the stakes.

2.3.6 AI competency as a new transversal requirement

A consequence of the developments set out in 2.2 to 2.3 is that **artificial intelligence (AI) competency** is now a transversal requirement on the agrifood and veterinary workforce alongside the established domain competencies. The term competency is used here in preference to literacy advisedly. **Literacy frameworks** address basic familiarity and access; the requirement on the present workforce is operational competence in deciding when and how to use AI within professional tasks, and in justifying those decisions to colleagues, employers, regulators and clients. The distinction matters in adult vocational education and training (VET) because the learners are not encountering AI as an abstract subject; they are encountering it embedded in the tools of their working day.

AGRIFOODSKILLS does not propose to develop a stand-alone AI module or a separate AI literacy framework. A module taught in isolation from domain practice would have limited value for an adult VET learner, and the embedded presence of AI within sector tools makes domain-integrated treatment more useful than free-standing instruction. The project instead adopts a four-element working construct for AI competency that may be applied within any module in which AI plays a role. The construct is referred to here as **QUSC: Question, Use, Scrutinise, Communicate** (Figure 2).

- **Question:** The user asks whether the AI output is true. An informed user assumes that generative AI output may be inaccurate, may be confidently wrong, and may contain hallucinations. The user interrogates the output before relying on it, and recognises the conditions under which hallucination is most likely.
- **Use:** The user uses AI tools effectively. Effective use includes prompt design, the choice of model, an understanding of when local computation is appropriate, and an understanding of what is lost when AI is not used. Competence under this heading covers both deployment and informed abstention.
- **Scrutinise:** The user asks whether the tool is the right tool, trained on the right data, fine-tuned in an appropriate way for the task at hand. Scrutiny extends to the provenance of training data, to known biases, and to the alignment of the tool's outputs with sector

norms. Scrutinise is of particular importance in the agrifood and veterinary context: unlike human medicine, animal medicine and farm production are not subject to **medical-device regulation**. There is no regulatory floor beneath the use of AI in these settings, and the floor must be constructed within professional practice.

- **Communicate:** The user discloses that AI was used, the purpose for which it was used, and the nature of the AI-generated content. Disclosure is treated as a positive expression of professional standards rather than as an admission. Open communication about AI use supports peer review, supports regulatory transparency, and contributes to the development of shared professional norms.

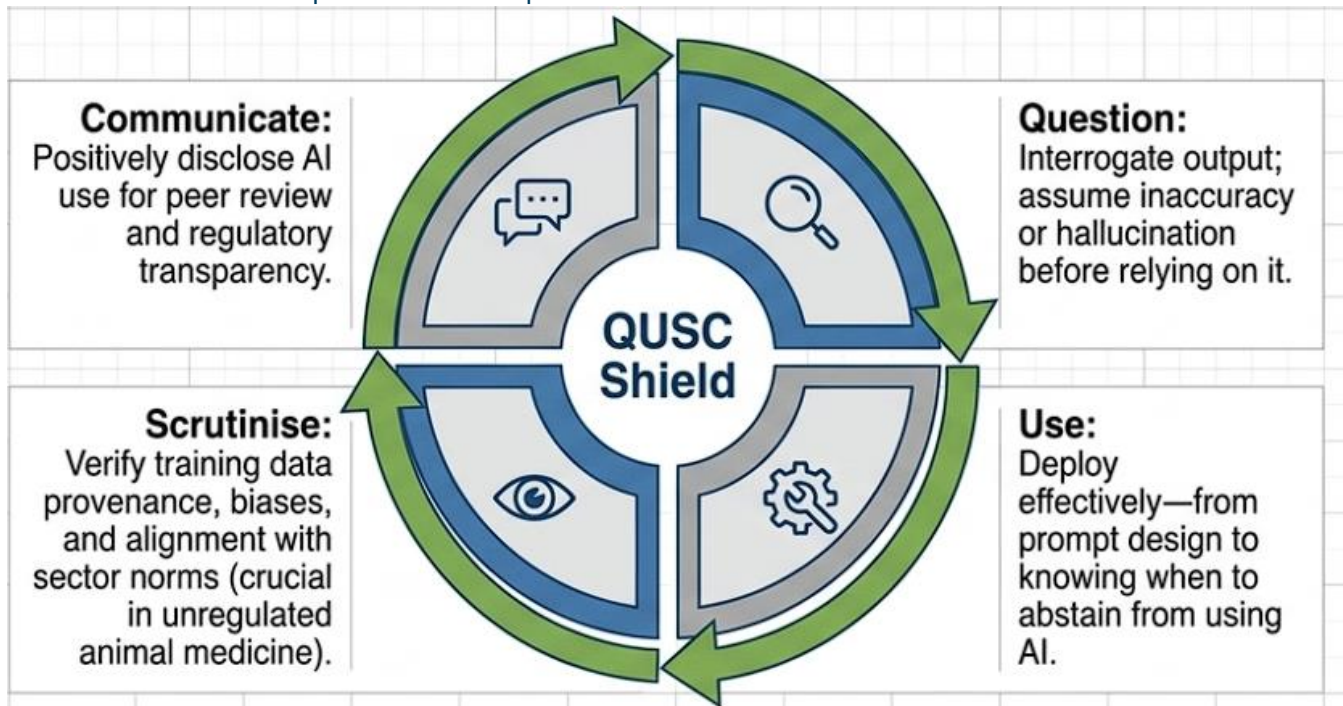


Figure 2: Proposed competency framework for Artificial intelligence skills in adult learning under the AGRIFOODSKILLS

The QUSC construct is offered as a **working framework** rather than as a complete competency taxonomy. Each of the four elements maps to existing and emerging concepts within the European Skills, Competences, Qualifications and Occupations classification (ESCO) and may be referenced in micro-credentials at a level of granularity appropriate to the European Qualifications Framework (EQF) level of the occupational profiles. The AGRIFOODSKILLS occupational profiles' curricula to be developed in Task 5.2 should incorporate one or more elements of QUSC as transversal elements within their domain-specific competency declarations. Both sectoral employer requirements and the entry into application of the AI Act indicate that AI-related competencies will be required for professional recognition during the period 2026 to 2030.

Literature in brief: AI in adult education

Four intersecting strands of evidence inform this section. (1) AI in adult education and lifelong learning: conceptual and review work, including Storey and Wagner on large language models (LLMs), adaptive systems, virtual and augmented reality, and learning analytics in adult learning. (2) Digital and open education: design heritage from connectivism, scenario-based learning and open pedagogy on which current AI capability builds. (3) Learning analytics and data-informed pedagogy: dashboards for tutors becoming real-time adaptive experiences for learners. (4) Ethics, equity and critical engagement with AI: policy and academic work on integrity, bias, privacy and the digital divide. Policy frames: the AI Act (Regulation (EU) 2024/1689); the European Quality Assurance in Vocational Education and Training framework (EQAVET); the European Skills Agenda. Field-practice references: the European Platform for Adult Learning (EPALE); the Digital Skills and Jobs Platform.

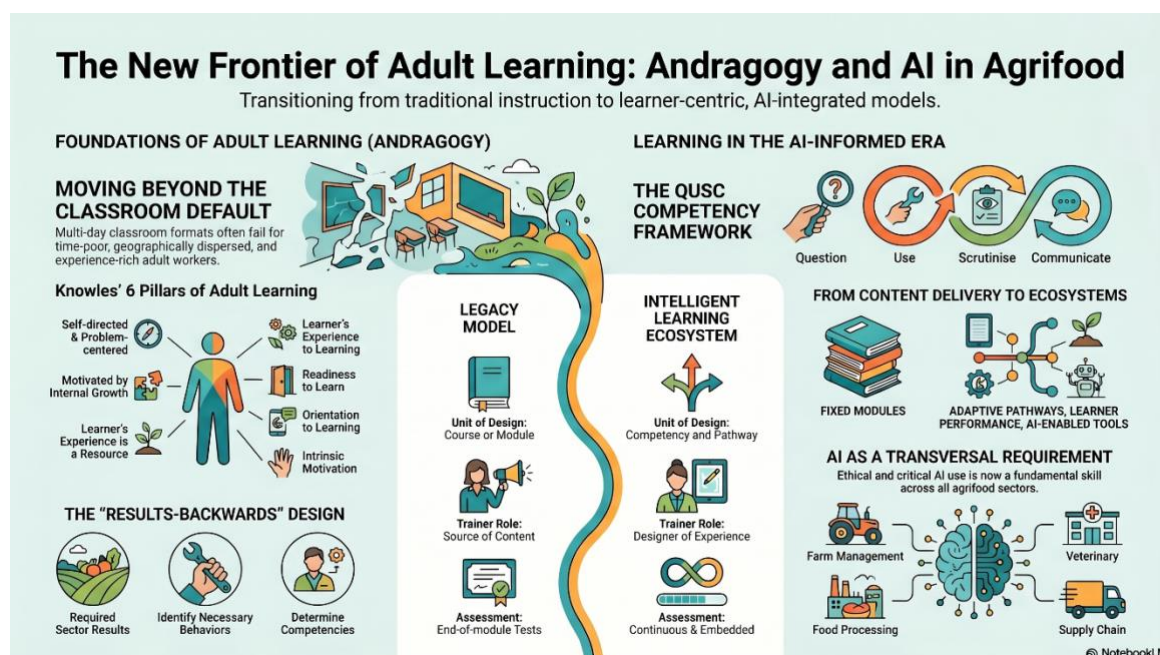
Design summary: what to do today

- When designing a micro-credential or learning activity that involves AI, consider at least one element of the QUSC framework explicitly in the design. The four elements are not all required for every design; consideration of at least one is.
- Pick a micro-credential you are already designing and rewrite its opening as a learner-led problem statement that an AI assistant could negotiate against.
- Incorporate one or more QUSC elements as a transversal strand of every occupational profile you author, at a granularity appropriate to the EQF level.
- State the AI Act and General Data Protection Regulation (GDPR) posture for the design in plain language before construction begins.
- Record which parts of the learning are AI-mediated; make this visible in the credential metadata at issuance.

2.3.7 Implications for educators and for the project

In an AI-informed learning ecosystem the role of the educator **extends beyond content delivery**. The educator's responsibilities include the **design of learning experiences**, the **curation of resources** and AI tools, the provision of **ethical oversight**, the **verification of evidence of practical competence** and the **scaffolding of learner-AI dialogue**. The competencies required of trainers are correspondingly different from those required under content-delivery models, and the AGRIFOODSKILLS train-the-trainer programme (Task 6.1) should specify those competencies explicitly. At institutional level the role of the provider extends from the delivery of content to the coordination of an ecosystem of contributing providers, assertions and credentials. The Agrifood Pact for Skills is positioned to perform this coordination function. Section 9 sets out one possible architectural approach.

SUMMARY INFOGRAPHIC



3. EQF differentiation: what level means for the learner

AGRIFOODSKILLS designs micro-credentials for two bands of the European Qualifications Framework (EQF): levels 3-5 (vocational, including adults employed in sector roles) and levels 6-8 (higher education students, postgraduates and qualified veterinarians) (Figure 3). The same Knowles foundation applies; the engagement at each band looks materially different. This section sets out the descriptor logic and turns it into engagement implications.

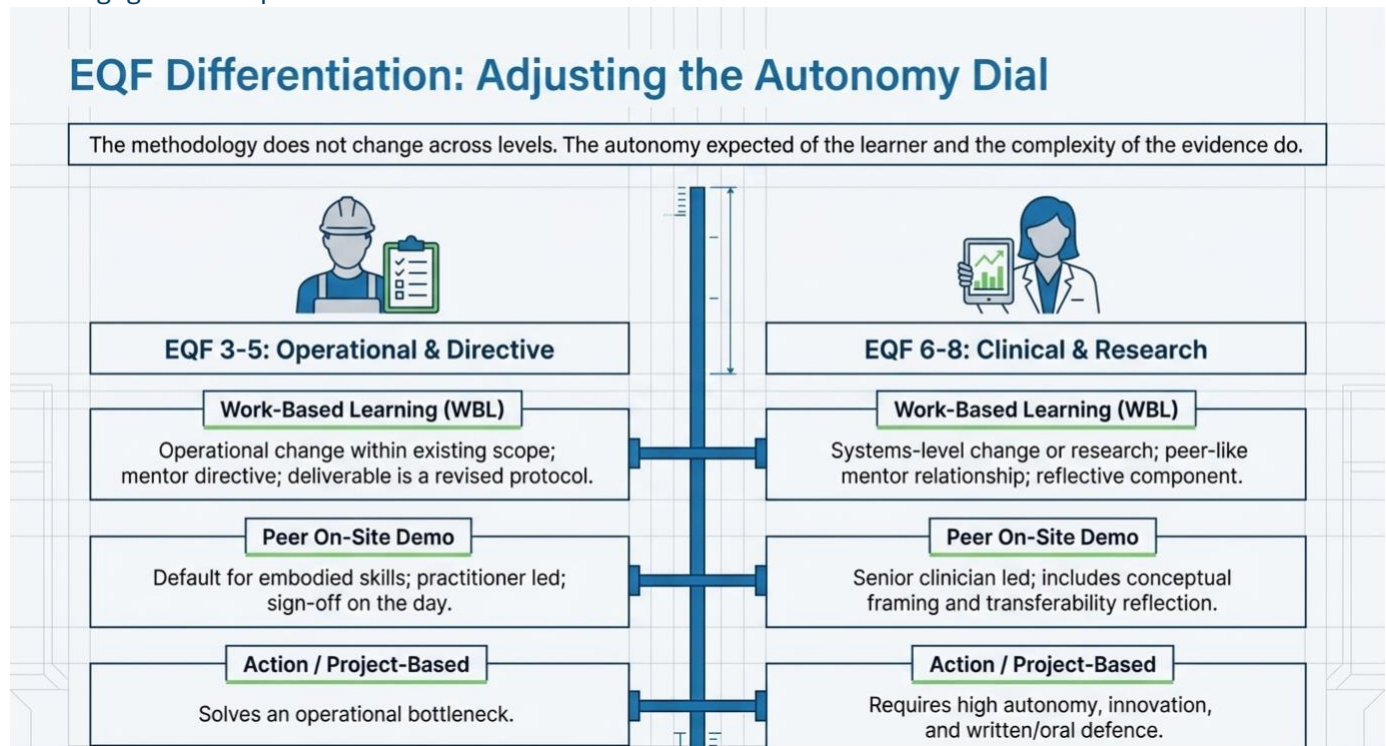


Figure 3: Overview of the key differentiators of learning design as they apply to different levels of European Qualifications Framework.

To maximize participation, learner profiles are defined by their structural barriers:

- **EQF 3–5 (The Frontline Workforce):** These learners are characterized by **'time poverty'** and a **'confidence gap'**. They cannot afford operational downtime. Therefore, the methodology prioritizes **mobile-first, low-text-density modules (5–10 minutes)** delivered through existing workplace tools like farm-management platforms.
- **EQF 6–8 (Specialists and Managers):** These learners require a **'Systems Thinking' approach**. Their learning paths utilize **flipped-classroom designs** and **AI-enhanced co-learning** to link clinical or technical tasks directly to broader sectoral resilience and **One Health outcomes**.

3.1 The three EQF descriptor strands

The EQF describes each level along Knowledge (factual and theoretical), Skills (cognitive and practical), and Responsibility and Autonomy (independence, breadth of context, leadership of others). The strand that shifts most sharply across the 3-5 / 6-8 boundary is responsibility and autonomy. The same methodology, such as project-based learning, is deployed in materially different forms at EQF 3 and at EQF 7. The methodology itself does not change; the autonomy and complexity expected of the learner do.

EQF	Knowledge	Skills	Responsibility & autonomy
3	Knowledge of facts, principles, processes and general concepts in a field.	Range of cognitive and practical skills to accomplish tasks and solve problems by selecting and applying basic methods, tools, materials and information.	Take responsibility for completion of tasks; adapt own behaviour to circumstances.
4	Factual and theoretical knowledge in broad contexts.	Range of skills required to generate solutions to specific problems.	Self-management within usually predictable contexts; supervise routine work of others.
5	Comprehensive, specialised, factual and theoretical knowledge; awareness of boundaries.	Comprehensive range of cognitive and practical skills to develop creative solutions to abstract problems.	Management and supervision in contexts of unpredictable change; develop performance of self and others.
6	Advanced knowledge involving critical understanding of theories and principles.	Advanced skills demonstrating mastery and innovation; solve complex and unpredictable problems.	Manage complex technical or professional activities; take responsibility for decision-making in unpredictable contexts.
7	Highly specialised knowledge, some at the forefront of a field, as basis for original thinking and/or research.	Specialised problem-solving skills in research and/or innovation; integrate knowledge across fields.	Manage and transform complex, unpredictable contexts requiring new strategic approaches; review strategic performance.
8	Knowledge at the most advanced frontier; interface between fields.	Most advanced and specialised skills including synthesis and evaluation; extend and redefine practice.	Substantial authority, innovation, autonomy, scholarly and professional integrity; sustained commitment to developing new ideas.

3.2 EQF 3-5 learner profile in agrifood and veterinary

Learner profile. For adults in the agrifood and veterinary sectors at EQF levels 3 to 5, such as farm operatives, milking-parlour operators, processing-line supervisors, quality-assurance technicians, veterinary nurses and agricultural contractors, working days are typically long, physical and structured around time-bound operational tasks. Most learners have completed their initial formal education. Working knowledge in these populations is generally extensive but tacit, accumulated through years of practical experience rather than through formal study. Preferences expressed in consortium surveys and in prior Erasmus+ Blueprint pilots are consistent: short, applied learning units that acknowledge existing experience are associated with higher engagement and higher completion than extended classroom formats.

Implications for engagement design.

- Author online components in short modular units, typically 10 to 30 minutes in length.
- Limit synchronous on-site components to half a day at most.
- Use peer-to-peer demonstration on the work site and work-based learning (WBL) as the principal modalities for embodied skills.
- Draw examples and assessment items from sector practice rather than from abstract cases.
- Rely on performance evidence rather than written examination where the competence is

operational.

- Link the credential to recognisable changes in professional standing or scope of practice.

3.3 EQF 6-8 learner profile in agrifood and veterinary

Learner profile. Learners at EQF levels 6 to 8 in these sectors are typically students in higher education, veterinarians undertaking clinical specialisation or continuing professional development (CPD), and research-active professionals. They have established self-management capabilities and habits of independent study, and they have the capacity for conceptual and research-engaged work. Time constraints in this population originate principally in competing intellectual and professional commitments rather than in physical working patterns.

Implications for engagement design.

- Author asynchronous components at a length appropriate to the topic; substantial durations are credible at this band.
- Use flipped-classroom designs that presume substantial preparatory work by the learner.
- Use project-based learning of comparable scope to research or clinical questions encountered in professional practice.
- Use AI-enhanced learning in which AI tools are treated as peer interlocutors subject to critical evaluation by the learner.
- Include written and oral defence alongside performance evidence in assessment.
- Align credentials with recognised professional progression frameworks, including veterinary specialisation pathways, higher-education module banks and industry certifications.

Literature in brief: EQF and the responsibility-and-autonomy strand

Official source: [Europass](#) description of the eight EQF levels and the underlying 2017 Council Recommendation. Practical commentary: European Proceedings (2019) on the responsibility-and-autonomy domain, the strand that shifts most across the 3-5 / 6-8 boundary and that the methodologies in Section 9 must adapt to. National qualifications frameworks differ in their alignment with EQF; the AGRIFOODSKILLS architecture treats EQF as the canonical reference and lets national instruments map onto it.

Design summary: questions to answer before you start designing

- What is the learner's working context: what shifts, what travel, what physical demands, what device access?
- Where in the day or week can learning happen at all? In ≤ 30 minute blocks on a phone? Half-day on site? An evening online?
- What prior experience must I assume present, and how will I respect it in the opening of the module?
- What is the EQF level, and what does that imply about the autonomy I expect from the learner and the evidence I require from them?
- What ESCO competencies are the prerequisites (in) and what ESCO competencies are the outgoing claims (out)? Write these before you write the learning outcomes.
- What evidence will I produce for each of the three assertions (engagement, assessment, can-do-the-role)?

4. Learning design principles

Section 4 sets out the design frameworks the project uses to translate theory and EQF level into the design of training. The section combines two strands: established instructional-design frameworks (blended pedagogy and andragogy, ABC, ADDIE, Universal Design for Learning, and Bloom's Revised Taxonomy) on the one hand, and the move from learning outcomes to a stackable competency architecture on the other. Together these supply both the process for authoring modules and the architectural commitment that AGRIFOODSKILLS modules are designed at the competency level.

4.1 A blended approach to pedagogy and andragogy

Recognising the unique characteristics of our diverse learner base, from experienced professionals to university students, the AFS project has adopted a blended approach that intelligently combines elements of both pedagogy and andragogy in our course design and development. This blended methodology will ensure that:

- Foundational knowledge and complex new concepts are delivered through clear, structured, and engaging methods (pedagogical elements), providing a strong base for all learners. Learners can then transition to self-directed, problem-based, and experiential activities (andragogical elements), allowing them to apply knowledge, draw on their experiences, and engage in peer learning.
- Ensuring that training is flexible and modular, accommodates different learning styles and allows participants to personalise their learning paths to address specific needs, whether for ECVET certification or targeted skill acquisition.
- By strategically integrating the best practices from both pedagogical guidance and adult-centered learning, we aim to create highly effective training programmes that empower all learners, maximise knowledge transfer, and cater to their varied contexts and goals. This ensures engagement and relevance for experienced professionals, while also providing necessary structure and support for those building new foundational knowledge.

4.2 Learning design models (ABC, ADDIE, UDL)

When designing and developing the AFS project training materials, we carefully considered various learning design models to ensure they are robust, engaging, and effective for our diverse audience, which includes experienced adult professionals like farmers, foresters, and farm advisors, as well as university students. These models provide structured frameworks that guide the creation of high-quality learning experiences. The three key models that informed our approach are:

4.2.1 ABC Model – Quick & Collaborative Design

The ABC Learning Design Model is a hands-on, workshop-based approach that helps design teams quickly develop a 'storyboard' for a learning programme or module. Design teams will sketch out the main plot points of a story before writing the full script. In a relatively short session, often using physical or digital cards, designers map out different types of learning activities that will take place in the course. These activities are categorised into six types:

- Acquisition:** Learners consume information (e.g., reading, watching videos).
- Investigation:** Learners explore and research (e.g., finding resources online).
- Discussion:** Learners communicate and debate with peers or instructors.
- Collaboration:** Learners work together on shared tasks.

- e. **Practice:** Learners apply skills in controlled environments (e.g., quizzes, simulations).
- f. **Production:** Learners create something new (e.g., a report, a plan, a demonstration).

The ABC Model is particularly beneficial for its speed and collaborative nature.

Andragogical Links: Its emphasis on a variety of activities naturally supports adult learners' preferences for active, hands-on, and problem-oriented learning. The design process encourages the inclusion of collaborative and discussion-based elements, which are vital for adults who learn well through interaction and sharing experiences. It also allows for rapid prototyping of modules that can be quickly refined to ensure relevance to the practical needs of farmers and professionals.

Pedagogical Links: While quick, it still provides a structured overview, helping to ensure a logical flow of learning. This is useful for establishing clear foundational learning pathways for topics where students might need more direct guidance and for ensuring a balance of different activity types to cater to varied learning styles.

4.2.2 A.D.D.I.E. Model: A systematic approach to Instructional Design

The ADDIE Model is a widely recognised, systematic instructional design framework that guides the development of learning programmes through five sequential phases:

A - Analyse: This phase involves understanding the learners (their existing knowledge, skills, needs, and learning context), the learning objectives, and the resources available.

D - Design: Here, the learning objectives are translated into a detailed plan, including content outlines, instructional strategies, assessment methods, and media choices.

D - Develop: The actual learning materials are created or assembled based on the design plan (e.g., presentations, handouts, online modules, activities).

I - Implement: The training programme is delivered to the learners.

E - Evaluate: The effectiveness of the training is assessed against the initial learning objectives, often leading to revisions and improvements.

ADDIE provides a thorough and structured methodology, particularly valuable for larger, complex training initiatives.

Andragogical Links: The initial 'Analyse' phase is crucial for an andragogical approach, as it requires a deep understanding of the adult learners' existing experiences, their specific job-related problems, and their motivations for learning. This ensures the developed training is highly relevant, practical, and tailored to their needs. The 'Evaluate' phase allows for continuous feedback and improvement, aligning with adult learners' desire for useful outcomes and practical application.

Pedagogical Links: ADDIE's systematic nature ensures that all learning objectives are meticulously defined and that content is developed in a logical, progressive manner. This is ideal for establishing clear, foundational learning paths, especially for university students or individuals new to a subject, where a well-structured curriculum and comprehensive materials are essential for building core knowledge effectively.

4.2.3 Universal Design for Learning (U.D.L) - Learning for Everyone

Universal Design for Learning (UDL) is a framework that guides the design of learning environments and materials to be accessible and engaging for *all* learners, regardless of their individual differences in how they learn, their background, or any challenges they might face. Instead of adapting a course for specific needs *after* it's been designed, UDL aims to build flexibility and options into the curriculum from the outset. It's based on three core principles:

1. **Multiple Means of Representation (The "What" of Learning):** Providing information and content in different ways (e.g., text, audio, video, diagrams) to cater to diverse perceptions and learning styles.
2. **Multiple Means of Action & Expression (The "How" of Learning):** Offering various ways for learners to demonstrate what they know (e.g., written assignments, oral presentations, practical demonstrations, group projects) to accommodate different physical, cognitive, and communicative abilities.
3. **Multiple Means of Engagement (The "Why" of Learning):** Providing varied options to stimulate interest, motivation, and self-regulation (e.g., choice in activities, relevant content, collaborative opportunities, timely feedback) to appeal to different emotional and motivational needs.

UDL ensures that our training programmes are inclusive and highly adaptable.

Andragogical Links: UDL strongly supports an andragogical approach by offering flexibility and choice, allowing adult learners to select methods of representation, action, and engagement that align with their preferred learning styles, prior experience, and practical needs. Providing multiple ways to engage directly addresses adults' internal motivation and desire for relevance, ensuring they can connect with the material in personally meaningful ways. The emphasis on problem-solving and collaboration fits well with adults' practical orientation.

Pedagogical Links: For foundational learning or for university students, UDL ensures that new or complex information is presented in various formats, helping to clarify concepts for all. By offering different ways to practice and demonstrate understanding, it supports diverse cognitive processing and allows instructors to assess comprehension effectively, even within a structured learning environment. It helps prevent barriers to learning from the start, benefiting all students, especially when encountering new subject matter.

5. Programme development: how to write a curriculum from an occupational profile

Section 5 sets out the process the project uses to translate the skills intelligence produced in WP2 and WP3 into the curriculum that will be delivered through the pilot programme. The process maps onto the Analyse, Design, Develop, Implement and Evaluate phases of ADDIE (Section 4.2) but is described here with the operational granularity that the project needs. Sub-sections cover assessing skills needs, translating occupational profiles into curricular frameworks, the curricular design workshop method, piloting and delivery, and the accessibility and scalability features that allow the project's outputs to be adopted across Member States.

5.1 Assessing skills needs and translating profiles

This section provides a comprehensive overview of the systematic approach adopted for the Learning Design, Development, and Delivery of the training programme. It outlines the crucial stages involved in transforming identified skill needs into robust, engaging, and impactful educational experiences. From the foundational principles guiding the structure and conceptualisation of learning activities (Design), through the meticulous creation and adaptation of educational content and materials (Development), to the practical implementation and strategic rollout of the training initiatives (Delivery), this section details the methodologies and considerations that ensure the programme is effective in enhancing the learning experience for all stakeholders across the agri-food value chain. The aim is to maximise learner engagement and facilitate the achievement of desired learning outcomes, contributing to the overall objectives of addressing current and future skill needs in agriculture.

5.1.1 Assessing the Skills Needs

Following the design process, the AGRIFOODSKILLS (AFS) Project partners will progress to the development of training materials, taking into account several key considerations. A systematic approach was adopted to understand the current and future skills needed within the sector. This began with insights derived from the skills intelligence conducted in Work Package 3 (WP3) of the AGRIFOODSKILLS Project, specifically the skills gaps analysis and the comprehensive skills mapping across the ESCO, GreenComp, and DigiComp frameworks. Further understanding was gained through the mapping of existing occupational profiles from the FIELDS and I-RESTART Projects, alongside the CEDEFOP Occupational Frameworks.

These frameworks consistently delineate the essential skills, competencies, knowledge, and optional skills and knowledge bases required for occupational profiles throughout the agricultural sector. Additional critical skills insights were gathered via the Agri-Food Pact for Skills and a series of skills-specific reports, all contributing to the research on identifying the sector's current and future skill requirements. Furthermore, Task 3.3 of the AGRIFOODSKILLS Project involved mapping existing training provisions across Europe to aid in the precise identification of skills needs. Collectively, these comprehensive resources and activities formed the robust foundation for developing training materials, specifically addressing the knowledge, skills, and competences required for 8 of the 12 new occupational profiles developed, as per the requirements of Task 5.4

5.1.2 Translating Occupational Profiles into Design & Development Frameworks

Subsequent to the detailed assessment of skills needs, the AGRIFOODSKILLS Project undertook the development of 12 new occupational profiles. The 12 profiles were validated and improved through a series of focus groups involving a broad range of stakeholders, ensuring their relevance and accuracy. The strength of these frameworks lies in their structured outline of essential skills,

competencies, and knowledge, as well as optional skills and knowledge bases pertinent to various occupational roles within the agricultural sector. This granular detail is crucial for translating these profiles directly into actionable design and development frameworks for training. By systematically breaking down each occupational profile into its constituent knowledge, skills, and competences, a clear pathway is established for defining specific learning outcomes. These defined outcomes then serve as the blueprints for the creation of targeted and effective training materials. This ensures that the developed content is directly aligned with the identified needs of the sector and the requirements of the new occupational profiles.

5.1.3 Curricular Design Approach

In designing the curriculum for the training content, the AGRIFOODSKILLS Project partnership will convene a dedicated design workshop. During this workshop, partners will collaboratively develop content specifically tailored to address the identified gaps in skills, competencies, and knowledge base, aligning with the prioritised 8 of the 12 new occupational profiles. These occupational profiles are structured into four sub-categories, with three profiles in each: veterinary practices, animal production, plant production, and the food industry. Additionally, these profiles are linked to the European Qualifications Framework (EQF) at levels 3-5 and 6-8 respectively. While the exact EQF level for each occupational profile is yet to be precisely determined, this will be a fundamental consideration during the curriculum design phase. The designed curriculum will then form the basis for training delivery. A strategic emphasis has also been placed on drawing upon existing training materials. The partnership has agreed to prioritise the development of new and innovative materials for missing content or to enhance existing provisions, thereby maximising the return on investment. The overarching goal is to ensure that the training materials are practical, skills-based, and highly adaptable, capable of being tailored to different learner needs and various training approaches.

5.2 Delivery: lesson plans, pilots and e-learning

Building upon the meticulously designed training methodologies and curricula, this section outlines the comprehensive strategy for the effective **delivery** of our training programmes. The primary objective is to ensure a high-quality, consistent, and impactful learning experience for all participants, applying a blend of innovative pedagogical approaches. This involves the systematic development of detailed lesson plans to guide trainers, the rigorous validation through pilot training sessions, and the strategic implementation of e-learning to ensure widespread accessibility and scalability. Our approach integrates methods such as **online learning, microlearning, gamification, flipped classroom, blended learning, peer learning, on-farm demonstration activities, and participatory learning**, all aimed at enhancing learning while cultivating real behavioural change as discussed in section 6. This structured framework ensures that the training not only imparts knowledge but also contributes to reskilling and upskilling of the workforce in the plant, animal production, agrifood, and veterinary sectors with innovative and future-proof skills for agriculture.

5.2.1 Lesson Plans

To provide a standardised approach to delivery of training programmes, the project will develop comprehensive lesson plans for each module and individual lesson. These meticulously designed plans will serve as a foundational guide for trainers, ensuring uniformity and quality across all learning sessions and creating an effective learning environment. However, they should not be seen as restricting either the order of learning, where pre-requisites permit, nor limit the value of self-paced adult education and AI tailored learning as discussed above.

Each lesson plan will systematically outline the following key elements:

- **Module Title:** Clearly identifying the overarching subject area of the training.
- **Lesson Title & Number:** Providing a specific identifier and sequential placement within its respective module.
- **Prerequisite competencies:** These define what a learner must be able to do to benefit from the materials. They are the unique structure that facilitate self-paced and self-ordered learning and are the signals that can inform AI when it creates tailored learning plans.
- **Learning Objectives/Key Learning Points:** Defining what learners are expected to know or be able to achieve by the conclusion of the lesson. **Competency statements:** Covering, where relevant the competencies required to benefit from a module of training, and the competencies expected at the end of the training.
- **Content Coverage:** Detailing the specific topics and information to be presented and discussed.
- **Delivery Methodology:** Specifying the pedagogical approaches to be recommended for each segment, such as online learning, microlearning, gamification, blended learning, or participatory activities, aligning with the project's defined training strategies. These are provided for guidance but should not limit modern approaches to andragogy.
- **Delivery Timing:** Allocating a timeframe for each distinct element or activity within the lesson, ensuring efficient use of learning time. While also allowing flexibility
- **Resources Needed:** Listing all necessary materials, including presentations, handouts, digital tools, equipment, or external links.
- **Activities to be Utilised:** Outlining practical exercises, group discussions, role-plays, case studies, or other interactive elements designed to foster active learning.
- **Assessment Methods:** Describing how learning will be evaluated for that specific lesson, which could include quizzes, practical tasks, discussions, or peer feedback.

This structured approach to lesson planning will facilitate effective programme execution, support trainers in maintaining high instructional standards, and ultimately enhance the learning experience for all participants.

5.2.2 Pilot Training Sessions

Following the meticulous design of our training methodologies and the development of detailed lesson plans, the next crucial step in ensuring the efficacy and impact of our programmes will be the execution of Pilot Training Sessions, as outlined in Work Package 6, Task 6.2. These sessions are designed to rigorously test and validate all training materials and delivery approaches in a practical setting and form a vital part of the evaluation cycle within the EQAVET framework.

The primary objective of the pilot sessions is to provide invaluable feedback, enabling us to refine both the content and the specified delivery methodologies – including online learning, microlearning, gamification, flipped classroom, blended learning, peer learning, on-farm demonstrations, and participatory learning ensuring they are robust, efficient, and highly engaging.

This iterative process is vital for optimising the learning experience and guaranteeing effective knowledge transfer for all participants.

Crucially, the pilot training sessions will not only assess the acquisition of knowledge but also evaluate their capacity to deliver real behavioural change. By observing how participants apply newly acquired skills and competences within simulated or actual scenarios, we can ensure that our training goes beyond theoretical understanding, encouraging practical application and contributing to the project's overarching goal of reskilling and upskilling people across the agricultural sector. The insights gained from these pilots will be instrumental in making necessary adjustments to achieve optimal learning outcomes and promote lasting impact.

5.2.3 E-Learning

E-learning, or electronic learning, fundamentally represents a continuing education process that leverages digital technologies and networks to design, deliver, select, manage, and extend learning experiences. It's a dynamic approach that facilitates access to resources and services, encouraging remote collaboration and improving the quality of learning. The core elements for effective e-learning content design include interactivity (promoting learning by doing), dynamism (acquiring targeted skills just-in-time), and modularity, which allows course content to be organised according to specific training objectives and user needs. This modularity also means that individual learning blocks, or 'learning objects', can be easily reused and reassembled to form new courses, enhancing flexibility.

In line with this modern pedagogical approach and building upon the principles of Open Access and Open Licence requirements stipulated by Erasmus+, all developed training materials, encompassing diverse methodologies such as microlearning, gamification, flipped classroom, blended learning, peer learning, and practical on-farm demonstrations will be made readily available via a dedicated digital repository. This central repository will be hosted on the Moodle platform, which was selected specifically for its open-source nature, robust multilingual capabilities supporting the project's targeted languages, and high customisation potential. Furthermore, its compatibility with smartphones, tablets, and personal computers ensures universal accessibility.

This comprehensive digital resource represents a significant added value and an advancement in our training delivery. Unlike previous projects where outputs, even if free, might be overlooked or not regularly updated once the consortium disbanded, our approach prioritises **long-term sustainability and usability and adaptability to changing patterns and needs**. The integration of a handbook on learning methodologies directly into the platform, coupled with the application of a Creative Commons license (CC BY-NC 4.0) during and four years post-project, ensures that these valuable Open Educational Resources (OERs) remain relevant and usable.

For **learners**, access to E-Learning platforms provides continuous, on-demand access to a personalised learning path compatible with the concepts of andragogy, allowing them to focus only on modules relevant to their skill needs, whether or not they pursue ECVET certification. The content, designed for interactivity and active learning, should be presented through easily navigable sections with clear icons, supporting engagement and catering to different learning paces. While the AGRIFOODSKILLS project does not provide learner facing content, the resources are structured to allow VET providers to administer these as relevant to their learners needs.

For **trainers and facilitators**, the repository provides a rich, organised bank of materials to aid in the preparation and delivery of training sessions. It ensures consistency, offers supplementary resources to enrich the learning journey, and supports the project's goal of promoting creative,

entrepreneurial, and critical thinking skills across diverse target groups. This strategic investment in an accessible, open-source, and continuously supported digital repository aims not only to meet immediate training needs but also to establish a lasting, adaptable resource for future VET and education providers, driving innovation in agricultural training.

5.2.4 Accessibility & Scalability

Our comprehensive approach to training design and delivery, as detailed throughout this section, is fundamentally rooted in ensuring broad **accessibility** and robust **scalability** across the entire agri-food sector aligned to Web Content Accessibility Guidelines. This strategic focus aims to provide unparalleled access to high-quality learning opportunities for as many individuals as possible, encompassing both trainers and learners across all four critical sub-sectors: animal production, plant production, veterinary practices, and the food industry.

By employing a diverse array of modern training methodologies, including online learning, microlearning, gamification, flipped classroom, blended learning, peer learning, on-farm/on-site demonstration activities, and participatory learning, the project's training materials are designed to be adaptable for all varieties of training contexts. This flexible and modular content structure serves as a powerful enabler of effective knowledge transfer and promotes tangible behaviour change, thereby supporting crucial development opportunities throughout the agri-sector. The digital repository, hosted on a user-friendly, multilingual Moodle platform, further amplifies this accessibility, ensuring that resources are readily available on multiple devices and in various languages, catering to the diverse needs of our European audience, thereby strategically contributing to a community of practice and the Observatory for the Agrifood Pact for Skills.

Crucially, the inherent **scalability** of this educational framework is a cornerstone of its long-term impact. The project's outputs and methodologies are specifically designed to facilitate widespread adoption through initiatives such as the Agri-Food Pact for Skills. This will enable a seamless rollout of training programmes and resources at regional, national, and ultimately, European levels. By providing open educational resources (OERs) under appropriate open licences we empower Vocational Education and Training (VET) providers and Higher Education (HE) institutions and Industry Partners to integrate, adapt, and utilise these materials within their own curricula, supporting a collaborative ecosystem of continuous learning and development.

This forward-thinking strategy promotes the idea that the project's resources will aid significantly in the long-term sustainability of the agri-food sector well into the future. By equipping the workforce with essential skills, we directly support advancements in key areas such as competitiveness, digitalisation, and overall sustainability of the sector. This not only translates into enhanced productivity and innovation for businesses but also provides invaluable upskilling and reskilling pathways for individuals, ensuring their relevance and professional growth in a rapidly evolving landscape. Ultimately, this comprehensive, accessible, and scalable approach benefits everyone: learners gain new competencies, trainers have rich resources at their disposal, businesses thrive with a skilled workforce, and VET and HE providers are bolstered with cutting-edge, adaptable training content, collectively strengthening the resilience and future prosperity of the European agri-food sector.

5.3 From learning outcomes to a competency stack

The argument set out in Section 3 against reliance on the multi-day classroom format has a parallel at the level of curricular design. The traditional approach to learning design means that sessions are grounded in learning outcomes formulated in the **Bloom's taxonomy tradition** (Anderson and Krathwohl, 2001). These are **course-internal statements** that describe what the learner is expected to

know or to be able to do on completion of a specific module. They remain necessary and will continue to be authored for every AGRIFOODSKILLS module. They are however insufficient as the principal organising unit for a stackable, portable microcredential architecture. That role requires **competency statements** that compose across modules and providers.

5.3.1 Why learning outcomes alone do not stack

A learning outcome such as 'the learner will be able to explain the principles of Hazard Analysis and Critical Control Points (HACCP)' tells a prospective learner, an employer or a credentialing body very little about **what comes before or after**. It does not say what the learner needs to already know to benefit from the course; it does not say what the learner can do next as a consequence of completing it; and it does not connect to any classification that another training provider, employer or qualification authority can read in the same way. Two modules with similar learning outcomes can produce learners with very different competencies. That is fine for a single course inside a single institution, but is inadequate for an ecosystem of stackable microcredentials.

5.3.2 ESCO as the lingua franca of competence

The European Skills, Competences, Qualifications and Occupations classification (ESCO) supplies a shared multilingual machine-readable vocabulary of skills and competencies, mapped to occupations across the European labour market. Each AGRIFOODSKILLS microcredential should declare its prerequisite and outgoing competencies with reference to ESCO concepts where applicable. The phrase 'where applicable' carries weight. AGRIFOODSKILLS has, in WP2 and WP3, identified **emerging occupational profiles and competence areas** for the agrifood and veterinary sectors that are not at present represented in the ESCO classification. For those competencies the project should publish working concept identifiers in a form aligned with ESCO conventions where possible (ESCO-aligned rather than ESCO-referenced), and should submit them for inclusion in ESCO through the established Commission process. In the interim the assertion-based model described in Section 6 supports composition even where no settled ESCO concept yet exists: assertions accumulate against the working concept identifier, and the credential composition rule references the same identifier. The result is consistent interpretation of a credential across providers and Member States, whether or not the underlying competency has been admitted to the ESCO classification.

5.3.3 Competencies in, competencies out: the architectural commitment

Each AGRIFOODSKILLS microcredential should, where possible, publish at the four items listed below.

1. **Prerequisite competencies**, aligned with ESCO concepts where applicable and otherwise published as ESCO-aligned working concept identifiers, together with the EQF level. These should be specified with sufficient precision to allow another provider's outgoing competencies to be tested against them.
2. **Outgoing competencies**, aligned with ESCO concepts where applicable and otherwise published as ESCO-aligned working concept identifiers, with the EQF level and an indication of the evidence base supporting each claim. AGRIFOODSKILLS will, where it defines new sector-specific competencies, submit them for inclusion in ESCO through the established Commission process; in the interim the assertion model permits credential composition against the working identifier.
3. **Institutional learning outcomes**, formulated in the Bloom's-informed tradition for the purposes of pedagogical design and assessment authoring. These remain part of the design documentation and are not displaced by the competency-level declarations.
4. **The EQF level** at which the competencies are demonstrated, and the credit or notional learning hours associated with the module (European Credit Transfer and Accumulation System (ECTS) for higher-education-aligned provision; notional hours otherwise).

When microcredentials publish their interface this way, occupational profiles curricula (T5.2) can be composed transparently: module B's prerequisites are satisfied by module A's outgoing competencies. Learners and employers see why the order matters. Other providers can substitute their own modules in, provided they declare the same outgoing competencies. The system becomes genuinely federated.

Literature in brief: outcomes, competencies, ESCO

Bloom's (1956) revised taxonomy (Anderson & Krathwohl, 2001) remains useful for authoring measurable learning outcomes at the module level. The case for ESCO competencies as the lingua franca of stackability rests on the European Skills Agenda, the 2022 Council Recommendation on microcredentials, and ESCO's own documentation on integration with the European Learning Model. The MICROBOL final reports (2022) frame the interaction between qualifications frameworks (EQF / NQFs) and microcredentials; their HE-centric conclusions are precisely what the post-MICROBOL VET agenda must now extend.

Design summary: the four interface declarations every microcredential must publish

- Prerequisite competencies (ESCO-referenced, with EQF level).
- Outgoing competencies (ESCO-referenced, with EQF level, and the evidence base for each).
- Institutional learning outcomes (Bloom's-informed, course-level, these sit inside the competency interface).
- Workload or duration in a recognised unit (ECTS for HE; notional learning hours for VET; clear in either case).

5.3.4 Learning outcomes or competency statements: which leads?

Both learning outcomes and competency statements are required components of an AGRIFOODSKILLS module. They are not alternatives, but operate at **different levels of the curricular architecture** and serve different purposes for different audiences. The practical question for a designer is therefore not which to write but which to write first, and which to use as the anchor for the rest of the design. The choice should be made consciously, on the basis of context and purpose, rather than by habit.

The **competency statement** describes what a professional can do in an integrated real-world practice context. It is broad enough to encompass a whole module, programme strand or occupational role domain. It is externally anchored, to a **professional body, a regulatory standard, or a classification** such as ESCO, and it speaks to employers, regulators and credential systems. The **learning outcome** describes what a learner will be able to demonstrate on completion of a specific module or teaching event. It is authored by the course designer in the Bloom's-revised tradition and serves as the internal organising unit of pedagogical and assessment design. It speaks to learners choosing between courses and to assessors designing tasks.

Since both are needed, the choice as to whether to lead with the outcome or competency will be influenced by a number of factors including the **educator, the session, the EQF and the sector**. The guidance below is just that, and should not be applied rigidly in curriculum design.

Use a **competency statement** as the lead when the module satisfies one or more of the following conditions:

- It contributes to a defined occupational or professional role with an existing competency framework (ESCO, sectoral, regulatory, or professional-body defined).

- It is designed to stack within a credential pathway in which interoperability with other providers' modules is required.
- Its principal audience is employers or regulatory bodies who need to know what a graduate can do in the workplace.
- Assessment will be largely performance-based or workplace-based.

Where competency statements lead, the design sequence should first identify the competencies the module is intended to develop, expressed with reference to ESCO concepts or equivalent sector-defined standards. Derive from each competency the learning outcomes that describe what the module specifically contributes toward that competency. The competency statements form the public credential interface; the outcomes and objectives are the internal design documentation.

Use a **learning outcome** as the lead when the module satisfies one or more of the following conditions:

- It is standalone or sits within a self-contained institutional programme without a formal external competency framework.
- Its principal audience is learners choosing between courses rather than employers or regulators assessing workforce readiness.
- Its content is knowledge-rich and foundational, where the pathway to practice performance is indirect or extended.
- Assessment is largely summative and based on knowledge and comprehension.
- No settled occupational or professional competency framework yet exists for the subject domain.

Where learning outcomes lead, the design should focus with authoring in the Bloom's-revised tradition, anchored to the content and skills the module is designed to develop. Once the outcomes are settled, map them to the nearest relevant ESCO concepts (where relevant) or sector-defined competency descriptors at the appropriate EQF level and use the result to produce the competency declarations required for credential publication. The mapping should be reviewed against the assessment design to ensure that the competency claim is proportionate to the evidence available. In all cases the relationship between learning outcomes and competency statements should be made explicit in the module documentation, so that learners, employers and credential systems can read both and understand what connects them. Details of how to construct competency statements are included in Appendix A.

5.3.5 Limitations of competency-based design

Competency statements provide the architectural foundation for the stackable, interoperable Verifiable Learning Credentials described in the rest of this report, but their use introduces a set of limitations that designers and quality assurance processes need to manage actively. These are set out below, paired with the mitigation already available within the EQAVET-aligned evaluation and quality assurance framework set out in Section 5; none of the limitations is a reason to abandon competency-based design, but each must be addressed if the architecture is to operate as intended.

Granularity risk. Competency statements may be written at very different grain sizes by different designers and providers. Two statements that appear to match at the level of an ESCO concept or EQF level may describe performance at substantially different levels of sophistication or contextual complexity. ESCO classification and EQF level are coarse instruments: they indicate domain and broad level but do not specify assessment standards or performance benchmarks. There is a real risk that a learner's outgoing competencies from one module are accepted as satisfying the prerequisites of another when the performance standard differs significantly. Mitigation: the EQAVET-aligned quality assurance cycle set

out in Section 5, with assessor moderation, inter-rater calibration (see 11.6 on the Brinkerhoff Success Case Method where variation is observed) and sector-body validation, is the principal safeguard. The architecture in this report does not, on its own, supply that safeguard.

ESCO lag and coverage gaps. ESCO is a living classification but it is not a real-time one. Emerging occupational profiles, new technology-driven competencies and sector-specific performance areas, particularly in agrifood and veterinary contexts, are routinely under-represented or absent. Credentials built on ESCO-referenced concepts inherit this limitation. Working concept identifiers, introduced in Section 5.2 and developed in Appendix A, are a pragmatic interim solution but they carry their own risks. Identifiers minted by one project may not be adopted by others. Credentials issued against a working identifier may require retrospective restatement if the concept is later admitted to ESCO in a modified form. The submission and adoption process for new ESCO concepts is slow relative to the pace of curriculum development. Mitigation: the shared register of working concepts proposed in Appendix A.8, stewarded across the consortium, reduces fragmentation; submission to ESCO through the established Commission process closes the longer-term gap.

Mapping validity. Where competency statements are derived retrospectively from learning outcomes, as the outcome-led design pathway in 6.4 permits, the quality of the competency claim depends entirely on the rigour of the mapping process. Poorly executed outcome-to-competency mapping may produce declarations that overstate or misrepresent what a module actually develops. The risk is highest where designers are working without external moderation, without reference to a settled competency framework, or under time pressure. Mitigation: peer review of the mapping, expert validation by sector body representatives, and alignment of the mapping against the institutional EQAVET internal review process.

Assessment alignment. Declaring an outgoing competency creates an implicit commitment about what the assessment has demonstrated. If the assessment design does not genuinely test the competency declared, for example a competency claiming workplace behavioural performance evidenced only by a knowledge-recall test, the credential is misleading. Competency declarations and assessment design must be authored and reviewed together. The evidence base for each outgoing competency declaration must be stated in the credential metadata, and it is the responsibility of the EQAVET-aligned quality assurance process (see 5.3 on Kirkpatrick Levels 2 to 4 and 5.7 on Thalheimer's LTEM) to verify that the evidence is proportionate to the claim.

Designer capacity. The three-tier distinction between competency statements, learning outcomes and session-level objectives is not universally understood, even among experienced educators. Designers without a background in competency-based education or credentialing frameworks may find the architecture unfamiliar and may default to writing all three as variants of the same kind of statement. The present report and Appendix A provide guidance and worked examples, but neither substitutes for training, mentoring and quality review of module documentation before publication. Mitigation: explicit incorporation of competency authoring within the train-the-trainer programme in Task 6.1, paired with quality review of module documentation as a checkpoint in the occupational profile learning design process in Task 5.2.

None of these limitations is unique to AGRIFOODSKILLS, and none is grounds for abandoning the architecture set out in this report. They are the working risks of operating a competency-and-assertion model at sectoral scale, and they are the risks that the EQAVET-aligned evaluation cycle,

the shared register of working concepts and the train-the-trainer programme are designed jointly to mitigate.

Design summary: managing the limitations

- Calibrate competency granularity by peer review within the consortium before publication; treat first-draft statements as drafts.
- Use the shared register of working concepts (Appendix A.8) to detect and resolve duplicate or overlapping competencies before submission to ESCO.
- Author the competency declaration and the assessment design together; do not allow either to be finalised in the absence of the other.
- Document the evidence base behind each outgoing competency claim in the credential metadata.
- Build competency authoring into the Task 6.1 train-the-trainer programme; do not assume the construct is already familiar to designers.

6. Training methodologies

This section does not propose a new catalogue of training methodologies. The methodologies referenced in the AGRIFOODSKILLS pilot programme were identified and described in I-RESTART D4.1 (Erasmus+, 2023), which surveyed online learning, microlearning, gamification, flipped classroom, blended learning, peer learning, on-site demonstration and action-based and participatory learning. The present section adds to that work the contextual material required for application within AGRIFOODSKILLS: the adaptation of each methodology across the EQF bands 3 to 5 and 6 to 8; the alignment of each methodology with the principles of adult learning set out in Sections 2 and 3; the time-pattern and learner-constraint each is suited to in the agrifood and veterinary working context; and the evidence each methodology produces against the three assertion categories identified in Section 6. When formulating learning sessions, the designer must identify the most appropriate methodology for the material and EQF. Educators may still choose alternative delivery platforms relevant to their cohort of learners.

Each methodology is presented as a single table. The same row structure is used in every entry to support comparison. Where a methodology has a well-developed ESCO concept correspondence on the outgoing side, established training materials are typically already available for it from existing providers; the added value of AGRIFOODSKILLS lies in the contextual adaptation across EQF bands and in the methodology choices that suit the time and work patterns of the target learner populations, rather than in the reauthoring of methodology descriptions or in the duplication of available training content.

Work-based learning (WBL) is treated in this section as a methodology in its own right, consistent with its role in the AGRIFOODSKILLS design. Each occupational profile's curricula developed in Task 5.2 incorporates sixty hours of WBL alongside sixty hours of formal learning and ten hours of assessment. WBL is therefore not an adjunct to the methodologies that follow; it is half of the training time. The defining feature of WBL is that the learner produces real workplace outputs under the supervision of an industry mentor: WBL is doing the work, not observing it.

6.1.1 Peer-to-peer on-site demonstration

Aspect	Description
Definition	Learning delivered in the work environment by a peer practitioner, such as a fellow farmer, processor, technician or clinician, who demonstrates a technique, procedure or piece of equipment to a small group of learners. The peer is not a professional trainer; authority derives from doing the work well. Sessions are short (60 to 180 minutes), local, hands-on, and recur across an operational cycle.
Andragogy fit	Strongest fit in the catalogue. Honours prior experience, is intrinsically problem-centred, triggers at the moment of readiness, and respects the location of the learner.
Sector-fit and time profile	Default modality for EQF 3-5 sector practitioners. One to three hours within a working day on a real site. Appropriate for agronomic, biosecurity, processing-line and clinical-handling topics in which the competence is fundamentally embodied.
Worked example	A demonstration of correct teat-end disinfection technique pre- and post-milking, delivered by a farmer who has reduced bulk somatic cell count by 30 per cent over twelve months, on her own parlour, to a group of eight neighbouring farmers and dairy workers. Ninety minutes. Each learner performs the technique under observation, with phone-recorded video reviewed in a brief debrief.
EQF 3-5 adaptation	Default form. Peer practitioner as demonstrator; sign-off on the day against an explicit rubric.
EQF 6-8 adaptation	Demonstrator is a senior clinician, advisor or researcher; demonstration paired with conceptual framing and a reflective component on transferability.
Three-assertion evidence	- Engagement: attendance and active participation logged by the demonstrator. - Assessment: performance checklist scored against the rubric. - Performance attestation: follow-up on-site visit by a mentor four to eight weeks later, or supervisor sign-off during work-based learning.

Design summary: applying peer-to-peer on-site demonstration in AGRIFOODSKILLS

- Choose the demonstrator on the basis of credibility with the audience, before credentials.
- Brief the demonstrator on the single competence the session must evidence, not on the topic in the abstract.
- Cap the group at 8 to 12 so each learner has hands-on practice.
- Build in a structured practice phase; demonstration without learner practice is observation, not training.
- Plan the four to eight week follow-up before the session ends; it carries the performance attestation.

6.1.2 Work-based learning (WBL)

Aspect	Description
Definition	A learning modality in which the learner accomplishes structured tasks within real working contexts under the supervision of an industry mentor or workplace supervisor. The defining feature of WBL is that the learner does the work and produces real workplace outputs; observation, demonstration and shadowing are not, on their own, WBL. WBL is implemented in AGRIFOODSKILLS under the framework set out in the COM(2017) 563 Recommendation on a European Framework for Quality and Effective Apprenticeships and within the WBL scheme developed in Task 5.3.
Andragogy fit	Maximal. WBL leverages prior experience, requires self-direction, is triggered by genuine workplace need, and is intrinsically problem-centred. The learner's existing or assigned role in the workplace is itself the curricular vehicle.
Sector-fit and time profile	Sixty hours per occupational profile, distributed over a typical period of eight to twenty-four weeks within normal working time. WBL is applicable at every EQF band; the form changes with band but the modality does not. Two delivery schemes are recognised in the AGRIFOODSKILLS WBL design: WBL for workers, in continuing vocational training and mentored by an industrial or VET mentor; and WBL for students, undertaken within an industry project supervised by an HEI or VET tutor and an industrial mentor.
Worked example	A quality-assurance technician completes sixty hours of WBL alongside a sixty-hour formal module on traceability systems. The WBL comprises three workplace deliverables: a baseline process map of the technician's own line, a redesigned process, and a measured comparison of the two. Each deliverable is signed off by the QA manager (mentor) against a published rubric.
EQF 3-5 adaptation	WBL is sized to operational changes within the learner's existing scope of work; mentor authority over context is high; the deliverable is operational evidence, such as a revised protocol, a new record system or a measured improvement in a defined process.
EQF 6-8 adaptation	WBL extends to research, innovation or systems-level change. Mentor relationship is more peer-like and includes a reflective and analytical component alongside the workplace artefact.
Three-assertion evidence	- Engagement: mentor-attested time-in-WBL and checkpoint submissions. - Assessment: rubric-scored deliverables. - Performance attestation: the workplace artefact itself is the can-do-the-role evidence. WBL is the strongest single source of performance-attestation evidence in the AGRIFOODSKILLS architecture.

Design summary: applying work-based learning (WBL) in AGRIFOODSKILLS

- Pair the learner with a mentor whose workplace role gives them authority over the context.
- Define the WBL deliverable as a workplace outcome, not as a learning activity.
- Set checkpoints at two to four week intervals with required evidence at each.
- Treat the workplace artefact as the primary deliverable; reflection is supplementary.
- Record the mentor sign-off as a performance attestation in the credential metadata.

6.1.3 In-person learning (short, intentional)

Aspect	Description
Definition	Synchronous, face-to-face instruction in a shared physical space, used sparingly and with explicit justification. AGRIFOODSKILLS does not adopt the multi-day classroom format as a default; in-person time is reserved for content that requires physical co-presence.
Andragogy fit	Weak in conventional long-form. Acceptable when reconfigured: short (≤ 90 minutes), problem-centred, highly interactive, and tied to a specific decision the learners must make.
Sector-fit and time profile	Used only where the learning benefits from physical co-presence (a regulatory briefing requiring collective interpretation; a strategy workshop with a group decision; a high-bandwidth case discussion). Sixty to one hundred and eighty minutes. Multi-day in-person formats reserved for combined train-the-trainer or consortium events that bundle several objectives.
Worked example	A two-and-a-half-hour regional workshop for fourteen large-animal veterinarians on the implementation of a new EU on-farm animal welfare standard. Pre-work: a twenty-minute asynchronous summary. In-person: structured case discussion of three borderline farms, consensus drafting of a regional interpretation note, and a role-played client conversation.
EQF 3-5 adaptation	Limited to ≤ 90 minutes; co-located with existing operational meetings where possible; single decision focus; travel-heavy half-day formats avoided.
EQF 6-8 adaptation	Half-day acceptable for high-complexity topics; seminar, journal-club or strategy workshop formats with a deliverable artefact. Multi-day in-person reserved for purpose-built blocks such as a residency case-conference week.
Three-assertion evidence	- Engagement: presence and contribution to in-session tasks. - Assessment: artefact produced in the session, such as a decision document or a drafted protocol. - Performance attestation: subsequent implementation in the learner's own setting, evidenced through follow-up reporting or audit.

Design summary: applying in-person learning (short, intentional) in AGRIFOODSKILLS

- Make pre-work non-negotiable; state the consequence of skipping it.
- Design synchronous time as a sequence of decisions or tasks, not as a discussion of the pre-work.
- Build an interaction at intervals of no more than fifteen minutes.
- Produce an artefact for each learner during the session.
- Capture a follow-up commitment before the session ends; this is the bridge to the performance attestation.

6.1.4 Action-based and project-based learning

Aspect	Description
Definition	Learning organised around the completion of a real, consequential project or action over a defined period (typically weeks to months). The learner produces an artefact or implements a change in the workplace; learning emerges from doing. In AGRIFOODSKILLS, action-based and project-based learning overlap substantially with WBL; the distinction is one of scale and structure rather than of principle.
Andragogy fit	Maximal. Self-direction, prior experience, readiness, problem-centred orientation and intrinsic motivation are all active.
Sector-fit and time profile	Strong at both EQF bands. Distributed effort over four to twenty-four weeks, absorbed mostly within working hours.
Worked example	A ten-week project run by a meat-processing line supervisor: redesign the changeover protocol between two product lines to reduce cross-contamination risk. The learner gathers baseline microbiological data, designs and implements a revised protocol with the QA manager as mentor, measures post-change data, and submits the revised protocol and the data set.
EQF 3-5 adaptation	Projects sized to an operational change within existing scope of work; mentor authority more directive; assessment weights operational evidence.
EQF 6-8 adaptation	Projects extend to research, innovation or systems-level change; learner autonomy is high; assessment includes written or oral defence alongside the artefact.
Three-assertion evidence	- Engagement: checkpoint submissions and mentor-attested time. - Assessment: rubric-scored deliverables. - Performance attestation: the workplace artefact itself is the evidence. The methodology is uniquely strong on the third assertion.

Design summary: applying action-based and project-based learning in AGRIFOODSKILLS

- Frame the project as an operational outcome, not a learning topic.
- Name a workplace mentor with authority over context before enrolling the learner.
- Set checkpoints at two to four week intervals with required evidence at each.
- Make the workplace artefact the primary deliverable; the reflective component is secondary.
- Reference the outgoing competencies before the project starts so the credential composes automatically when it ends.

6.1.5 Online guided learning

Aspect	Description
Definition	Asynchronous, self-paced learning delivered through a learning management system (LMS) or equivalent platform, with structured navigation, embedded assessment, and human availability (mentor, tutor, peer cohort) at defined points. 'Guided' carries design weight: a pile of PDFs in a Moodle folder is not online guided learning.
Andragogy fit	Strong on self-direction and on respect for learner time. Weaker on experience and social motivation unless explicitly designed in.
Sector-fit and time profile	Workhorse modality for both bands where the content is propositional knowledge, procedural rules or interpretive framework. Total load typically four to twenty hours, in units of fifteen to forty-five minutes. Synchronously supported asynchronous design: asynchronous work, with named human points of contact at defined intervals.
Worked example	An eight-hour LMS-hosted module on Integrated Pest Management regulation and decision frameworks, in sixteen units of approximately thirty minutes, with embedded checks, decision scenarios at each section, a weekly mentor question session and a moderated peer-discussion thread. Designed for plant-production technicians at EQF 4.
EQF 3-5 adaptation	Mobile-first; visual; low text density. Decision scenarios drawn from sector practice. Mentor availability prioritised over peer discussion.
EQF 6-8 adaptation	Heavier text; conceptual framing; reading-list extension. Substantial peer discussion. Assessment includes written or oral defence alongside embedded checks.
Three-assertion evidence	- Engagement: LMS activity log, unit-completion record, peer-interaction log. - Assessment: embedded checks plus a summative item. - Performance attestation: weak unless paired with WBL or with an on-site demonstration; this methodology supports the other two assertions but does not deliver performance attestation on its own.

Design summary: applying online guided learning in AGRIFOODSKILLS

- Cap unit length at forty-five minutes and label the expected time on every unit.
- Embed assessment in every unit, not only at the end of the module.
- Publish a named synchronous human-availability schedule.
- Use LMS analytics to identify disengagement at week one rather than week four.
- For EQF 3-5 audiences, design mobile-first.
- Pair with at least one demonstration or WBL component before the credential is considered complete.

6.1.6 Gamification

Aspect	Description
Definition	The application of game-design elements (challenges, levels, narratives, feedback loops, cooperative or competitive mechanics) to non-game learning contexts to support engagement and retention. Gamification is a design layer over other methodologies; it is not a standalone methodology.
Andragogy fit	Conditional. Adults respond well where gamification amplifies intrinsic motivation (mastery, identity, autonomy) and poorly where it substitutes extrinsic incentives.
Sector-fit and time profile	Most useful as a layer on online guided learning and micro-learning. Limited direct use in on-site demonstration. Per-interaction one to five minutes; engagement distributed across module lifetime.
Worked example	An online module on biosecurity uses a narrative in which the learner manages an escalating disease incursion across six scenarios. Each scenario unlocks the next; successful management earns a microcredential. No leaderboard or points; the progression itself is the engagement device.
EQF 3-5 adaptation	Narrative grounded in operational reality; rewards tied to recognisable role progression; cooperative mechanics in preference to competitive; short-form.
EQF 6-8 adaptation	Complex simulation with branching scenarios and reflective decision rationale; competition between teams is acceptable in short form.
Three-assertion evidence	- Engagement: strong (gamification's principal purpose). - Assessment: as for the underlying module. - Performance attestation: weak unless paired with WBL or with on-site demonstration.

Design summary: applying gamification in AGRIFOODSKILLS

- Identify the intrinsic motivator that the game amplifies before selecting mechanics.
- Favour narrative and role-progression over points and leaderboards.
- Use cooperative mechanics with sector audiences; ranking colleagues against each other often backfires.
- Tie every in-game achievement to a real competence the learner is acquiring.
- Test with the actual target audience before scaling.

6.1.7 Micro-learning

Aspect	Description
Definition	Learning delivered in short focused units (typically two to ten minutes), each addressing a single defined competence or sub-competence. Consumed in the working day, frequently on a mobile device, frequently at the moment of need.
Andragogy fit	Excellent. Respects time, supports self-direction, triggers at readiness, is inherently problem-centred.
Sector-fit and time profile	One of the strongest-fit modalities for adult sector learners, particularly at EQF 3-5. Two to ten minutes per unit. Best deployed as a stream of related units (a pathway) rather than as one-off items.
Worked example	A pathway of twenty micro-learning units on common reproductive health issues in dairy cows, each ≤5 minutes, delivered through a sector-trusted mobile application. Completion of all twenty units, paired with an on-site demonstration session, awards an EQF 3 microcredential in dairy reproductive management.
EQF 3-5 adaptation	The natural home of the methodology. Visual; vernacular language; embedded in tools the workforce already uses (sector applications, messaging channels, farm-management platforms).
EQF 6-8 adaptation	Used principally as a reference and refresher layer; just-in-time access to specialist content during practice.
Three-assertion evidence	- Engagement: per-unit completion, pathway-level streaks, mobile analytics. - Assessment: per-unit checks and a pathway-level summative item. - Performance attestation: weak unless explicitly paired with demonstration or WBL.

Design summary: applying micro-learning in AGRIFOODSKILLS

- One competence per unit; if a unit covers two, split it.
- Lead each unit with the operational problem it solves, not with the topic.
- Mobile-first format; assume phones in the field.
- Include an immediate practice or check at the end of every unit.
- Curate units into a visible pathway that maps to a recognisable occupational profile.

6.1.8 Flipped classroom

Aspect	Description
Definition	Inversion of the conventional sequence: the learner encounters content asynchronously in advance, and the synchronous component (in-person or live online) is reserved for application, problem-solving and discussion. The synchronous time is short, high-bandwidth and learner-active.
Andragogy fit	Strong. Learner controls the pace of content acquisition; synchronous time is problem-centred and respects experience.
Sector-fit and time profile	Strong at both bands, provided the synchronous component is short. Pre-work thirty to ninety minutes; synchronous component sixty to one hundred and twenty minutes. For EQF 3-5 audiences, the synchronous component should usually be on-site rather than in a classroom.
Worked example	A flipped session on antimicrobial stewardship for mixed-practice veterinarians. Pre-work: forty-five-minute online module covering antimicrobial resistance (AMR) data, the legal framework and a prescribing decision tree. Synchronous: ninety-minute case-based workshop applying the decision tree to four real cases from the participants' caseloads, with a committed practice-level change. Follow-up at thirty days: report on the change.
EQF 3-5 adaptation	Pre-work short, mobile-friendly, visual. Synchronous component preferably on-site and demonstration-led.
EQF 6-8 adaptation	Pre-work substantial (60 to 90 minutes asynchronous); synchronous component can extend to a half-day for high-complexity topics; learner extension expected to be high.
Three-assertion evidence	- Engagement: pre-work completion and synchronous participation. - Assessment: synchronous output (decision document, case write-up, role-play assessment). - Performance attestation: the follow-up reported change in own practice, with mentor or supervisor sign-off.

Design summary: applying flipped classroom in AGRIFOODSKILLS

- State a clear consequence for skipping pre-work and enforce it on the day.
- Design the synchronous session as decisions and tasks, not as discussion of the pre-work.
- Produce an artefact in the room or on-site for each learner.
- Define the follow-up action and its evidence requirement before the session ends.

6.1.9 Blended learning

Aspect	Description
Definition	Deliberate combination of asynchronous and synchronous components designed as a coherent whole. Blended learning is less a single methodology than a design pattern that combines several of the methodologies in this catalogue, including WBL.
Andragogy fit	Strong by construction. A well-designed blend uses the andragogy-strong features of each component and compensates for the weaknesses.
Sector-fit and time profile	Dominant design pattern for AGRIFOODSKILLS training. Each profile is in practice a blend: approximately sixty hours of formal learning (online guided plus micro-learning plus occasional in-person or flipped session), sixty hours of WBL, and ten hours of assessment.
Worked example	An occupational profile (module might fit better) in sustainable on-farm nutrient management (EQF 4) delivered as a five-month blend: eight weeks of online guided learning (four units, approximately fifteen hours), interspersed with three on-site peer demonstrations of ninety minutes each, culminating in a twelve-week WBL project. Continuous assessment across the blend, with the WBL artefact carrying the performance-attestation evidence.
EQF 3-5 adaptation	Asynchronous components weighted toward micro-learning and mobile delivery; synchronous components weighted toward on-site demonstration; WBL is core, not optional.
EQF 6-8 adaptation	Asynchronous components can carry substantial reading and conceptual framing; synchronous can include seminar formats; WBL takes the form of a research or innovation project.
Three-assertion evidence	All three assertion categories are evidenced across the blend, with each component contributing to the assertion category it most strongly supports.

Design summary: applying blended learning in AGRIFOODSKILLS

- Design the whole blend first; design the components only after the blend is coherent.
- Make transitions between components explicit so that the learner can see what is next and why.
- Use each component for what it does best.
- Use one LMS-of-record so that learner progress is visible in one place.
- Map each component to the assertion category it evidences; rebalance if any category is empty.

6.1.10 AI-enhanced learning (co-learning)

Aspect	Description
Definition	The deployment of generative and predictive AI tools as active components within a learning experience: as adaptive tutors, simulation partners, feedback engines and drafting and critique partners. The term co-learning denotes that the AI is treated as an interlocutor subject to critical evaluation, not as an oracle.
Andragogy fit	Potentially strong (see Section 2). Realisation depends entirely on design.
Sector-fit and time profile	Underused at present. Typically a layer within other methodologies rather than a standalone format. Useful for drafting under critique, simulation of complex cases (with the caveats in Section 2.3), the generation of personalised practice scenarios, and first-line feedback on learner output.
Worked example	A flipped clinical-reasoning module for final-year veterinary students at EQF 7. Pre-work: an AI-driven simulation generates a unique case for each student, requiring history-taking, differential diagnosis, diagnostic plan and treatment plan, and critiquing the learner's reasoning. Synchronous: ninety-minute small-group session in which students present the AI-generated case to peers and a clinician, defending where they agreed with the AI's critique and where they did not. Assessment scores both clinical reasoning and the quality of critical engagement with the AI.
EQF 3-5 adaptation	AI used as a personalised practice generator and as a first-line feedback engine; the trainer remains primary authority; data handling and trust framings are made simple and explicit.
EQF 6-8 adaptation	AI used as a co-learning peer in higher-order tasks. Critical engagement with AI is itself a transversal outgoing competence (the QUSC framework from Section 2.6).
Three-assertion evidence	- Engagement: AI interaction logs and tutor-attested critical engagement. - Assessment: scored output covering both the learner's product and their critique of the AI. - Performance attestation: evidenced in subsequent WBL or practice in which the learner uses AI responsibly in their own context.

Design summary: applying ai-enhanced learning (co-learning) in AGRIFOODSKILLS

- Specify the AI's role explicitly per use: drafter, simulator, feedback engine or study partner.
- Build assessment around the learner's critical engagement with the AI, not around either party's standalone production.
- State the AI Act and GDPR posture before construction begins.
- Record in the credential which parts of the learning were AI-mediated and how.
- Test for hallucination and bias before the design encounters a learner.

6.2 Integrating methodologies

This section provides four cross-cutting tools that integrate those individual entries into instruments for the routine design of a microcredential or module. The first tool maps each methodology against the six principles of andragogy set out in Section 3, indicating where each methodology is at its strongest. The second tool sets out, in summary form, the practical differences in the way each methodology is deployed at EQF 3-5 and at EQF 6-8. The third tool reverses the usual design starting point: it begins from the learner's dominant constraint and identifies the methodologies most likely to succeed under that constraint, on the principle that design should start from the learner's working life rather than from the trainer's preferred mode. The fourth tool maps each methodology to the three assertion categories from Section 6, on the principle that a credential whose evidence map leaves any category empty is by definition incomplete and should be redesigned before issuance. Work-based learning is included in each of the four tools, in recognition of its place as half of every occupational profile's designed in Task 5.2.

6.2.1 Knowles × methodology engagement matrix

Each cell rates how strongly the methodology, when designed well, leverages the principle.

Strong = ●●●; Moderate = ●●; Conditional = ●.

Methodology	Need to know	Self direction	Experience	Readiness	Problem centred	Intrinsic motivation
Peer on-site demo	●●●	●●	●●●	●●●	●●●	●●●
Work-based learning	●●●	●●●	●●●	●●●	●●●	●●●
In-person (short)	●●	●	●●	●●	●●	●●
Action / project-based	●●●	●●●	●●●	●●●	●●●	●●●
Online guided	●●	●●●	●●	●●	●●	●●
Gamification (layer)	●●	●●	●	●●	●●	●●●
Micro-learning	●●●	●●●	●●	●●●	●●●	●●
Flipped classroom	●●●	●●●	●●	●●	●●●	●●
Blended (pattern)	●●●	●●●	●●●	●●●	●●●	●●●
AI-enhanced co-learning	●●	●●●	●●	●●●	●●●	●●

6.2.2 EQF 3-5 vs EQF 6-8 adaptation table

Concise reference for how each methodology shifts across the two EQF bands. The methodology does not change, the autonomy expected of the learner, and the form of evidence, do.

Methodology	EQF 3-5	EQF 6-8
Work-based learning	Operational change within existing scope of work; mentor directive; deliverable is operational evidence (revised protocol, new system, measured improvement).	Research, innovation or systems-level change; peer-like mentor relationship; reflective and analytical component alongside the workplace artefact.
Peer on-site demo	Default for embodied skills; peer practitioner as demonstrator; sign-off on day.	Used with conceptual framing and reflection on transferability; demonstrator is senior clinician/researcher.
In-person (short)	≤90 min, co-located with operational meetings where possible; single decision.	Up to half-day for high-complexity topics; seminar / case-conference / workshop formats.
Action / project-based	Operational change in own scope; mentor directive; weight on operational evidence.	Research, innovation or systems change; high autonomy; written/oral defence in addition to artefact.
Online guided	Mobile-first, visual, mentor-availability prioritised; short units (≤30 min).	Heavier reading and conceptual framing; peer discussion substantive; longer units acceptable.
Gamification	Narrative + role-progression rewards; cooperative mechanics; short-form.	Branching simulation; reflective decision rationale; competition acceptable in short bursts.
Micro-learning	Primary delivery vehicle; mobile; vernacular language; embedded in sector tools.	Reference and refresher layer; just-in-time access to specialist content.
Flipped classroom	Pre-work mobile-friendly; synchronous on-site demo-led where possible.	Substantial pre-work; synchronous can extend to half-day; high learner extension.
Blended (pattern)	Async = micro + mobile; sync = on-site demo; WBL core.	Async = reading + concept; sync = seminar; WBL = research/innovation project.
AI-enhanced co-learning	AI as practice generator + first-line feedback; trainer primary authority.	AI as co-learning peer in higher-order tasks; critical-AI-use becomes an explicit outgoing competence.

6.2.3 Method selection by learner constraint: a decision aid

Start design from the learner's constraint, not from the trainer's preference. Maps dominant constraint to methodologies most likely to succeed.

If the learner's dominant constraint is...	Then prefer...
Cannot leave the worksite during operational hours	Peer on-site demo; work-based learning embedded in the working day; micro-learning on mobile.
Available only in fragmented short windows	Micro-learning pathway; flipped classroom with short on-site sync; online guided in short units.
Strong prior experience that must be respected	Peer on-site demo; project-based learning anchored in own operation; flipped classroom that opens with case interrogation.
Geographically dispersed and travel-constrained	Blended pattern with on-site local sync; online guided + micro-learning + WBL; AI-enhanced co-learning.
Needs to apply learning to an immediate workplace problem	Work-based learning; project-based learning; flipped classroom paired with on-site demo at the moment of need.
Higher-education context with high autonomy	Online guided + flipped classroom + AI-enhanced co-learning + research project.
Regulatory or interpretive content where collective judgement matters	In-person workshop (short, focused) supported by asynchronous pre-work and follow-up reporting.

6.2.4 The three-assertion evidence map by methodology

Each methodology produces evidence more naturally for some assertions than for others. Use this when composing a blend; a micro-credential whose evidence map leaves any assertion empty is incomplete.

Methodology	Engagement	Assessment	Performance attestation
Peer on-site demo	Strong	Strong	Strong
Work-based learning	Strong	Strong	Strong
In-person (short)	Adequate	Adequate	Adequate
Action / project-based	Adequate	Strong	Strong
Online guided	Adequate	Strong	Weak
Gamification	Strong	Adequate	Weak
Micro-learning	Strong	Adequate	Weak
Flipped classroom	Adequate	Strong	Adequate
Blended (pattern)	Strong	Strong	Strong
AI-enhanced co-learning	Adequate	Adequate	Adequate

Three observations follow. First, on-site demonstration and the blended pattern are the only methodologies that score strongly on all three assertions; they should be the backbone of the AGRIFOODSKILLS training. Second, the asynchronous methodologies are weak on can-do-the-role and must be paired with a demonstration or WBL component. Third, in-person learning on its own is the weakest pattern - which is the empirical foundation of this report's argument against the multi-day classroom default.

Design summary - how to use these tools in practice

- Start with the decision aid in 6.2.3: which learner-constraint applies?
- Look at 6.2.4 to check which assertions the methodologies you have shortlisted will and will not evidence.
- Use 6.2.1 to confirm the andragogy fit of the design - make sure no Knowles principle is being trampled.
- Use 6.2.2 to adapt the design for the correct EQF band.
- Sketch the blend on a single sheet of paper with the three-assertion evidence map alongside; if any column is empty, redesign.

7. Assessment strategy: evidencing mastery in an AI-informed learning ecosystem

Assessment is the evidential bridge between learning design, competency claims and credential issuance. In an AI-informed adult-learning ecosystem, assessment must serve two related but distinct functions: it must support learning by identifying gaps, misconceptions and developmental needs, and it must provide proportionate evidence of mastery where claims of competence are made. Artificial intelligence changes the evidential value of unsupervised product-based assessment and increases the importance of authentic, constrained, triangulated and well-documented assessment designs. A Learning Record System strengthens this model by capturing granular assessment evidence at item, task and competency level, allowing educators to target blended and flipped learning around demonstrated learner need. For AGRIFOODSKILLS, assessment is designed with the competency claim from the outset, interpreted through the relevant EQF level, captured in a form suitable for assertions and credentials, and reviewed within the EQAVET quality cycle.

7.1 Why assessment requires separate treatment

Assessment is the evidential bridge between five upstream design elements (learning outcomes, competency statements, EQF level, training methodology and credential architecture) and the downstream output the credential architecture depends on. Assessment is therefore distinct from evaluation (Section 8) and from credentialing (Section 9), although all three interact. Assessment produces evidence about an individual learner. Evaluation interprets the quality of the provision as a whole. Credentialing packages evidence into recognised claims. The three should not be conflated in design documentation or in credential metadata.

7.2 Assessment for learning and assessment of mastery

Assessment serves a dual role in adult learning. Both roles are required; both must be designed for explicitly. The narrower phrase 'assessment of knowledge' is avoided in this report except where the assessment is genuinely knowledge-only, because the phrase carries the unhelpful implication that knowledge testing equals low-level assessment. A well-written selected-response item can assess judgement, prioritisation and decision-making; a poorly designed workplace sign-off can assess little more than attendance. The two uses of assessment are distinguished by purpose rather than by format. Assessment for learning identifies gaps and directs feedback. Assessment of mastery supports progression, credentialing and competence claims. Assessment as learning develops the learner's own self-monitoring and judgement.

7.3 AI and the changing evidential value of assessment

AI has not destroyed assessment. It has changed the evidential value of different assessment formats. Outside secure, proctored or otherwise authenticated conditions, AI weakens the evidential value of unsupervised product-only assessment. This does not make assessment obsolete. It requires assessment to become more formative, more authentic, more constrained, more triangulated and more explicit about what level of mastery it can support. Unsupervised essay submissions and reflective accounts are weak as standalone summative evidence unless defended, authenticated or process-tracked. Selected-response, viva, workplace-based and portfolio formats retain or increase their evidential value when designed well.

Where AI is part of the professional task, assessment should not merely prohibit AI. It should assess whether the learner can question, use, scrutinise and communicate AI use appropriately. This is the QUSC framework (introduced in Section 2.4) applied directly to assessment construction: the

construct under test is no longer the production of an artefact in isolation, but the learner's critical engagement with the AI tools they will use in practice.

7.4 The Learning Record System as assessment-for-learning infrastructure

A conventional Learning Management System (LMS) records course completion and test scores. A Learning Record System (LRS) captures granular learning and assessment events that can be used to improve learning, guide educators and support credential claims. The educational claim is that the LRS makes assessment for learning operational. In a flipped or blended design, item-level and competency-level evidence can show what learners did not master before the synchronous session. The educator can then use scarce face-to-face time for the areas where learners struggled, rather than repeating material they already understood. Adaptive sequencing of any kind depends on the presence of an LRS, because the required data do not accumulate in interoperable form without one. Section 10 develops the full conceptual proposal for an Observatory-hosted LRS as the cross-provider and cross-border architecture for AGRIFOODSKILLS; the present section is concerned with the LRS in its narrower role as assessment-for-learning infrastructure.

7.5 Catalogue of assessment types

The seven assessment families used by AGRIFOODSKILLS are presented below, each as a single entry with a common row structure to support comparison. The structure mirrors the training methodologies catalogue in Section 6: each entry covers the definition, the principal evidential role, how the format is constrained, the EQF adaptation, AI-era considerations, the three-assertion evidence map, and a worked AGRIFOODSKILLS example. The catalogue is the assessment counterpart to the delivery catalogue in Section 6: each delivery methodology produces evidence; the assessment method is the instrument that captures that evidence and translates it into the assertion the credential depends on.

7.5.1 Selected-response assessment (MCQ, SBA, extended matching)

Aspect	Description
Definition	Items that require the candidate to choose the best of several options. The construct is judgement under constraint, not pure recall.
Principal evidence	Applied knowledge, judgement, prioritisation, decision-making. Strong on Kirkpatrick Level 2 evidence under secure conditions.
Where the constraint sits	Options, lead-in and single-best-answer logic. Common failure modes: cueing, implausible distractors, length cues, convergence flaws.
EQF adaptation	At EQF 3-5, blueprint against operational decisions the learner makes in their role. At EQF 6-8, items extend to judgement under uncertainty and decisions with competing constraints.
AI-era considerations	Still strong evidence under secure conditions. Weak when sat unsupervised online without AI prohibitions.
Three-assertion evidence	Engagement: weak. Assessment: strong. Performance attestation: weak (paired with workplace assessment for the third assertion).
AGRIFOODSKILLS worked example	A 20-item SBA paper on UAV legal framework and GDPR for the drone-survey microcredential (Appendix A.5.1), blueprinted against the regulation and standard-set by Angoff procedure.

7.5.2 Constructed-response assessment (short-answer questions, structured written response)

Aspect	Description
Definition	Items that require the candidate to author a constrained written response. Used for explanation, interpretation, calculation, prioritisation or justification under written constraint.
Principal evidence	Reasoning, written argument, applied calculation, structured judgement.
Where the constraint sits	Task wording, response scope, expected elements, marking rubric. Short-answer questions fail not because the format is weak but when the task is unconstrained. A candidate cannot be assessed fairly against an answer the examiner only discovers after reading the scripts.
EQF adaptation	At EQF 3-5, scope and required elements are tightly specified. At EQF 6-8, scope opens to allow reasoning under partial information; rubrics specify the evidence of reasoning rather than the conclusion alone.
AI-era considerations	Vulnerable to AI assistance if sat unsupervised. Constraint shifts to in-person or proctored sitting, or to oral defence of a written response, or to tasks where the construct includes the disclosed and assessed use of AI.
Three-assertion evidence	Engagement: weak. Assessment: strong under supervised conditions. Performance attestation: weak.
AGRI FOODSKILLS worked example	A structured written assessment on a sample farm dataset for the farm-emissions microcredential (Appendix A.5.2), with defined data inputs, defined methodology choice, 1500-word response cap, and rubric specifying required elements.

7.5.3 Oral and defended assessment (viva, oral defence, case discussion)

Aspect	Description
Definition	Synchronous oral examination in which the candidate explains, justifies or defends a position, an artefact or a body of work, with the assessor able to probe and challenge.
Principal evidence	Ownership, reasoning, synthesis, response to challenge, the candidate's authentic understanding of their own work.
Where the constraint sits	Defined task brief, defined panel, rubric for the marking dimensions, time. Borderline regression where the panel is large enough.
EQF adaptation	At EQF 3-5, oral assessment is most useful as defence of a workplace artefact. At EQF 6-8, oral defence extends to research, innovation and strategic decision contexts.
AI-era considerations	One of the strongest formats in the AI era because ownership is exposed by direct probing. Where AI has been used in the underlying work, the candidate's ability to explain and justify the AI's contribution is part of the construct.
Three-assertion evidence	Engagement: adequate. Assessment: strong. Performance attestation: adequate.
AGRIFOODSKILLS worked example	A 30-minute oral defence of the integrated crop-livestock system design from Appendix A.5.3, with a panel of two assessors and a rubric for ownership, reasoning and synthesis.

7.5.4 Simulated performance assessment (OSCE, simulation, role-play, scenario station)

Aspect	Description
Definition	Assessment of performance in a controlled artificial setting against a defined task and rubric, typically across a sequence of stations.
Principal evidence	Performance under standardised conditions. Useful where workplace performance cannot be observed at acceptable cost or risk.
Where the constraint sits	Station brief, standardised participant, checklist or rubric, time, station sampling.
EQF adaptation	At EQF 3-5, simulation focuses on the safe execution of defined procedures. At EQF 6-8, stations extend to clinical reasoning, communication under pressure and decisions under uncertainty.
AI-era considerations	AI-driven simulations (Section 2.3) extend the format toward living cases that adapt to the candidate; the AGRIFOODSKILLS pilot does not assume the capacity to deliver such simulations, but documents them as an opportunity.
Three-assertion evidence	Engagement: adequate. Assessment: strong. Performance attestation: adequate (the performance is in a simulated context rather than authentic practice).
AGRI FOODSKILLS worked example	An OSCE station for veterinary trainees on antimicrobial prescribing decisions, with a standardised client, a defined case, a checklist for the decision-making process and a rubric for client communication.

7.5.5 Workplace-based assessment (DOPS, mini-CEX-style observation, supervisor sign-off)

Aspect	Description
Definition	Assessment of performance in authentic practice. The candidate performs the task in the real working context; the assessor observes against a defined rubric and signs off the entrustment level.
Principal evidence	Authentic performance evidence. The strongest evidence available for the performance-attestation assertion.
Where the constraint sits	Defined task, observation criteria, assessor judgement, multiple observations across cases, assessor calibration. Halo effect is the principal failure mode, mitigated by behaviourally anchored rating scales and brief moderation sessions among assessors.
EQF adaptation	At EQF 3-5, sign-off is for the safe execution of defined procedures within scope of work. At EQF 6-8, sign-off extends to entrustment with progressively greater autonomy, using EPA-compatible scales.
AI-era considerations	Robust to AI use because the performance is observed in the workplace. Where AI is part of the workplace task, the assessment includes the candidate's QUSC-compliant use of AI in performing the task.
Three-assertion evidence	Engagement: strong. Assessment: strong. Performance attestation: strong. The format is uniquely strong on the third assertion.
AGRIFOODSKILLS worked example	Direct observation against a flight-operations checklist for the drone-survey microcredential (Appendix A.5.1), with the signed flight log entered into the LRS as a performance attestation.

7.5.6 Portfolio and artefact assessment

Aspect	Description
Definition	Longitudinal collection of authentic workplace artefacts, mapped to competencies, with reflective statements and defence, demonstrating progression and current standard of practice.
Principal evidence	Long-arc evidence of practice, reflection and output quality. Captures progression that point-in-time assessment cannot.
Where the constraint sits	Required artefacts, mapping rules, reflection prompts, authentication mechanism (mentor attestation, oral defence, process tracking via LRS).
EQF adaptation	At EQF 3-5, portfolios capture the workplace artefacts produced through WBL with mentor attestation. At EQF 6-8, portfolios include scholarly outputs, reflective writing and a defence of practice.
AI-era considerations	Vulnerable to AI-authored artefacts where authentication is weak. Authentication is achieved through mentor attestation, viva defence and LRS-tracked production. AI use within the artefacts is recorded in artefact metadata under the QUSC framework.
Three-assertion evidence	Engagement: adequate. Assessment: strong. Performance attestation: strong where artefacts are authenticated.
AGRIFOODSKILLS worked example	The Clinical Educator MC1 portfolio (Appendix A.5.5), including a teaching session observation, an AI-governance note authored under QUSC, and a defended reflective entry.

7.5.7 AI-mediated assessment

Aspect	Description
Definition	Assessment in which the candidate uses AI tools as part of the task, with the construct including the candidate's critical engagement with the AI under the QUSC framework.
Principal evidence	Critical AI use as a professional competence in its own right; the candidate's ability to direct, scrutinise and validate AI output in the context of the domain task.
Where the constraint sits	Defined task, declared AI role, rubric covering both the candidate's product and their critique of the AI's contribution.
EQF adaptation	At EQF 3-5, AI-mediated assessment uses AI as a practice generator and first-line feedback engine; the trainer remains the primary authority. At EQF 6-8, AI-mediated assessment extends to higher-order tasks in which critical AI use is itself an outgoing competence.
AI-era considerations	The format is enabled by current AI capability. Hallucination and bias must be tested for before the design encounters a learner. The AI's role is recorded in the credential metadata.
Three-assertion evidence	Engagement: adequate. Assessment: adequate. Performance attestation: adequate (subject to evidence of transferred AI use into practice).
AGRI FOODSKILLS worked example	The AI-governance note authored under QUSC for the Clinical Educator MC1 (Appendix A.5.5): the candidate produces a written governance note for an educational use of AI in their own setting, with the rubric specifying the QUSC elements that must be addressed and the standard against which the note is judged.

7.6 Constraints as a core assessment principle

All defensible assessment constrains the candidate. The difference between formats lies in where the constraint sits. MCQ items constrain through options and the single-best-answer logic. Short-answer questions must constrain through task wording, response scope and marking rubric. OSCE stations constrain through brief, time, checklist and standardised participant. Workplace-based assessment constrains through defined task and observation criteria. Portfolios constrain through required artefacts and mapping rules. The constraint is the assessment-science contribution that distinguishes a defensible assessment from a learner experience that produces unscored material.

7.7 EQF level as a modifier of assessment complexity

EQF level appears in assessment design as a modifier, not as a classification. The level affects what factual, theoretical or conceptual material the learner is expected to bring; what cognitive or practical operations they must perform; how independently, safely and adaptively they must act; whether the task is predictable or uncertain; and what level of evidence is proportionate to the claim. At EQF 3 to 5 the assessment emphasis is reliable performance of defined tasks, applied understanding, safe execution and supervised autonomy. At EQF 6 to 8 the emphasis shifts to judgement, synthesis, adaptation, leadership, defence of decisions and handling uncertainty. The same assessment format can serve both bands; what changes is the task complexity, the autonomy expected, and the standard against which performance is judged.

7.8 From assessment evidence to assertions

An assessment result is not automatically a credential. It becomes credential-relevant only when the assessment method, standard, context and evidence are sufficient to support a defined claim. Every assessment assertion produced under AGRIFOODSKILLS should carry metadata that records: the competency or learning outcome assessed; the EQF level; the assessment method; the assessment condition (secure, supervised, workplace, simulated, AI-assisted, open-resource); the standard applied (pass mark, rubric threshold, competency standard); the result; the evidence strength (formative, summative, high-confidence, triangulated, limited); the assessor or issuer; the date and validity; and the AI-use status (prohibited, permitted, required, assessed). The metadata is what supports the transition from assessment in Section 7 to assertion-and-credential composition in Section 9.

7.9 Design implications for AGRIFOODSKILLS

Each assessment is authored together with the competency claim it supports. The format is chosen on the basis of the evidence it can provide, not on the basis of habit. The constraint is stated explicitly: candidates are not assessed against answers the examiner only discovers after reading scripts. The EQF level shapes the complexity and the autonomy required, not the format. AI use is either prohibited under supervision, or permitted and disclosed, or required and assessed under the QUSC framework. Evidence is captured in the LRS at the granularity at which it will be needed for adaptive sequencing and credential composition. The assertion metadata is authored before the credential is issued, and reviewed within the EQAVET quality cycle.

Operational guidance on item construction (selected-response stems and distractors, short-answer rubrics, OSCE checklists, DOPS rubrics, portfolio authentication, standard-setting procedures) sits in Appendix C.

8. Evaluation and quality assurance

This section sets out the framework that AGRIFOODSKILLS adopts for the **evaluation** and **quality assurance** of the learning provision designed within the project. Evaluation is treated here as distinct from assessment. Assessment, addressed in Sections 5 to 6, concerns the determination of whether an individual learner has acquired a specified competence; it produces assertions that compose a credential. Evaluation, addressed in this section, concerns the determination of whether the learning provision itself, at programme and at system level, is achieving its stated purposes; it produces evidence that informs **continuous improvement, governance and the allocation of resources**. Both are required for an operationally complete system, and both feed into the credential architecture set out in earlier sections, but they answer different questions and are served by different instruments.

8.1 EQAVET as the overarching framework

The **European Quality Assurance in Vocational Education and Training** framework (EQAVET) is adopted as the overarching reference for the quality assurance and evaluation of the AGRIFOODSKILLS pilot programme. EQAVET supplies the quality cycle, the indicative descriptors at provider and system level, and the ten reference indicators against which a vocational provider may monitor performance over time.

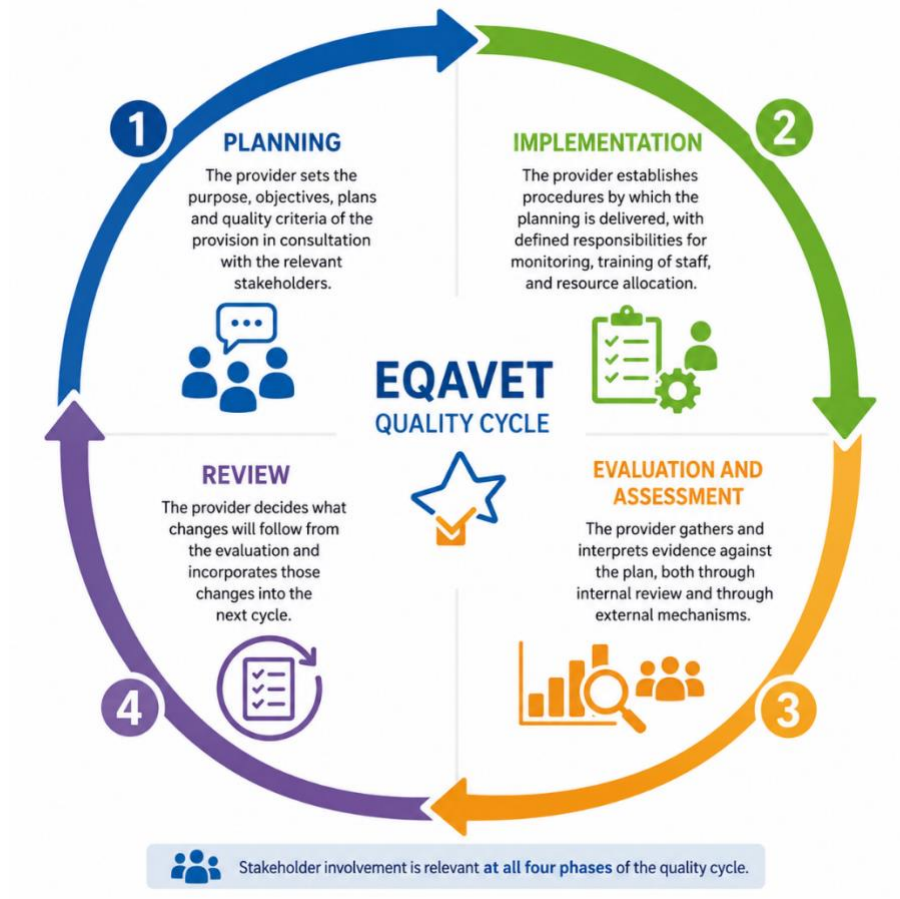
1. Relevance of quality assurance systems for VET providers: Whether providers have quality systems in place, how far they are implemented, and how useful they are in practice.
2. Investment in training of teachers and trainers: The level of resources and support devoted to developing teaching and training staff.
3. Participation rate in VET programmes: How many people enter vocational education and training programmes.
4. Completion rate in VET programmes: The proportion of learners who successfully finish their programmes.
5. Placement rate of graduates from VET programmes: How many graduates move into employment, further education, or other positive destinations.
6. Utilisation of acquired skills at the workplace: The extent to which graduates actually use the skills they gained in training in their jobs.
7. Unemployment rate in the country or region: The overall unemployment level, used as a contextual indicator for the environment in which VET operates.
8. Prevalence of vulnerable groups: The participation and outcomes of learners from groups at risk of exclusion or disadvantage.
9. Mechanisms to identify training needs in the labour market: The processes used to detect current and emerging skills needs.
10. Schemes to promote better access to VET and provide guidance: Initiatives and services that improve access to VET and offer guidance to potential and current learners.

EQAVET treats quality assurance as **continuous and evidence-informed** rather than as a one-off accreditation event. The framework operates as a toolbox: providers select the indicators, descriptors and methods that are relevant to their context and adapt them to their setting. EQAVET does not prescribe the instruments by which evidence is generated; instrument selection sits with the provider.

The EQAVET quality cycle comprises four phases. In the **Planning phase**, the provider sets the purpose, objectives, plans and quality criteria of the provision in consultation with the relevant stakeholders. In the **Implementation phase**, the provider establishes procedures by which the planning is delivered, with defined responsibilities for monitoring, training of staff, and resource allocation. In the **Evaluation and Assessment phase**, the provider gathers and interprets evidence against the plan, both through internal review and through external mechanisms. In the **Review phase**, the provider decides what changes will follow from the evaluation and incorporates those changes into the next cycle.

8.2 Evaluation models as tools within the EQAVET cycle

The evaluation models described in 5.3 to 5.7 are treated in this report as instruments that plug into specific points within the EQAVET cycle rather than as competing alternatives to EQAVET. Each model is suited to a particular kind of question and to a particular phase of the cycle. The models do not displace the framework; they supply the methods through which an EQAVET-aligned provider generates the evidence that the framework expects. The models are introduced individually below and integrated in 5.8.

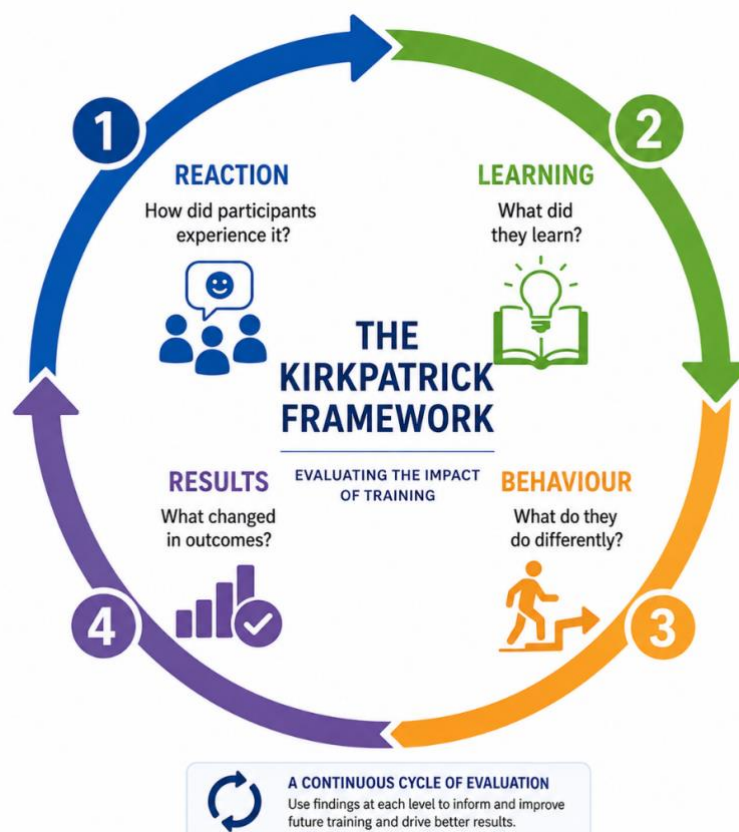


8.3 The Kirkpatrick model

The Kirkpatrick model (Kirkpatrick, 1959, with subsequent revisions) provides a four-level structure for the evaluation of a specific learning programme. The four levels are summarised below.

Level	Question	Typical instruments
Level 1: Reaction	How did participants respond to the training? Did they find it relevant, well-paced, well-pitched?	Post-session questionnaires, structured feedback, focus groups. Thalheimer-informed instrument design (see 5.7) is recommended to avoid weak proxy metrics.
Level 2: Learning	To what extent did participants acquire the intended knowledge, skills and attitudes during the training?	Pre and post tests; demonstrations; observed performance against rubric; assessment items linked to the outgoing competencies declared for the module.
Level 3: Behaviour	To what extent do participants apply what they learned when they return to the workplace?	Workplace observation; mentor or supervisor attestation; performance data over a four-to-twelve-week window; structured self-report against a defined behavioural rubric.
Level 4: Results	What organisational, sectoral or systemic outcomes follow from the behaviour change?	Operational performance data such as disease incidence, animal welfare scores, traceability compliance, food safety incident rates, training completion rates across a cohort.

Within the EQAVET cycle, Kirkpatrick feeds principally into the **indicators on completion, utilisation of acquired skills at the workplace, and internal effectiveness**, and supplies structured evidence for the Evaluation and Review phases.



8.3.1 Limitations of the Kirkpatrick model

The Kirkpatrick model has been subject to sustained and substantive critique since its original formulation, and a methodologically complete framework must acknowledge these limitations explicitly. Seven lines of criticism are of particular relevance to the AGRIFOODSKILLS context.

The event assumption. The model's architecture presupposes that learning originates in a discrete, bounded training event with an identifiable start and end. In a programme that incorporates on-farm experiential learning, peer knowledge exchange, AI-mediated coaching, and continuous workplace practice, this presupposition does not hold. Learning that was never a course, and never had a start date, is structurally invisible to an unmodified Kirkpatrick evaluation.

Invisibility of informal and social learning. The 70:20:10 framework, whatever the precision of its proportions, draws attention to the fact that the majority of workplace learning occurs through experience and social interaction rather than through formal instruction. Kirkpatrick's four levels were developed to evaluate the formal portion and offer no evaluative vocabulary for incidental knowledge sharing, community of practice participation, peer attestation, or AI-assisted reflective practice: all of which are components of the AGRIFOODSKILLS learning model.

Absence of a verified causal chain between levels. The model implies a sequential, causal relationship in which positive reaction produces learning, learning produces behaviour change, and behaviour change produces organisational results. Meta-analytic evidence does not support this chain, particularly at the Level 1 to Level 2 and Level 2 to Level 3 transitions. Satisfaction with a learning experience is a weak predictor of knowledge acquisition; knowledge acquisition measured at end of session is a weak predictor of workplace transfer. An engaging digital or AI-mediated intervention may generate high Level 1 scores while producing negligible Level 3 transfer, and the model as originally formulated provides no mechanism to detect or explain this.

Absence of contextual and motivational variables. The model collects no data on learner motivation, prior knowledge, self-efficacy, workplace climate, or organisational support for behaviour change, yet these variables are among the strongest empirically established predictors of training transfer. When Level 3 data indicate that behaviour change has not occurred, the model identifies the failure but cannot distinguish between inadequate programme design, workplace conditions that blocked transfer, and absence of learner motivation. This explanatory gap is directly relevant in agrifood contexts where workplace culture, resource constraints, and production pressures are known barriers to adoption of new practices.

Insufficient resolution at Level 2. The model treats learning as a single construct. Contemporary learning science distinguishes between recognition, recall, procedural skill, decision-making under realistic conditions, near transfer, and far transfer: distinctions with direct implications for assessment design and for predictions about workplace application. Collapsing these into a single level obscures the most diagnostically important differences in what has actually been acquired.

Poor fit for digital and adaptive learning environments. The model was designed for synchronous, cohort-based, classroom delivery. Modern digital learning environments, especially in adult education, are asynchronous, self-paced, multi-modal, and capable of generating granular, continuous learner data. Attempting to represent xAPI statement data, adaptive pathway decisions, or LRS analytics within a four-level post-training survey schema results in substantial loss of evaluative signal.

No needs analysis or design front end. The model is evaluative in scope: it has no planning or needs assessment component and does not require alignment between programme goals and evaluation design at the outset. It can indicate whether stated outcomes were achieved; it cannot indicate

whether the right outcomes were targeted in the first place.

8.3.2 Mitigation within the integrated framework.

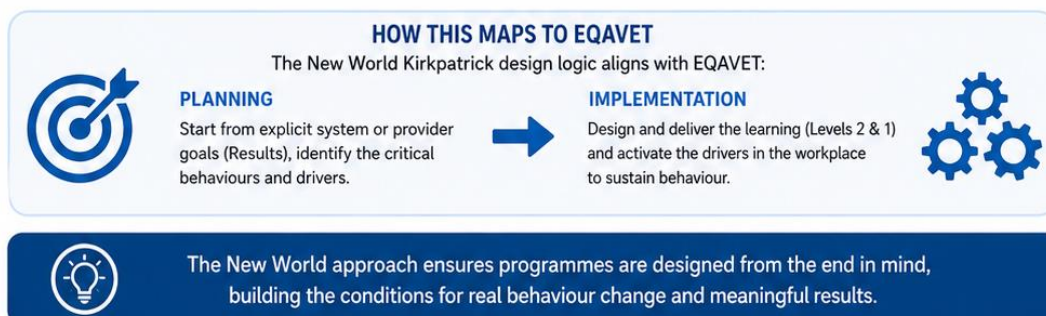
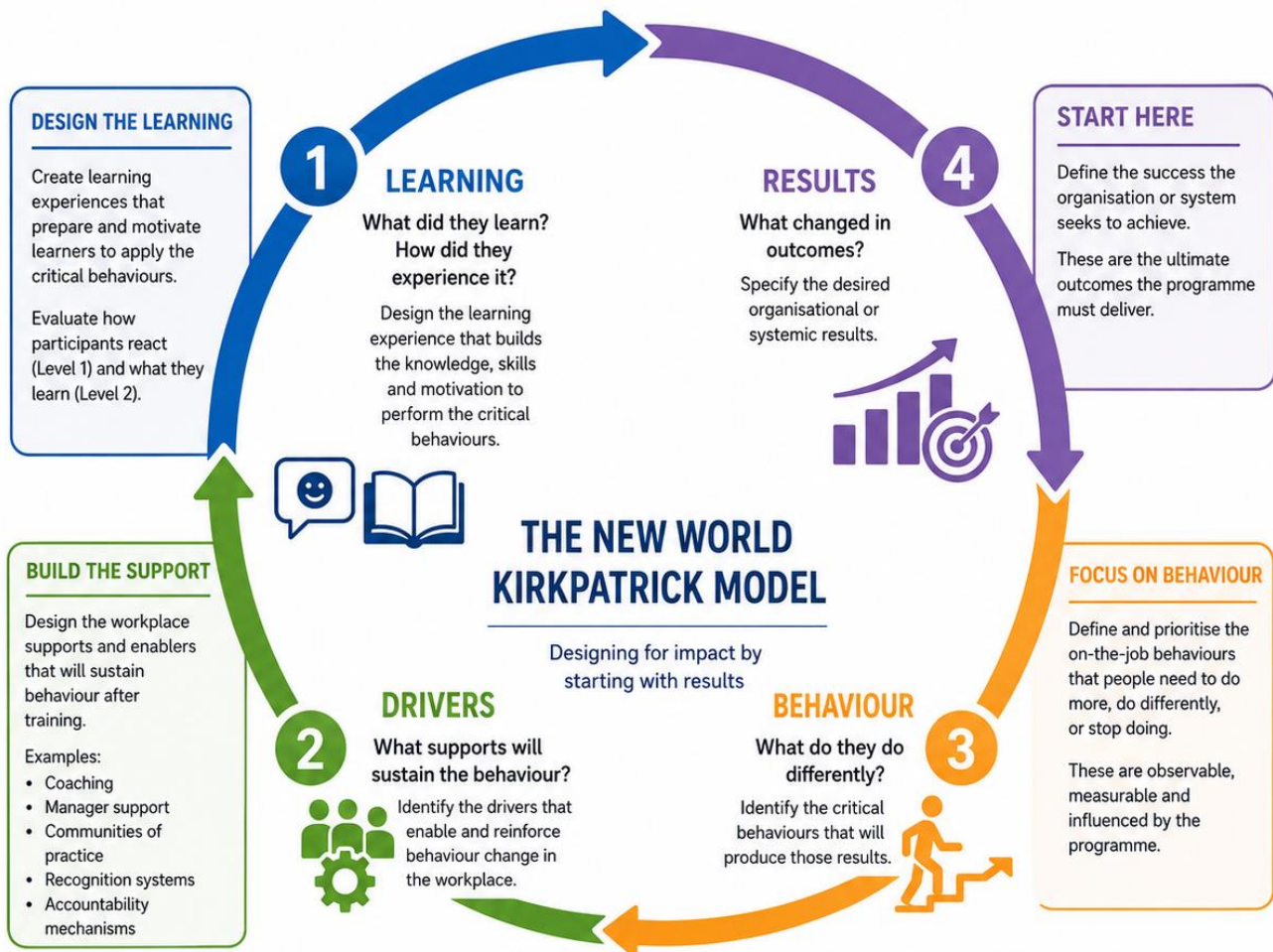
These limitations are treated in this document not as arguments against including Kirkpatrick but as design constraints that the integrated evaluation toolkit must address. Specifically:

1. the EQAVET Planning and Review phases supply the governance, needs assessment, and outcome-alignment functions that Kirkpatrick lacks;
2. Thalheimer's LTEM (see 5.7) strengthens the resolution and evidential quality of Level 1 and Level 2 instruments;
3. Brinkerhoff's Success Case Method (see 5.5) provides the contextual and motivational explanatory capacity absent from Level 3 evaluation;
4. the explicit inclusion of informal and social learning within the evaluation frame, supported where possible by xAPI-compatible data architectures, extends coverage beyond the formal event boundary.

The critique of the model thus becomes part of the justification for the integrated toolkit rather than a reason to exclude it.

8.4 New World Kirkpatrick

The New World Kirkpatrick model (Kirkpatrick and Kirkpatrick, 2016) is a revision of the classic framework that **reverses the conventional design sequence**. Programme design begins by **specifying the desired organisational or systemic results** (Level 4), then identifying the critical workplace behaviours (Level 3) that will produce those results, and then determining the required drivers such as workplace supports, coaching arrangements, communities of practice, recognition systems and accountability mechanisms. Those behaviours will be sustained once formal training has ended. Only when these elements are defined **is the learning programme itself** (Levels 2 and 1) designed. This results-backwards framing maps directly onto the Planning and Implementation phases of EQAVET, since it **requires programmes to start from explicit system or provider goals and from realistic workplace conditions**, rather than from course-level learning outcomes considered in isolation. At the same time, it is strongly congruent with adult learning principles: it anchors provision in problems and outcomes that matter to learners and their organisations, recognises the workplace as the primary site of learning and transfer, and positions formal instruction as a targeted support rather than the centre of gravity.



8.5 Phillips ROI methodology (Level 5)

Phillips (2007) extends the Kirkpatrick model with a fifth level addressing **financial return on investment** (ROI). Level 5 translates the Level 4 results into monetary terms (cost-benefit, cost-effectiveness and ROI percentage) so that decisions about programme **funding, scaling and continuation** can be made on the same basis as decisions about other forms of organisational investment. Within EQAVET, ROI analysis informs decisions on **resource allocation between training options**, on the case for investment in the training of teachers and trainers, and on the relative effectiveness of programmes within a portfolio. ROI is not appropriate to every programme; it is most useful where strategic or budgetary decisions depend on the comparative value of provision across options.

8.6 Brinkerhoff Success Case Method

The Success Case Method (Brinkerhoff, 2003) is a qualitative method that complements the quantitative levels of Kirkpatrick. The method identifies the **most and least successful cases** within a programme cohort, investigates them in depth through interview and document review, and produces **narrative explanations of what produced success and what prevented it** in each case. Within the EQAVET cycle the Success Case Method sits in the **Evaluation and Review phases**. It supplies the causal explanation that quantitative indicators rarely provide on their own and is particularly useful where Kirkpatrick or EQAVET indicator data show substantial variation across participants but where the variation is not accounted for by demographic or other obvious factors. The method is also a strong fit for programme improvement, as it surfaces design strengths and weaknesses with sufficient granularity to act on.

8.7 Thalheimer's LTEM and performance-focused evaluation

The Learning-Transfer Evaluation Model (LTEM; Thalheimer, 2018) and related performance-focused approaches refine the design of evaluation instruments to reduce reliance on weak proxy measures. LTEM distinguishes, in ascending tiers of evidential strength, between **attendance, completion, satisfaction, recall of taught content, demonstrated decision-making and transfer to authentic task performance**, on the principle that not every measurable artefact is informative about learning. Within EQAVET, LTEM and similar approaches strengthen the validity of the evidence that populates the indicators on achievement, utilisation of skills at the workplace and relevance. They are particularly useful for the design of Level 1 and Level 2 instruments and for the avoidance of satisfaction-only metrics that correlate weakly with subsequent transfer.

8.8 The integrated evaluation toolkit for AGRIFOODSKILLS

The AGRIFOODSKILLS pilot programme adopts EQAVET as the overarching framework for evaluation and quality assurance and this will be developed further in T7.4 (Quality assurance EQAVET and ESG) Within that framework the project adopts a defined methodological toolkit for programme-level evaluation:

- Kirkpatrick and New World Kirkpatrick will provide the default structure for programme-level evaluation across Levels 1 to 4, with backward design from desired outcomes and explicit identification of required workplace drivers.
- Phillips ROI methodology will be applied selectively where decisions on resource allocation, scaling or continuation depend on the economic value of provision.
- Brinkerhoff's Success Case Method will be used to investigate variation in pilot outcomes and to surface contextual factors that quantitative data cannot reveal.
- Thalheimer-informed instrument design will be applied to strengthen Level 1 and Level 2 measurement and to avoid weak proxy metrics.

Together these models supply the methods through which the AGRIFOODSKILLS provider community satisfies the EQAVET requirement for continuous, evidence-informed improvement. The instruments are not treated as alternative frameworks to EQAVET.

Evaluating Quality: The EQAVET Toolkit

Evaluation determines if the ecosystem is working.
Assessment determines if the individual is competent.

Brinkerhoff Success Case

Investigate extreme variations in pilot outcomes to surface contextual design flaws.

New World Kirkpatrick

Start at Level 4 (Sector Results) and define required workplace drivers before designing the learning.

Thalheimer's LTEM

Strip away weak proxy metrics (like satisfaction surveys) in favour of demonstrated decision-making.

Kirkpatrick L1-L4

Measure reaction, learning, behaviour transfer, and systemic results.

NotebookLM

Literature in brief: evaluation models

Kirkpatrick D. L. (1959 onwards) and Kirkpatrick J. D. and Kirkpatrick W. K. (2016) on the four-level evaluation model and its New World revision. Phillips J. J. (2007) on the ROI extension. Brinkerhoff R. O. (2003) on the Success Case Method. Thalheimer W. (2018) on the Learning-Transfer Evaluation Model (LTEM) and on performance-focused evaluation instruments. Policy frame: EQAVET (Recommendation of the European Parliament and of the Council of 18 June 2009 on the establishment of a European Quality Assurance Reference Framework for Vocational Education and Training; subsequent revisions and the EQAVET Work Programme 2024 to 2026). System-level commentary: Cedefop and OECD reports on quality assurance and on the evaluation of vocational provision.

Design summary: applying the toolkit within an EQAVET-aligned design

- **Planning.** Use New World Kirkpatrick to define the desired Level 4 results, the Level 3 critical behaviours, and the required drivers for the workplace. Map these to the relevant EQAVET indicators.
- **Implementation.** Use the required drivers identified in Planning to design supports in the workplace (coaching, prompts, audit, communities of practice) that will produce the L3 behaviour change.
- **Evaluation.** Use Kirkpatrick L1 to L4 to structure outcome data collection. Use Thalheimer-informed instrument design for L1 and L2. Use Brinkerhoff to investigate variation where the indicator data show it.
- **Review.** Use Brinkerhoff narrative findings to guide design changes. Use ROI and New World Kirkpatrick to argue for reallocation of resources to what works.
- **Reporting.** Report evidence against EQAVET indicators alongside the credential composition data described in Sections 5 to 8.

9. Credentials and recognition

Section 9 sets out the architecture by which the assessment evidence produced under Section 7 and the evaluation cycle in Section 8 is composed into credentials that are portable, machine-readable and recognised across providers and Member States. The section is organised in two halves: 9.1 sets out the assertion-and-credential model under the European Learning Model; 9.2 sets out the practical credential infrastructure (ELM, EQAVET, ESCO, eIDAS, EUDI Wallet) and the post-MICROBOL VET opportunity.

9.1 Assertions vs credentials: a vocabulary the sector needs

In order to develop the deliverable of stackable competencies that deliver on the needs of adult learners, the field requires a vocabulary that distinguishes the **atomic claims (assertions)** from the composed claims (**credentials**) built from them. The MICROBOL discussion and the 2022 Council Recommendation have, on the whole, blurred this distinction by treating microcredentials as the atomic unit. However, the atomic unit is the assertion. Credentials, including microcredentials, are compositions of assertions. Making that distinction operational supports portability across providers, mobility across borders, and stackability across a career. The European Learning Model (ELM) has created new purpose to the concept of atomic assertions that can be summarized as

1. Claims that a person **has achieved** something (an achievement/credential assertion).
2. Claims that a person has **completed/participated in** a specific learning activity or opportunity.
3. Claims about the person's **entitlements/rights or status** that follow from that achievement (e.g. access, professional rights, recognition).
4. Claims about the person's **identity or affiliation** (who they are / how they are linked to a provider or context).

Throughout this section claims 1 and 2 will be the focus of learning assertions.

9.1.1 Definitions

Term	Definition	Examples
Assertion	An atomic, signed, machine-readable claim about a learner made by a qualified issuer at a point in time. Each assertion stands or falls on its own evidence and signature.	'Maria attended demonstration X on 2026-04-15.' / 'Maria passed assessment Y (rubric Z) with score 78%.' / 'Maria's supervisor signs off that she performed protocol P correctly on three occasions across April-May 2026.'
Credential (incl. microcredential)	A composition of assertions, issued by a recognised authority, claiming that a learner has acquired a named competency or set of competencies. The credential's validity rests on (i) the validity of its constituent assertions and (ii) the rule that composed them.	'Maria holds the AGRIFOODSKILLS microcredential 'Biosecurity Lead, EQF 4'.' This credential is composed of assertions of attendance, assessment, and mentor sign-off of can-do-the-role behaviour.
Issuer	A qualified party that signs assertions or credentials under a recognised trust framework. Under eIDAS 2.0, qualified trust service providers underwrite the signature.	A VET provider, an HEI, a sector training body, a veterinary specialisation college, an employer with assessor authority.
Wallet	A user-controlled instrument in which the learner holds assertions and credentials, and from which they can present them to relying parties.	The European Digital Identity Wallet (EUDI Wallet) under eIDAS 2.0.

9.1.2 Why the distinction matters

Three architectural consequences follow from treating the assertion as the atomic unit of recognition.

1. Multiple issuers can contribute to a single credential. A learner can collect an **attendance assertion** from one provider, an **assessment assertion** from another, and a **competency assertion** from a workplace mentor. A recognised authority then composes the credential.
2. Credentials can be recomposed by different recognising authorities for different purposes. The same set of assertions about a learner might support a national VET credential in one country, an HEI module credit in another, and a sector certification recognised by an industry employer.
3. **Learners can move freely across providers, countries and modalities** without losing prior evidence. The assertion travels in the learner's wallet; the credential is a view onto that evidence.

The three-assertion framework introduced earlier in the report may therefore be reinterpreted as the three principal categories of assertion from which a complete microcredential should be composed: engagement assertions, assessment assertions and competency attestations. A microcredential composed only of engagement and assessment assertions does not evidence performance in a work context. A microcredential that incorporates evidence under all three categories supports recognition by employers and by regulatory bodies.

9.1.3 How assertions map to ELM and ESCO

The **European Learning Model (ELM)** already supports this distinction. ELM is built for verifiable claims about learning achievements that integrate ESCO competency references and reference the EQF level. A well-designed ELM credential is a **structured composition of attestations**, each of which is itself verifiable. The architecture already exists in specification form and now requires consistent use across providers. AGRIFOODSKILLS's contribution is to apply this architecture to VET in agrifood and veterinary education in a way that other Pact for Skills partnerships can adopt.

Literature in brief: assertions, badges and verifiable credentials

The atomic-assertion model traces back to the Open Badges specification (Mozilla, 2011, 2013) and the later W3C Verifiable Credentials data model (2019; updated 2022). The European Learning Model adopts the verifiable-credentials approach and integrates ESCO at the competency level. The xAPI (Tin Can) specification provides a long-standing schema for atomic learning records and is widely used as an LRS transport layer. Recent commentary (ENQA 2023; EQAVET PLA 2022) addresses quality assurance specifically for microcredentials and is converging on the assertion-and-composition view.

Design summary: how to author an assertion that will compose

- State at least one claim per assertion (one attendance, or one assessment, or one performance attestation; not several claims in one record).
- Reference the ESCO concept the assertion concerns by identifier where one exists; where no settled ESCO concept exists, use the AGRIFOODSKILLS working concept identifier.
- Author the assertion in a form aligned with the European Learning Model. The Model defines the data structure; a learning event that does not record assertions in this structure is not part of the European Learning Model and the European Commission position requires forward-looking design.
- Note that issuer signing under qualified eIDAS trust services is anticipated but not yet operationally available at the time of this report. AGRIFOODSKILLS designs should be prepared for that signing layer when it is in place, and should record provisional issuer identification in the interim.
- Attach the evidentiary basis: a rubric reference, an assessment record, a workplace artefact reference, a mentor attestation, or other material that supports the claim.
- Author the assertion in both human-readable and machine-readable forms; the European Learning Model schema provides both.

9.2 The credential infrastructure:

The architecture in Sections 6 and 7, ESCO-aligned competencies, atomic assertions, composed credentials, has until very recently lacked the public infrastructure to make it portable in practice. This section sets out the policy and technical environment in which AGRIFOODSKILLS will operate and argues that agrifood and veterinary VET is well-positioned to use it first.

9.2.1 The MICROBOL legacy and the higher-education adoption pattern

Between 2020 and 2022 the **MICROBOL project** ('Microcredentials linked to the Bologna key commitments') worked through whether microcredentials could fit inside the European Higher Education Area using the existing Bologna toolkit, qualifications frameworks, ECTS, the Diploma Supplement, recognition. Its conclusion was, in essence, that the toolkit was largely sufficient. The 2022 Council Recommendation on a European approach to micro-credentials translated this into EU-level policy and extended scope beyond higher education to lifelong learning and employability.

In the years that followed, member states moved fastest in the HE lane. National HEIs adopted microcredential policies, attached them to existing modular ECTS structures, and began issuing them, often as European Digital Credentials for Learning (EDCs). VET has lagged. Microcredentials in vocational training have been piloted but at nothing like the same density. The reason is partly cultural, VET had less modular tradition to graft onto, but it is also partly infrastructural: the **trust and portability** layer that EDCs provided for HE was harder to use for VET, where issuers are more numerous, more fragmented and less institutionally homogeneous. In some jurisdictions, micro credentials are restricted entirely to HEIs.

9.2.2 eIDAS 2.0 and the EUDI Wallet: the missing trust layer

The revised eIDAS Regulation (EU 2024/1183) places a legal obligation on every member state to provide citizens with a **European Digital Identity Wallet** (EUDI Wallet). The Wallet is, among other things, a qualified container for verifiable attestations of attributes, including **educational** and **professional** credentials, issued under a **public trust framework**. From the citizen's perspective it is a single, cross-border instrument in which credentials from any qualified issuer can be held, presented and verified. From the issuer's perspective it removes the need for bilateral recognition agreements: if a credential is issued by a qualified issuer in a conformant data format, it is portable by default. Together with ELM and ESCO, this layer makes a federated microcredential ecosystem realistic, and it closes the gap between HE and VET, because it does not depend on the issuer being an HEI.

9.2.3 ELM-conformant assertions under EQAVET quality

The European Learning Model (ELM) is the EU's open data model for learning achievements. It supports **formal, non-formal and informal learning** across general education, VET, higher education, adult learning and labour-market training. It integrates ESCO directly so competencies and skills referenced in a credential are machine-readable in 31 languages. Credentials designed in alignment with the European Learning Model, by a qualified issuer, presentable through the EUDI Wallet, are the technical realisation of the assertions/credentials model from Section 6. The European Learning Model is **agnostic** as to whether the issuer is a **higher education** institution or a **vocational education and training provider**. The principal historical obstacle to its use in vocational settings has been the absence of an applicable **trust framework** for non-HEI issuers, and the revised eIDAS Regulation provides that framework. The quality assurance dimension for vocational provision is now supplied by EQAVET, which has been extending its scope to microcredentials since 2022 through the EQAVET network's peer-learning activity on quality assurance of microcredentials.

Taken together these instruments form a layered European architecture in which EQAVET supplies the quality assurance frame for vocational provision, ESCO supplies the shared competence vocabulary, the European Learning Model supplies the credential schema, and the European Digital Identity Wallet

operating under the revised eIDAS Regulation supplies the portability and trust layer.

The architectural intent is that a vocational provider, whether a farm advisory service, a sector training organisation, a food-industry trainer or a veterinary specialisation body, will be able to issue microcredentials that are quality-assured under EQAVET, expressed in the European Learning Model with ESCO concept references where available, and portable across Member States through the EUDI Wallet, without recourse to higher-education intermediation. Several elements of this architecture are not yet operationally available at the time of writing. The EUDI Wallet is at varied stages of rollout across Member States; qualified trust services for educational credentials are not yet in routine operation; and ESCO does not yet contain concept identifiers for several of the competencies that AGRIFOODSKILLS will define. The AGRIFOODSKILLS approach is therefore to design in alignment with the architecture rather than to assume immediate issuance under it, with the expectation that issuance under each element of the architecture will become possible as the relevant operational infrastructure matures.

9.2.4 Reclaiming the badge principle

The original **Open Badges movement** (Mozilla, 2011 onwards) was correct in that small, declarative and stackable units of achievement are the natural shape of lifelong learning. It was wrong about the trust infrastructure: without a public, qualified trust framework, badges drifted toward participation trophies. The realistic course now is to reclaim the badge principle on top of an infrastructure that finally exists. **Small units of learning that declare what they require and what they evidence; that compose into occupational profile;** that are quality-assured under EQAVET; that are signed under eIDAS 2.0; that are portable in EUDI Wallets; and that are searchable in ESCO-tagged form across the European labour market. AGRIFOODSKILLS should design its microcredentials to this specification from day one.

9.2.5 What this means for AGRIFOODSKILLS occupational profiles

Each occupational profile designed in Task 5.2 (eight profiles, approximately 130 hours each) should:

1. be decomposable into smaller microcredential units that may be earned independently;
2. publish, for each unit, its prerequisite competencies and its outgoing competencies, each aligned with an ESCO concept identifier where one exists and otherwise with a working concept identifier published by the project;
3. evidence the three assertion categories (engagement, assessment and performance attestation) that the European Learning Model treats as the constituent achievements of a credential;
4. be designed in alignment with the European Learning Model, in anticipation of issuance under qualified eIDAS trust services and presentation through the European Digital Identity Wallet as those services become operationally available;
5. be quality-assured under EQAVET indicators, with the quality assurance evidence linked from the credential metadata;
6. be designed using at least one engagement methodology from Section 6, selected on the basis of fit to the learner's time pattern and EQF level rather than trainer preference.

Literature in brief: credential infrastructure

Regulation: Council Recommendation 2022/C 243/02 on a European approach to micro-credentials; Regulation (EU) 2024/1183 (eIDAS 2.0) on the European Digital Identity Framework. Data model: European Learning Model documentation and the ELM Browser (Europass). Vocabulary: ESCO classification and its integration with ELM. Quality: EQAVET 2024-2026 work programme on QA of micro-credentials; ENQA WG report on quality assurance of micro-credentials (2023). Background: MICROBOL final reports (2022) for the HE-side conclusions that the VET agenda now extends.

Opportunities: for AGRIFOODSKILLS and the Pact for Skills

- Design at least one AGRIFOODSKILLS microcredential in alignment with the European Learning Model before M30, with a view to issuance under qualified eIDAS trust services as those services become operational.
- Use it as a reference implementation for other Pact for Skills partnerships. The target date for issuance under signed eIDAS infrastructure is set in alignment with the European Digital Identity Wallet rollout milestones and may be missed where upstream Member State or Commission infrastructure is delayed.
- Engage the Pact's communications channels to position agrifood and veterinary VET as one of the sectors to adopt portable VET microcredentials at the earliest practicable point, and coordinate with the EQAVET peer-learning activity on quality assurance of microcredentials to share project learnings.

Design summary: minimum-viable alignment for an AGRIFOODSKILLS microcredential

- Prerequisite and outgoing competencies published, aligned with ESCO concept identifiers where available and otherwise with project working concept identifiers, together with the EQF level.
- Three-assertion evidence model implemented, with explicit sources for each assertion category and the assertions structured in alignment with the European Learning Model schema.
- Credential design aligned with the European Learning Model, prepared for issuance under qualified eIDAS trust services and presentation through the EUDI Wallet as those services become operationally available.
- Quality assurance applied under EQAVET indicators, with the QA evidence linked from the credential metadata.
- Multilingual rendering supported, in English and in the country language of the issuing provider, using the European Learning Model's native multilingual support.

10. A Learning Record System in the Observatory: an innovative conceptual proposal

Status note: conceptual proposal

This section sets out a conceptual proposal for discussion within the AGRIFOODSKILLS consortium and the Agrifood Pact for Skills. The proposal is exploratory rather than a project commitment. Its purpose is to indicate an architectural possibility that the Observatory function might plausibly host. Implementation would require separate governance arrangements, funding and a comprehensive data-protection impact assessment, none of which are within the scope of Deliverable 5.1.

10.1 The mobility problem we are trying to solve

Career trajectories in the agrifood and veterinary workforce are commonly **distributed across multiple training providers** and Member States. A representative trajectory might combine initial training with a local advisory service in one Member State, an online module from a vocational provider in a second, a peer-demonstration day organised by a cooperative in a third, a six-month placement in a processing plant in a fourth, and a specialisation course in a fifth. Each of these touchpoints may produce **evidence of competence**. At present, each produces evidence in a different form within a different system, and there is **no shared mechanism** by which the evidence may be composed into a credential recognisable by an employer in a further Member State.

The credential infrastructure described in Section 7 addresses portability for credentials and assertions once they have been issued. It does not, on its own, address the question of **composition**: that is, the mechanism by which dispersed assertions issued by multiple providers across multiple Member States may be composed into recognised credentials, and the mechanism by which learners, employers and recognising authorities may **obtain an integrated view of a learner's accumulated evidence**. A **Learning Record System (LRS)** provides the architectural element that addresses these questions.

10.2 What an LRS is, conceptually

A Learning Record System is, in essence, a **learner-centric ledger** of assertions. Each assertion arriving at the LRS is structured (typically conforming to a schema such as xAPI/Tin Can or to **ELM**), references the ESCO competency it concerns, identifies its issuer, carries its evidence and its signature, and is logged against the learner's identifier. The LRS **does not itself issue credentials**; it is a record-keeping infrastructure that lets credentialing authorities compose credentials from the assertions, lets learners see their own evidence, and lets employers verify what they are being shown.

10.3 Why the Observatory is a natural home

The AGRIFOODSKILLS Observatory function is positioned exactly where such a system would need to sit: sector-wide, cross-border, neutral between providers, focused on long-term skills intelligence and credential coherence. The Observatory already gathers anonymised data about skills demand and training supply across the agrifood value chain. Adding an LRS function would **extend that intelligence** from the supply side (what training is offered) to the evidence side (what competence is actually being acquired, where, and by whom). This is not a function any single provider, country or HEI can deliver. It needs the neutrality and longevity of a sector-wide observatory. The Agrifood Pact for Skills already commits to that role.

The Observatory-hosted LRS serves as the sector's '**Single Digital Gateway**'. By integrating real-time job postings with training supply data, it creates a **searchable environment** for agrifood expertise. This allows the Observatory to identify precisely **where the supply-and-demand mismatch is concentrating** and provides the learner with **industry-validated content** through the professional channels they already use, such as LinkedIn

10.4 Anonymised data as the analytics layer

An Observatory-hosted LRS offers a powerful potential source of anonymised analytics about skills acquisition in the sector. It could, with appropriate privacy and aggregation rules, tell the Pact for Skills community things that are otherwise unknowable: which competencies are being acquired across the sector and which are not, where stacking patterns are emerging, which providers are producing strong can-do-the-role evidence, where the supply-and-demand mismatch is concentrating geographically. This is skills intelligence at a granularity that no individual provider or national qualifications authority can provide alone.

Two principles would be foundational.

1. **Individual learner data is the learner's;** the LRS holds evidence on the learner's behalf, with their consent, and the learner controls what is presented to third parties.
2. **Aggregated analytics are anonymised by design,** with k-anonymity or differential privacy thresholds enforced before any cross-sector view is produced. Neither principle is exotic; both are well-understood in the European data-protection tradition and consistent with GDPR.

10.5 Cross-border, cross-provider mobility by design

Under the proposed architecture, two distinct learner journeys are supported by the same machinery.

Journey	What happens
Whole credential with one provider in one country	All assertions for the credential are issued by one provider, signed under that country's qualified trust service, deposited in the LRS, composed into a credential by the recognised authority, and held in the learner's EUDI Wallet.
Assertions collected across providers and borders	Assertions are issued by multiple providers in multiple countries, each signed under their own qualified trust service, each deposited in the LRS against the same learner identifier, then composed into a credential by a recognised authority whose composition rule recognises the satisfying assertion set.

From the perspective of the learner the experience is uniform: **evidence accumulates progressively** as the learner undertakes training, and the learner is credentialed when the evidence satisfies the rule for composition. From the perspective of the credentialing system, learner mobility across providers and Member States is supported as a default property of the architecture rather than as a case requiring bilateral recognition arrangements.

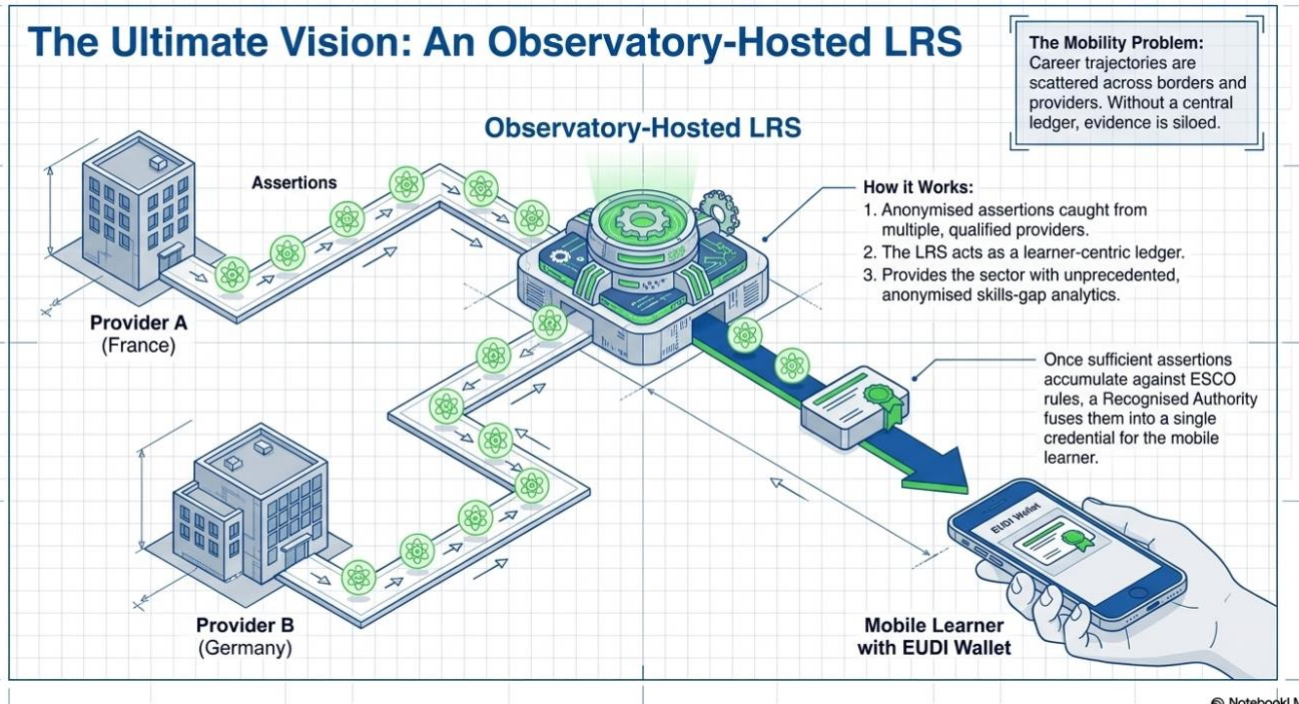
10.6 Governance, trust and what would need to be worked out

A substantive LRS proposal would need to address, at minimum, the following design questions.

- **Issuer accreditation:** which providers can deposit assertions, and under what quality assurance (presumably EQAVET indicators for VET, ESG for HE).
- **Identity binding:** how a learner's identifier in the LRS connects to their EUDI Wallet identity without becoming a pan-European personal identifier (a politically and legally sensitive question).
- **Composition rules:** who has authority to compose assertions into a recognised credential, and against what published rule.
- **Data residency and jurisdiction:** where the data sits, which member state's data-protection authority oversees it, how cross-border requests are handled.
- **Sustainability:** who funds the LRS over the long run, given that no single provider or national authority is the natural owner.
- **Open standards:** ELM, ESCO, W3C Verifiable Credentials, xAPI/Tin Can, the LRS should be schema-agnostic on the issuer side and ELM-canonical on the presentation side.

10.7 Why this is timely

The conjunction of post-MICROBOL VET microcredential adoption, the EUDI Wallet rollout obligation under eIDAS 2.0, the maturation of ELM and the EU AI Act creates a near-term policy window in which a sector-wide LRS could be designed, piloted and validated on the back of a Pact for Skills mandate. AGRIFOODSKILLS does not have to build it within the project's M13-M17 window, but the project should put the idea formally on the table, sketch the architecture, and recommend that the Agrifood Pact for Skills explore it as part of its long-term sustainability plan.



Literature in brief: learning record systems and federated credentials

Schema and transport: xAPI (Tin Can) specification 1.0.3 (ADL); W3C Verifiable Credentials data model 2.0. Architectural references: IMS Global / 1EdTech Comprehensive Learner Record Standard; Open Recognition Alliance literature on federated badge ecosystems. EU context: European Learning Model documentation; the EUDI Wallet Architecture and Reference Framework; the Common European Skills Data Space (proposed under the European Data Strategy). Privacy: GDPR Articles 5, 6, 9; ENISA guidance on pseudonymisation and k-anonymity in education data.

Opportunities: open questions for the consortium

Should the AGRIFOODSKILLS Observatory commission a feasibility study of the LRS concept in H2 2026, scoped against the EUDI Wallet rollout milestones and the Common European Skills Data Space timeline?

Which member state's data-protection authority would host the lead supervisory function?

Which existing Pact for Skills partnership in a related sector would be the natural co-pilot?

Can the EQAVET network and ENQA both be invited to advise on QA-of-LRS-deposited-assertions?

Is there an Erasmus+ Forward-Looking or Horizon Europe call that could fund the next-stage architecture work?

Design summary: what could happen between M18 and M24

- M18 (Dublin meeting): consortium endorsement, or rejection, of the conceptual proposal, go / no-go.
- M19: scoping note circulated to the Pact for Skills board for indicative interest.

- M20-M22: feasibility study commissioned, covering governance, data protection, technical architecture, sustainability.
- M23: stakeholder workshop with EQAVET, ENQA, Europass and one or two friendly member state authorities.
- M24: decision point on whether to move to a pilot under a separate funding instrument.
- Throughout: AGRIFOODSKILLS continues to issue ELM credentials through its providers as if the LRS already existed, i.e. in a form that the LRS could ingest at any later date.

11. Directional recommendations to the Agrifood Pact for Skills

The recommendations set out below summarise the positions developed in Sections 1 to 10 and are presented as proposals for adoption by the Agrifood Pact for Skills as standing practice.

R1. Move off the classroom default. Multi-day single-venue classroom training is reserved for cases where physical co-presence is demonstrably necessary. The default delivery format is a blended design combining short asynchronous units, on-site peer demonstration, and work-based learning.

R2. Design at the competency level, not the syllabus level. Every microcredential publishes prerequisite competencies (ESCO-referenced) and outgoing competencies (ESCO-referenced). Learning outcomes remain as the institutional design layer beneath the competency interface, but they are not the unit of stackability.

R3. Adopt the assertion-and-credential distinction. Treat assertions as the atomic unit. Every microcredential is a composition of assertions, each independently verifiable. Engagement, assessment and can-do-the-role are the three assertion categories every credential must support.

R4. Design credentials in alignment with the European Learning Model under EQAVET quality. All AGRIFOODSKILLS microcredentials are designed for alignment with the European Learning Model, are quality-assured under EQAVET indicators, and reference ESCO concept identifiers where available. Issuance under qualified eIDAS trust services and presentation through the EUDI Wallet will follow as the corresponding Member State infrastructure becomes operational.

R5. Treat the EUDI Wallet rollout as a strategic opportunity for VET. AGRIFOODSKILLS aims to be among the earliest sector consortia to issue VET microcredentials in the EUDI Wallet, and to publish a reference implementation other Pact for Skills partnerships can adopt.

R6. Reclaim the badge principle. Small, declarative, stackable units of competence - built on EQAVET quality, ELM data, ESCO vocabulary and EUDI trust - are an appropriate form for a sector with time-poor, mobile, lifelong learners. AGRIFOODSKILLS commits explicitly to this shape and uses it as the design constraint for occupational profile decomposition in T5.2.

R7. Frame AI as a transformation of the learning ecosystem. AI is not one new methodology; it is a transformation of what learning experiences are possible, what trainers do, and what durable competence looks like. AI literacy is added as a transversal competence to every occupational profile at level-appropriate granularity. AI Act and GDPR compliance is by design.

R8. Explore a Learning Record System within the Observatory. Place the LRS concept formally on the M18 consortium agenda. Commission a feasibility study in H2 2026 scoped against the EUDI Wallet rollout milestones and the Common European Skills Data Space timeline.

R9. Build the trainer profession to match. The methodologies in this report ask different things from trainers. The train-the-trainer programme (T6.1) develops explicitly the competencies of designing blends, briefing peer demonstrators, authoring micro-learning pathways, running flipped synchronous sessions and using AI co-learning critically. These competencies are themselves ESCO-tagged and credentialed under the same architecture.

R10. Adopt EQAVET as the overarching evaluation framework, with Kirkpatrick as the default spine. Pilot evaluation (Task 7.4) is structured by Kirkpatrick Levels 1 to 4 within the EQAVET cycle, with New World Kirkpatrick used at the planning stage to specify desired results, critical behaviours and required workplace drivers. Phillips ROI methodology is applied selectively where resource decisions depend on economic value. Brinkerhoff Success Case Method is used to investigate variation in outcomes. Thalheimer-informed instrument design is used to strengthen Level 1 and Level 2 measurement. Satisfaction metrics are treated as one input to Level 1 and are not relied upon as a primary measure of programme effectiveness.

R11. Transition to a 'Utility Model' for Sectoral Sustainability. To ensure the infrastructure survives beyond the initial project grant, the Pact should implement a '**servitization**' model. This includes generating non-grant revenue through **customized skills audits for SMEs** and **micro-credential certification fees** for VET providers. Recognition must be built on **standardized, industry-validated profiles** that offer **monetary and social/societal recognition**, allowing users to advertise their verified expertise directly to the labor market.

Design summary - what each stakeholder does next

- Consortium partners (this quarter): adopt R1-R10 in principle at M18; nominate one champion per recommendation.
- T5.2 lead (ICOS) and contributors: decompose occupational profiles to the architecture in Section 7.5.
- T6.1 lead and partners: build the train-the-trainer programme around R9.
- Pact for Skills board: receive R5, R6 and R8 as standing items; commission a feasibility study on the LRS concept.
- Communications (WP8): position AGRIFOODSKILLS publicly as a VET reference implementation for ELM-EQAVET-EUDI.
- External providers funded by the consortium: align with the minimum-viable conformance in Section 7's design summary.

Opportunities - open questions to bring back to the consortium

Which of the eight occupational profiles is the natural first candidate for a reference implementation of the full ELM-EQAVET-EUDI stack? Which member state's qualified trust service is the natural partner for the eIDAS 2.0 signature layer for our first cross-border microcredential? Can the veterinary representatives agree on a shared specialisation use case as the EQF 6-8 reference case? Is there an alignment opportunity with the Common European Skills Data Space pilot calls? What does the Observatory need to do by M30 to be ready to host an LRS function later?

The recommendations above describe an implementation path that is consistent with the present regulatory and technical environment and with the adult-learning evidence base summarised in Sections 2 and 3. The agrifood and veterinary sectors are well-positioned to adopt the path within the period of AGRIFOODSKILLS delivery.

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13. AI use declaration

This document was collaboratively formulated by three authors. The following AI-assisted processes were employed during its preparation:

- **Transcription assistance** — AI-assisted transcription of authored spoken content was performed using Anthropic Claude Sonnet.
- **Deduplication and restructuring** — AI-enhanced deduplication and structural reorganisation were applied to consolidate and refine the document architecture using Anthropic Claude Sonnet.
- **Linguistic and spelling review** — Linguistic refinement and spelling checking were performed using Anthropic Claude Sonnet.
- **Critical review and stress-testing** — The conceptual framework of a unified learning record system, including the competency stacking architecture and pathway logic was developed solely by the authors, based on a wider understanding of the literature and the underpinning technology. It was subjected to AI-assisted critique using Anthropic Claude Sonnet as an adversarial reviewer to interrogate assumptions, identify logical gaps, and stress-test the coherence of the proposed model. It surfaced the broad structural concerns that were then developed and evaluated by the authors and the framework was revised accordingly.
- **Image generation** — Images contained within this document were generated using Nano Banana 2 and Nano Banana Pro (Google DeepMind / Gemini) and NotebookLM.

All substantive content, professional judgements, competency statements, and educational design decisions were originated by and remain the intellectual work of the named authors. All AI-assisted outputs were reviewed, verified, and approved by the authors prior to inclusion. The authors accept full responsibility for the accuracy, integrity, and appropriateness of the content of this document.

This declaration is made in accordance with the authors' own AI competency framework principle:

Question, Use, Scrutinise, and Communicate described in section 2 and the use of AI tools in this work has been purposeful, critically evaluated, and openly disclosed.

Appendix A. Practical guidance: Formulating competency statements

This appendix is a practical companion to Section 6 of the report. Section 6 establishes why AGRIFOODSKILLS designs at the competency level rather than at the learning-outcome level; the appendix sets out how a competency statement is written and how it is connected to the European Skills, Competences, Qualifications and Occupations classification (ESCO). The appendix is scoped to the competency layer only. The two pieces of guidance are intended to be read together by trainers and curriculum designers working on the occupational profiles' learning design to be developed in Task 5.2.

A.1 Outcomes and competencies in practice

Learning outcomes describe what a learner is expected to know or to be able to do at the end of a specific module. Competency statements describe what a learner is able to do in a real working context, at a recognised EQF level, with reference to an external classification (here, ESCO). The two are complementary. The learning outcome is internal to the module; the competency statement is the module's public interface to other modules, to employers, to recognising authorities and to credentialing systems. The table below illustrates the practical difference for the same topic.

Topic	A learning outcome (module-internal)	A competency statement (external interface)
Milking parlour hygiene	On completion of this module, the learner will be able to describe the principles of teat-end disinfection and identify the situations in which routine and contingency hygiene protocols apply.	Apply hygiene protocols in milking parlour operations under routine and incident conditions, in accordance with farm standard operating procedures and applicable EU dairy hygiene regulation, to a standard verifiable by mentor sign-off. ESCO: aligned with 'apply hygiene rules' Occupations: livestock and dairy worker; EQF: 3 to 4
HACCP	On completion of this module, the learner will be able to identify hazards and critical control points within a defined food production process.	Conduct a hazard analysis and identify critical control points for a defined food production process , document them in a HACCP plan, and validate them against the current legislative framework. ESCO: aligned with 'apply HACCP' and related food safety concepts; EQF: 4 to 5.
Antimicrobial stewardship (veterinary)	On completion of this module, the learner will be able to apply the current legal and clinical framework for antimicrobial prescribing .	Make antimicrobial prescribing decisions in mixed-species clinical practice, in accordance with the current legal framework and with reference to the current antimicrobial resistance evidence base, in a manner that is auditable against peer-reviewed prescribing rubrics. ESCO: aligned with veterinary clinical and prescribing concepts; EQF 7.

A.2 Anatomy of a competency statement

A competency statement has four elements. 1) **The action verb** names what the learner does (apply, conduct, design, audit, supervise, lead, evaluate). 2) **The object** names the thing acted upon (a HACCP plan, a treatment regimen, a herd health protocol, a traceability record). 3) **The context** specifies the working environment in which the action is performed (in routine and incident conditions; in mixed-species clinical practice; on a primary-processing line; under farm SOPs). 4) **The standard** specifies how performance will be judged (to a specified rubric; against current legislation; with mentor sign-off; with auditable record-keeping). The four elements are written in continuous prose, not as a checklist, but each is recognisable in a well-formed statement.

Element	Question it answers	Examples
Action verb	What does the learner do?	Apply ; conduct; design; audit; supervise; lead; evaluate; instruct; commission.
Object	On what is the action performed?	HACCP plan ; treatment regimen; herd health protocol; traceability record; biosecurity audit.
Context	In what working environment?	On a primary-processing line ; in mixed-species clinical practice; under farm SOPs; during emergency response.
Standard	How is performance judged?	To a specified rubric; in accordance with current legislation ; with mentor sign-off; with auditable records.

A.3 Linking competencies to ESCO

Every competency statement should be tested against the ESCO classification of skills, competences and occupations. ESCO is the European Commission's multilingual classification, with stable concept identifiers and machine-readable relations to occupations and to qualification frameworks. The use of ESCO concept identifiers in AGRIFOODSKILLS competency statements is what makes them interoperable across providers and across Member States.

A practical procedure for the linkage runs as follows. **First**, search the ESCO portal for the action verb and the object of the draft competency statement. **Second**, examine the candidate concepts returned; preferred candidates are those whose definition matches the intended scope of the competency, whose skill type (knowledge, skill or transversal) is appropriate, and whose linked occupations include those addressed by the AGRIFOODSKILLS occupational profiles. **Third**, examine the broader and narrower relations of the candidate concept; competency statements often map to a concept that is one level more specific than the first-returned candidate. **Fourth**, where no appropriate concept exists in ESCO, define a working concept identifier under the AGRIFOODSKILLS namespace using the ESCO conventions (label, definition, broader/narrower relations, skill type) so that it can be submitted for inclusion in ESCO through the established Commission process.

Working concept identifiers should be treated as provisional and replaced by the ESCO identifier once it has been issued. The assertion-based architecture described in Section 7 supports composition against either identifier without disruption.

During the development of materials in WP5, New ESCO competencies should be listed in a shared document to develop overlap between occupational groups to prevent duplication. An example template for emerging competencies is shown below. This will allow the consortium to develop an outline of specific outcomes that can be aligned to verifiable learning credentials. The shared register serves five functions during the WP5 working period.

1. It prevents the reinvention of equivalent competencies under different wordings by separate

partners working on adjacent occupational profiles, by establishing a single working concept identifier that any partner may cite.

2. It surfaces transversal competencies as a by-product: where the register shows the same emerging competency cited by more than one occupational profile, that competency is by definition a candidate for a stand-alone transversal microcredential.
3. It strengthens the ESCO submission package by allowing AGRIFOODSKILLS to submit a curated, deduplicated and cross-referenced set of candidate concepts to the European Commission rather than a fragmented bundle of one-off proposals.
4. It supports stacking in practice, because the prerequisite and outgoing competencies of microcredentials issued by different partners can reference common identifiers from the register.
5. It enforces consistency in EQF level anchoring, by recording the level at which each working concept is expected to be demonstrated and exposing any inconsistencies between partners.

Ownership and stewardship of the register sit with EBVS in respect of its appendix authorship and ESCO-stewardship role within the project, in coordination with the Microcredential learning design working group of the Observatory. Working concept identifiers in the register are provisional. Each entry carries a status indicator that tracks its progression from draft, through working-group review, to submission to ESCO and ultimately to acceptance into the classification. Once a working concept has been accepted into ESCO, the ESCO identifier replaces the working identifier in subsequent AGRIFOODSKILLS material; the architecture in Sections 6 and 7 supports composition against either identifier without disruption during the transition.

Step	What you do	Output
1. Search	Use the ESCO portal or API to search for the action verb and the object of the draft competency.	A short list of candidate ESCO concept identifiers (URIs).
2. Examine	Read the concept definition, the skill type, and the linked occupations. Discard candidates whose definition does not match the intended scope.	A preferred candidate, or the conclusion that no candidate is appropriate.
3. Refine	Examine broader and narrower concepts in the ESCO hierarchy; the right candidate is often one step more specific than the first hit.	A refined identifier (URI) at the appropriate level of specificity.
4. Working ID	Where no candidate is appropriate, publish a working concept identifier under the AGRIFOODSKILLS namespace and submit it for inclusion in ESCO.	A provisional identifier suitable for use in microcredentials and occupational profiles in the interim.

Template for emerging AFS competencies

REF	EQF Level	Module	Competency statement	Profession	Occupation profile
001	5		HACCP Plan	Food	

A.4 EQF level anchoring

Each competency statement is anchored to an EQF level using the descriptor strands set out in Section 4. Anchoring is performed against the responsibility-and-autonomy strand in particular, because that strand is the most discriminating across the EQF 3-5 / EQF 6-8 boundary. A competency that requires the learner to apply an established protocol in a routine setting sits at EQF 3 or 4. A competency that requires the learner to manage variation, to supervise others or to develop their own practice sits at EQF 5. A competency that requires the learner to take responsibility for decision-making under conditions of uncertainty, or to lead the development of new practice, sits at EQF 6 to 8. The relevant Europass descriptors for the chosen level should be consulted directly when anchoring a competency.

A.4.1 Webb's Depth of Knowledge as an aid to EQF anchoring

Anchoring a competency to an EQF level is, at times difficult on the basis of the descriptor strands alone, because the strands are written in deliberately general terms in order to accommodate qualifications across all sectors. Webb's Depth of Knowledge (DOK) framework (Webb, 1997 and subsequent revisions) offers a complementary instrument that AGRIFOODSKILLS adopts for the purposes of competency-level anchoring. DOK is referenced in the EBVS Developing Mastery in Clinical Education programme as part of its intellectual framework, alongside Zones of Engagement and Domains of Awareness; the present appendix formalises that position for use across the AGRIFOODSKILLS occupational profiles.

The DOK framework distinguishes the cognitive demand that a task places on the learner, rather than the cognitive process the task uses. The distinction matters because two competency statements that use the same action verb may require materially different cognitive demand. Applying a known procedure in a routine setting is DOK level 2. Applying the same procedure in conditions of incomplete information, where the practitioner must decide which procedure is appropriate and at what point to escalate, is DOK level 3 or 4. The DOK framework is complementary to the Bloom's-revised taxonomy (Anderson and Krathwohl, 2001) used in the learning-outcomes guidance, not an alternative to it. Bloom's-revised is well suited to the formulation of learning outcomes within a module, where the verb category is the principal organising idea. DOK is well suited to the anchoring of a competency statement to an EQF level, where the depth of cognitive demand is the principal organising idea.

The four DOK levels map to EQF levels approximately as set out below. The mapping is interpretive rather than definitional: DOK and the EQF derive from different traditions (Webb originated DOK in the assessment of school-age learners; the EQF is a qualifications-framework instrument for European recognition). The mapping should therefore be used as a working aid to anchoring rather than as a substitute for the Europass descriptors, which remain authoritative.

DOK level	Cognitive demand	Approximate EQF correspondence	Indicative agrifood example
DOK 1 Recall and Reproduction	The learner retrieves known information, recognises a procedure, or follows a defined step-by-step routine without choice between procedures.	EQF 2 to 4.	Identify the standard operating procedure for a defined cleaning step on a processing line and complete it without variation.

DOK 2 Skills and Concepts	The learner applies a skill or concept to a defined problem, with some procedural choice between recognised alternatives, in a setting whose parameters are stable.	EQF 4 to 5.	Select the appropriate hygiene protocol from a defined set for the conditions encountered on the line, and apply it correctly.
DOK 3 Strategic Thinking	The learner reasons through a multi-step problem under conditions of incomplete information or competing constraints, justifies decisions, and adapts approach as evidence accumulates.	EQF 5 to 8.	Investigate a non-conforming microbiological result on a processing line, identify likely causal factors, recommend interventions, and justify the recommendation against the operational and regulatory context.
DOK 4 Extended Thinking	The learner sustains a complex investigation, synthesis, or innovation across an extended period, drawing on multiple sources, integrating across domains, and contributing to or redefining practice.	EQF 7 to 8.	Design and lead an integrated quality-improvement programme for a multi-line processing operation, with documented baseline measurement, intervention, outcome evaluation, and dissemination to peer practice.

DOK level is not a feature of the action verb in isolation; it is a feature of the whole competency statement, including the object, the context and the standard against which performance is judged. A useful diagnostic during competency drafting is to read the statement aloud and ask which DOK level the standard requires. If the standard requires only that the learner produced an output, the competency is likely at DOK 1 or 2. If the standard requires that the learner justified the output against alternatives or adapted it under variation, the competency is likely at DOK 3 or 4. The diagnostic is a practical shortcut to the EQF anchoring step.

A.5 Worked examples: from a domain task to a stack of competencies

The worked examples in this section share a common move. Each starts from a real workplace task, drawn from a proposed AGRIFOODSKILLS occupational profiles from WP3, and decomposes that task into a stack of smaller competencies that are independently meaningful and externally recognisable. Each smaller competency could be designed, taught and credentialed as its own microcredential. Together they compose the larger workplace capability described in the occupational profile. This is what stackability looks like in practice. The decomposition is also the point at which competency design becomes substantively different from outcome design: an outcome describes what a single course produces; a competency describes a unit of professional capability that can stand on its own and recompose with other units issued by other providers. It should be remembered that these competencies are developed before the WP3 has progressed to the point of selecting which profiles will be developed, and therefore may not be found later within WP5.

For each example, the report sets out the workplace task, sketches the stack of smaller competencies the task decomposes into, and then presents two or three of those smaller competencies in dedicated boxes. The boxes show the learning outcome as the ICOS-led learning-outcomes guidance would formulate it for the module, alongside the competency statement as the present guidance would formulate it for the credential's external interface. The contrast is the practical pay-off of working at the competency level.

A.5.1 Drone-based land surveys (Precision Crop and Irrigation Management Specialist, EQF 5)

The Precision Crop and Irrigation Management Specialist profile (WP3 draft) describes a professional who uses remote sensing, drones, GPS mapping and data analytics to monitor crop health, soil moisture and climatic conditions. One of the workplace tasks within this profile, and one that the WP2 employer consultation has identified as a near-term skills gap, is the undertaking of drone-based land surveys for site-specific crop management decisions. The task is ill-served by a single course called "drone surveys". It is would be better delivered by a stack of four small competencies that a practitioner accumulates over the course of a season or even between roles

1. Read and interpret an existing land survey (cadastral map, soil survey, drainage plan).
2. Operate within the legal framework for unmanned aerial vehicles (UAVs).
3. Explain the GDPR framework as it applies to drone imagery.
4. Plan and safely conduct a drone flight under the relevant national operating authorisation.
5. Process drone imagery into a site-specific management map that supports an agronomic decision.

Each of the four small competencies is independently meaningful. The first can be earned by a farm operative who never intends to fly a drone, but who needs to use the map a contractor produces. The third can be earned by a professional drone operator who never intends to process the imagery into agronomic advice. The four compose into the full Drone-based Land Survey capability described in the occupational profile. They also map cleanly onto distinct ESCO concepts and onto different EQF levels: 4 for the first and the third, 5 for the second and the fourth. The path from novice to competent operator maybe branched. Indeed, they may already have achieved competency 4 through non-commercial flight outside of the workplace.

Module 3 - Operate within the legal UAV framework (EQF 5)

Learning outcome. On completion of this unit, the learner will be able to:

- a) describe the EU and national legal framework for the operation of unmanned aerial vehicles in agricultural settings;
- b) describe the conditions under which an operating authorisation is required.

Competency statement. *Operate* as the **pilot-in-command of an unmanned aerial vehicle** for the purpose of **agricultural survey**, within the **current EU UAV regulation** (Implementing Regulation (EU) 2019/947 and successors), with documented pre-flight risk assessment, completed flight log.

ESCO alignment: anchored on concepts in the ESCO clusters for environmental monitoring and for regulatory compliance; a working concept identifier may be required.

Process drone imagery into a site-specific management map (EQF 5)

Learning outcome: On completion of this unit, the learner will be able to:

4. explain the principles of orthomosaic processing;
5. describe the principal vegetation indices used in precision agriculture;
6. demonstrate the production of a prescription map in a recognised farm management software package.

Competency statement. *Process* aerial **imagery** captured by an unmanned aerial vehicle into a site-specific prescription map for a **defined agricultural decision** (irrigation scheduling, variable-rate nutrient application, targeted scouting), **in accordance with the operator's quality assurance procedures** for prescription mapping, and present the map and supporting data to the holding operator in a form suitable for inclusion in a farm management system.

ESCO alignment: anchored on ESCO concepts for spatial data analysis and for the application of precision agriculture technologies.

A.5.2 Farm-level emissions reduction advisory (Farm Emissions Reduction Specialist, EQF 6)

The Farm Emissions Reduction Specialist profile (WP3 draft) describes a professional who quantifies, monitors and reports greenhouse gas emissions from livestock operations and advises on evidence-based mitigation strategies. The combined workplace task, the production of a farm-level emissions reduction plan that the operator can act on, is a substantial body of work. It decomposes into five small competencies:

1. Identify emission sources on a defined livestock operation.
2. Apply a recognised greenhouse gas accounting methodology to a defined operation.
3. Collect on-farm operational data of sufficient quality for emissions calculation.
4. Interpret emissions outputs and shortlist evidence-based mitigation options.
5. Communicate the assessment and the mitigation options to the operator in a form that supports decision-making.

An advisor who can perform all five at EQF 6 is the full Farm Emissions Reduction Specialist. An advisor who can perform only the first three is a farm-level emissions auditor; a useful occupational profile in its own right, recognisable to employers, suitable as a microcredential without dependence on the advisory layer. A farmer who can perform only the third is a competent data contributor to a third-party audit; useful, recognisable, microcredential-issuable on its own. The stack is the point at which the same content can serve materially different learner populations through different combinations of the same small competencies.

Apply a recognised greenhouse gas accounting methodology (EQF 6)

Learning outcome: On completion of this unit, the learner will be able to:

1. describe the principal greenhouse gas accounting methodologies in use for European livestock operations;
2. explain the assumptions and uncertainties associated with each;
3. complete a worked accounting exercise on a sample farm dataset.

Competency statement. *Apply a recognised greenhouse gas accounting methodology to a defined livestock operation*, with a **written assessment** that identifies the methodology used, the operational data inputs, the assumptions made, the uncertainty range, and the resulting emissions estimate, to a standard verifiable by external sustainability audit.

ESCO alignment: anchored on emerging concepts in environmental monitoring and in agricultural sustainability assessment; working concept identifier likely to be required because the methodology-application competency is not at present discretely represented in ESCO.

Communicate emissions findings and mitigation options to the operator (EQF 5 to 6)

Learning outcome: On completion of this unit, the learner will be able to:

1. describe the principal categories of mitigation option available to a livestock operation;
2. explain the relative evidence base supporting each;
3. produce a written advisory note for a sample farm.

Competency statement: *Communicate* greenhouse gas **emissions findings** and a shortlist of evidence-based mitigation options to the operator of a **defined livestock operation**, in a written and oral advisory format that is **appropriate to the operator's decision-making context**, with documented evidence base for each option and a costed implementation indication where one is requested.

ESCO alignment: anchored on ESCO concepts for technical advisory practice and for the translation of scientific information into operational decisions.

A.5.3 Design of an integrated crop-livestock system (Circular Livestock Systems Advisor, EQF 6)

The Circular Livestock Systems Advisor profile (WP3 draft) describes an advisor who designs and implements integrated livestock production systems that optimise resource efficiency by reusing waste streams and closing nutrient loops between animal and crop farming. The workplace task of producing a system design for a specific holding decomposes into four small competencies:

1. Map current nutrient flows, waste streams and resource use on the holding.
2. Identify circular-economy opportunities available within the holding's constraints.
3. Model the performance of an integrated system design against baseline.
4. Cost the integrated design and present it to the holding owner with an implementation plan.

The decomposition is again practically useful. A farm advisor with the first competency only is a competent nutrient-flow analyst; usable by sustainability-reporting initiatives. A consultant with the third competency only is a systems modeller; usable by agritech firms developing circular-system design software. The combination of all four is the full Circular Livestock Systems Advisor profile.

Map nutrient flows on a holding (EQF 5 to 6)

Learning outcome: On completion of this unit, the learner will be able to:

1. explain the principles of nutrient cycling in mixed crop-livestock systems;
2. identify the principal data required to construct a nutrient flow diagram;
3. complete a worked nutrient flow exercise for a sample holding.

Competency statement. *Construct* a documented **nutrient flow and material balance assessment** for a defined **mixed crop-livestock holding**, drawing on operational data supplied by the operator and on **recognised reference data where operational data is incomplete**, presented in a form suitable for submission to a regional sustainability scheme.

ESCO alignment: anchored on ESCO concepts for agricultural systems design and for environmental-impact assessment; precise identifiers to be confirmed in Task 5.2.

A.5.4 Climate risk assessment for an arable holding (Climate-Resilient and Regenerative Agriculture Specialist, EQF 6)

The Climate-Resilient and Regenerative Agriculture Specialist profile (WP3 draft) describes a professional who designs, implements and evaluates agricultural systems that enhance ecosystem health, improve soil fertility, increase biodiversity and build resilience to climate change. The task of producing a climate risk and resilience assessment for an arable holding decomposes into five small competencies:

1. Conduct a baseline soil health assessment of the holding.
2. Identify the principal climate risks to which the holding is exposed.
3. Select appropriate regenerative practices for the site, taking account of soil, climate and cropping system.
4. Plan a monitoring regime for soil carbon, biodiversity and water indicators.
5. Produce an integrated assessment and resilience plan in a form suitable for submission to a CAP eco-scheme.

The fifth competency is a useful test of the stack: it is the integration step that combines the outputs of the first four into a single decision-grade document. A learner could earn the first four as separate microcredentials issued by separate providers and present the integration competency to a fifth provider for credentialing. That kind of cross-provider stacking is what the architecture in Sections 7 to 9 is designed to support.

Identify the principal climate risks for a defined holding (EQF 6)

Learning outcome: On completion of this unit, the learner will be able to:

1. describe the principal categories of climate risk relevant to European arable production;
2. explain the principal data sources used in farm-level climate risk assessment;
3. complete a worked risk assessment on a sample holding.

Competency statement: *Identify and characterise* the principal **climate risks** to which a defined **arable holding** is exposed, using a **recognised assessment methodology** and the latest available climate projection data for the holding's region, and produce a written risk register in a form suitable for inclusion in the holding's sustainability plan.

ESCO alignment: anchored on ESCO concepts in environmental sciences and risk assessment; precise identifiers to be confirmed in Task 5.2.

A.5.5 Developing Mastery in Clinical Education (Veterinary Clinician-Clinical Educator, EQF 7 to 8)

The Veterinary Clinician-Clinical Educator profile describes a practising veterinarian who combines clinical practice with the supervision and education of postgraduate trainees. The workplace task of becoming a fully credentialed clinical educator is substantial. The programme prepared to greater detail in preparation for this workpackahe decomposes the role into three micro-credentials of four ECTS each. The micro-credentials are stackable: each can be earned independently, and the combination of all three produces a 12 ECTS portfolio that supports an Academy of Medical Educators Fellowship application as an external accreditation scheme

The three-micro-credential stack is the practical realisation of the same logic illustrated above for the other profiles. Each micro-credential carries two of the ten competency clusters set out in the occupational profile, each cluster has its ESCO anchors, and each micro-credential is independently recognisable in the labour market for veterinary specialisation.

- 3 Teaching and AI. Modules 1 (Educational Foundations) and 2 (AI in Clinical Education). These consider the delivery of inclusive clinical education cluster and the AI enhanced clinical learning cluster of the occupational profile.
- 4 Supervision and Assessment. Modules 3 and 4. Consider the clinical supervision and entrustment cluster and the assessing competence and assessment construction cluster.
- 5 Scholarship and Leadership. Modules 5 and 6. Consider quality improvement and educational scholarship cluster as well as the wellbeing and mattering cluster.

Deliver inclusive clinical education (MC1, Module 1, EQF 7)

Learning outcome: On completion of Module 1, the learner will be able to:

1. plan a learner-centred clinical teaching session for a defined group of postgraduate trainees,
2. describe the principles of inclusive practice in clinical education and adapt session design to diverse learner needs.

Competency statement. Plan and deliver inclusive **clinical teaching sessions** for **postgraduate veterinary trainees**, in routine clinical settings, with explicit outcome **alignment to the residency programme blueprint** and with evidence of adaptation to diverse learner needs, to a standard verifiable by peer-observation rating against the AoME Professional Standards (4th Edition).

ESCO anchors: train employees /e54ff029-1ce9-447d-a5b2-eb7283a23e6e, use learning strategies /a44c0ed6-68b1-457a-9c6d-ea8487bdfef2. The numeric codes are the ESCO URIs that map to these competencies

Facilitate AI-enhanced clinical learning (MC1, Module 2, EQF 7)

Learning outcome: On completion of Module 2, the learner will be able to:

1. describe the principal categories of AI tool currently in use in veterinary clinical practice and education;
2. apply the QUSC framework (Question, Use, Scrutinise, Communicate) to a chosen AI tool;
3. identify the principal risks associated with the use of AI in clinical education, including hallucination, bias, and the absence of medical-device regulation in animal medicine.

Competency statement: *Facilitate the responsible use of AI* in clinical education for **postgraduate veterinary trainees**, applying the **QUSC framework explicitly within session design, documenting AI use within the credential metadata**, and producing a written AI-governance note appropriate to the educational setting, to a standard verifiable by the institutional AI Act and GDPR posture statement.

ESCO anchors: use digital tools and technologies to create knowledge and to innovate (data.europa.eu/esco/skill/1a4cc54f-1e53-442b-a6d2-1682dc8ef8f9); teach digital literacy (data.europa.eu/esco/skill/c8fa4313-80b0-4f37-8b1b-1739707bc362).

Two general observations close the worked examples. First, the same architectural principle applies across all five examples, whether the source profile is a farm advisor, a remote-sensing specialist, a regenerative agronomist or a veterinary clinical educator. A workplace task decomposes into a stack of smaller competencies. Each smaller competency is independently meaningful in the labour market. The stack composes back into the full occupational competency. Second, the move from learning outcome to competency statement is consistent across the examples. The learning outcome describes what the module produces inside the institution; the competency statement describes the unit of capability that the credential issues into the wider system. The two are not in tension. They are written for different readers, and they are stewarded by different parts of Deliverable 5.1: ICOS for the learning outcomes, EBVS for the competency statements, with Task 5.2 publishing both together for every occupational profile in the AGRIFOODSKILLS programme. Importantly the stackable competencies decompose into individual, but valid learning or combine to create a number of potential agrifood roles.

A.7 Common pitfalls

Pitfall	What it looks like	Correction
Topic in place of competency	'Milking parlour hygiene.' This is a topic, not a competency statement.	Rewrite with action verb, object, context and standard: 'Apply hygiene protocols in milking parlour operations under routine and incident conditions, in accordance with farm SOPs and EU dairy hygiene regulation.'
Vague verb	'Understand antimicrobial resistance.' Understanding is not externally verifiable.	Replace with a verb that names an observable action: apply, conduct, decide, evaluate.
Over specification	Twenty-five words listing every possible step the learner must perform.	Promote sub-steps to associated learning outcomes within the module; keep the competency statement at the level of the externally meaningful action.
Under specification	'Apply HACCP.' Too generic to test against another provider's prerequisite competencies.	Add object, context and standard: 'Conduct a hazard analysis and identify critical control points for a defined food production process, document them in a HACCP plan, and validate them against the current legislative framework.'

ESCO drift	The statement uses the action verb from one ESCO concept and the object from another.	Re-anchor against a single ESCO concept, or publish a working concept identifier that the statement is faithful to.
Competency 'soup'	Several competencies bundled into one sentence.	Split into separate statements, one ESCO concept each, each independently testable.

A.8 ESCO stewardship and the submission of new concepts

AGRIFOODSKILLS will, in the course of Task 5.2, identify a number of competencies that are not at present represented in ESCO. The project will publish working concept identifiers for these competencies under the AGRIFOODSKILLS namespace and will submit them for inclusion in ESCO through the European Commission's established stewardship process. The submission package for each new concept should include: the preferred label and definition; the skill type (knowledge, skill or transversal); the relevant broader and narrower concepts in the existing ESCO hierarchy; the occupations to which the concept applies; the EQF level or levels at which the concept is expected to be demonstrated; and evidence from WP2 and WP3 of AGRIFOODSKILLS that the concept addresses an unmet sectoral skills need. Submitted concepts that are accepted into ESCO become the canonical identifier for the competency; until acceptance, the working identifier remains valid and the assertion model supports composition against it.

Design summary: writing a competency statement

- Start with the working context: what is the learner actually doing in their role that this competency captures?
- Choose an observable action verb. Avoid 'know', 'understand', 'be aware of'.
- Name the object precisely; if the object is sector-specific, use the sector terminology.
- State the context in which the action is performed.
- State the standard against which performance will be judged.
- Test the statement against ESCO. Adopt the closest existing concept identifier, or publish a working identifier and submit it for inclusion.
- Anchor the statement to an EQF level using the responsibility-and-autonomy strand.
- Show the statement to a non-trainer from the sector (an employer, a regulator, a mentor) before it is finalised. If they cannot tell from the statement whether a learner has demonstrated the competency, redraft.

Appendix B. Stackable lesson plans and pathway design

Appendix A sets out how a competency statement is written. The present appendix sets out how competency statements are used to make a learning programme actually stackable in practice. A stackable lesson plan is a discrete, self-contained unit of instruction that declares the competencies it requires of the learner on entry, and the competencies it produces in the learner on completion. This input-output architecture is the mechanism by which individual lessons chain into coherent learning pathways. Grounded in the principles of andragogy set out in Section 3, the learner, not the educator, navigates these pathways, using the curriculum as a self-directed instrument oriented to their own professional goals and context.

B.1 The stackable lesson plan template

Each lesson module is authored using a standardised template. The template ensures that every module declares its position in the stack with precision, so that the module is portable, auditable and combinable with other modules across different pathways. The template comprises seven blocks, each described below.

B.1.1 Identity block

Field	Description
Lesson ID	Unique identifier enabling cross-referencing across the module pool (for example, EIM-L-03).
Title	Concise, action-oriented descriptor.
Domain or module	The broader curricular area to which the lesson belongs.
Level	Aligned to a recognised taxonomy (for example, Bloom's-revised, Miller's pyramid, Webb's DOK as set out in Appendix A.4.1).
Estimated duration	Contact hours and independent study, stated separately.

B.1.2 Prerequisite competency block

A structured list of prerequisite competency statement identifiers (from the Appendix A format) that the learner must hold before entry to the lesson. Prerequisites may be none for foundational lessons, or one or many for downstream lessons. Each prerequisite is stated formally using the Appendix A competency statement format. An illustrative entry:

Learner must hold: [CompID] — ***Operate as the pilot-in-command of an unmanned aerial vehicle for the purpose of agricultural survey***

B.1.3 Learning objectives

Three to five session-level objectives written using Bloom's-revised verbs, each mapping directly to at least one output competency declared in B.1.7. The objectives are internal design documentation, in line with the three-tier distinction set out in Section 6.4, and are not exposed in the public credential descriptor. An illustrative format:

By the end of this lesson, the learner will be able to [Bloom's verb] [knowledge, skill or attitude content] in the context of [defined scenario].

B.1.4 Instructional content and methods

The block captures the core content outline as metadata rather than as a full script, identifies the principal delivery modality (didactic, case-based, simulation, clinical, asynchronous) and identifies the active learning strategy used (problem-based learning, flipped classroom, worked examples, peer demonstration, work-based learning, or an AI-enhanced approach in line with the QUSC framework from Section 2.6). Where multiple modalities are used, each is recorded so that the engagement profile of the module is visible from the documentation.

B.1.5 Assessment of mastery

The assessment must demonstrate attainment of the output competencies declared in B.1.7, not merely attendance or recall of taught content. The block records: whether the assessment is formative or summative; the assessment instrument (MCQ, OSPE, structured viva, portfolio entry, direct observation, mentor sign-off, workplace artefact); and the evidence base by which the output competency claim is supported. The block sits within the EQAVET-aligned evaluation cycle set out in Section 5 and should be authored in conjunction with the competency declarations, in line with the assessment-alignment limitation discussed in Section 6.5.

B.1.6 Output competency block

A structured list of output competency statement identifiers (Appendix A format) awarded on demonstrated mastery. These identifiers immediately become available as prerequisites for downstream lessons in the module pool. An illustrative entry:

Upon completion, learner is awarded: [CompID] — ***Process aerial imagery captured by an unmanned aerial vehicle into a site-specific prescription map for a defined agricultural decision.***

B.1.7 Stacking metadata

Field	Description
Feeds into	Lesson IDs for which this lesson's output competencies serve as prerequisites.
Fed by	Lesson IDs whose output competencies are prerequisites for this lesson.
Credential contribution	The microcredential, module or qualification unit to which this lesson contributes.

B.1.8 Complete lesson plan outline

The blocks combine into a single per-lesson record. An illustrative completed record for a veterinary respiratory medicine lesson:

Field	Example content
Lesson ID	PCS-L-03
Title	Plan and safely conduct a drone flight for agricultural survey
Domain	Precision Crop and Irrigation Management; Drone-based Land Survey microcredential stack
Level	EQF 5; Miller's 'Shows how' (applied operational skill); Webb's DOK 3
Duration	3 contact hours (1 hour pre-flight planning workshop, 2 hours supervised practical flight) plus 2 hours independent study (regulatory familiarisation and flight-plan drafting)
Prerequisites	PCS-L-01: Read and interpret an existing land survey (CompID-A from Appendix A.5.1). PCS-L-02: Operate within the EU UAV regulation (Implementing Regulation 2019/947) and the GDPR framework for aerial imagery (CompID-B from Appendix A.5.1).
Learning objectives	1. Conduct a documented pre-flight risk assessment for a defined agricultural survey site. 2. Draft a flight plan that meets the survey objectives and the conditions of the national operating authorisation. 3. Execute the flight safely under instructor supervision, with continuous airspace and ground-risk monitoring. 4. Complete and file a flight log and post-flight equipment checks.
Methods	Pre-flight planning workshop using a real holding's survey data; supervised on-site practical flight session; structured debrief against the flight log
Assessment	Direct observation against a flight-operations checklist by a qualified instructor; review of the learner-authored pre-flight risk assessment, flight plan and flight log; signed competence statement on file
Output competencies	CompID-C from Appendix A.5.1: Operate as the pilot-in-command of an unmanned aerial vehicle for the purpose of agricultural survey, within the current EU UAV regulation and within the GDPR framework for image data captured during the flight, with documented pre-flight risk assessment, completed flight log and a documented data-handling plan.
Feeds into	PCS-L-04: Process drone imagery into a site-specific management map (CompID-D from Appendix A.5.1). PCS-L-05: Conduct repeat surveys across a growing season for time-series analysis.
Credential contribution	Module 2 of the Drone-Based Land Survey microcredential within the Precision Crop and Irrigation Management occupational profile; one of the four competencies that compose the full Drone-Based Land Survey capability described in the source occupational profile.

Design summary: authoring a stackable lesson plan

- Use the eight blocks set out in B.1.1 to B.1.8 for every lesson; do not omit any.
- Reference prerequisite and output competencies by Appendix A competency statement identifier.
- Map every learning objective to at least one output competency.
- Author the assessment of mastery and the output competency declarations together; do not author one without the other.
- Record the stacking metadata even where the downstream lessons are not yet authored; leave the field blank rather than omitting it.

B.2 The pathway mapping matrix

The most practical instrument for showing how modules stack differently for different learners is a matrix. Columns represent each lesson module in the pool; rows represent distinct learner profiles or target outcomes. Each cell records whether that module is required, optional, waivable on the basis of prior learning, or not applicable for that profile.

B.2.1 Matrix key

Symbol	Meaning
Required	Core to this pathway.
Optional	Enriches but is not essential.
Waivable	May be bypassed where the learner can evidence the relevant prior competency.
Not applicable	Does not contribute to this pathway.

B.2.2 Example pathway mapping matrix

The illustrative example below continues the drone-based land survey stack introduced in Appendix A.5.1 and developed in the lesson plan in B.1.8. The module pool comprises six lessons covering survey interpretation, the legal and data-protection framework for drone operation, flight planning and execution, imagery processing, repeat surveying for time-series analysis, and the translation of survey output into an advisory note. The matrix below shows how four distinct learner profiles draw on the same pool of six lessons to reach materially different destinations.

Learner profile	L-01 Land survey interpretation	L-02 UAV and GDPR framework	L-03 Flight planning and execution	L-04 Imagery processing into management map	L-05 Repeat surveys and time-series	L-06 Advisory note from survey output
Farm operative using contractor-produced maps	Required	N/A	N/A	Optional	N/A	N/A
Agricultural contractor (drone operator only)	Waivable	Required	Required	Optional	Optional	N/A
Agronomy advisor (data consumer and advisor)	Required	Required	Waivable	Required	Optional	Required
Precision Crop and Irrigation Management Specialist (full)	Required	Required	Required	Required	Required	Required

The four profiles differ on the role destination dimension set out in B.3.2: the farm operative consumes survey outputs without creating them; the agricultural contractor creates flight outputs without interpreting them agronomically; the agronomy advisor consumes and interprets survey outputs and may delegate the flight execution; the Precision Crop and Irrigation Management Specialist holds the full set. Second, lesson L-03 is Required only for those profiles that include the act of flying the drone; it is Waivable for the agronomy advisor and Not Applicable for the farm operative. Third, L-01 is Required for every profile that uses survey

data and Waivable for the contractor profile, where prior surveying experience is common; the recognition of prior learning route is therefore particularly visible in this row. Fourth, no learner profile in this pool requires all six lessons except the full Specialist; the matrix surfaces the modularity of the stack and the different exit points it supports.

B.2.3 How to use the matrix

Constructing the matrix forces three design decisions on the curriculum designer.

- Coverage check. For every learner profile (row), the set of Required modules must between them deliver all of the output competencies the profile requires. If they do not, a module is missing.
- Redundancy check. If two Required modules in the same row both deliver the same output competency, one of them may be reclassifiable to Optional or removed from the row.
- Entry point clarity. The pattern of Waivable cells across the matrix reveals where a formal recognition of prior learning (RPL) process is necessary, so that the learner is not required to repeat content for which they hold prior evidence.

As new modules are added to the pool, a new column is inserted and each learner profile row is reviewed to determine its relevance. This sustains the modularity of the curriculum and prevents the programme from drifting back into a single fixed sequence.

B.3 Stacking for different outcomes

The same module pool does not lead to a single destination. The point of stackable design is that the same curricular components can be recombined to produce distinct outcomes for distinct learner populations. Curriculum design therefore shifts from asking what the course is, to asking what the possible journeys are and where each one ends.

B.3.1 Working backwards from outcomes

For each learner profile, the designer identifies the output competencies required at the destination, then traces back through the prerequisite chain to establish which modules must appear in that pathway. The reverse pathway mapping distinguishes a common core (modules that appear in every pathway) from specialist branch modules (modules that serve only one or two profiles). The architecture in Sections 7 to 9 of the report is designed to support this recombination across providers, not only within a single provider's offer.

B.3.2 What makes pathways genuinely different

Pathways that differ only in speed or depth are not genuinely distinct pathways. Meaningful pathway differentiation occurs across at least one of the following four dimensions.

- Role destination: the professional task the learner will perform on completion.
- Competency emphasis: diagnostic, therapeutic, managerial, educational or another principal orientation.
- Assessment context: clinical, academic, regulatory or workplace-based.
- Credential type: microcredential, certificate, specialist diploma, or CPD award.

B.3.3 Avoiding false stacking

The credential stack is not simply content piled higher. Each pathway should carry a coherent narrative: a learner completing it should be able to articulate what they can now do that they could not before, and that statement should be meaningfully different from what a learner on an adjacent pathway can do. Where the difference cannot be articulated, the pathways are duplicative and should be consolidated.

B.4 The learner as navigator

The pathway mapping matrix only functions as an andragogical instrument if the learner is the one who

reads across their own row and makes active choices. As set out in Section 3, Knowles' principle of self-direction holds that adult learners orient themselves toward learning that serves their own professional goals and life context. The educator's role shifts from gatekeeper to architect: the educator ensures that the prerequisite logic is sound, that the competency declarations are honest, and that the assessment of mastery is rigorous, but leaves the sequencing decisions, pace and exit point to the learner.

The matrix should therefore be presented to learners before enrolment, not after, with three explicit prompts. The prompts give the learner the active role described in Knowles' framework and convert the matrix from an internal curriculum-planning instrument into a learner-facing navigation instrument.

- Where are you going? Which outcome profile best describes your professional destination?
- Where are you now? Which competencies can you already evidence, making the corresponding modules waivable?
- What do you need next? Given the above, which module is your logical next step?

This is the goal that stackable microcredential frameworks at European level consistently identify as the purpose of modular design: a learner who can read the curriculum as a map and navigate it on their own terms, supported by educators who have made the map honest and navigable.

Design summary: operationalising stackable pathways

- Author every lesson plan to the eight-block template in B.1.
- Construct a pathway mapping matrix for the module pool; review it whenever a new module is added.
- Work backwards from outcomes for each learner profile; identify the common core and the specialist branches.
- Articulate the meaningful difference between adjacent pathways; consolidate where the difference cannot be stated.
- Present the matrix to learners before enrolment, accompanied by the three navigation prompts.

Appendix C. Practical Assessment Item Construction

AGRI FOODSKILLS Deliverable 5.1 — Companion Toolkit (Appendix C)

About This Guide

This guide is for practitioners across the agrifood sector who are writing multiple-choice questions (MCQs) to assess competencies linked to microcredentials and verifiable credentials. It covers the full span of the European Qualifications Framework (EQF) levels 4 through 8, and is designed for use across four agrifood domains:

- **Veterinary** (farm animal species: cattle, sheep, pigs, poultry, goats, and aquaculture)
- **Farming** (arable, livestock husbandry, and mixed systems)
- **Food** (food processing, safety, traceability, and supply chain)
- **Plant** (crop science, horticulture, and plant health)

The guide is built around the one-best-answer MCQ format. This format is recommended for all EQF levels, with the distinction between levels achieved primarily through the cognitive demand of the scenario and the nature of the decision being tested, not through switching to a different question type.

The principles here draw on established psychometric practice in health professions assessment, adapted for the agrifood context and for the specific purpose of supporting credential-linked assessment: assessments whose primary function is to provide valid evidence that a learner has achieved the competency described in a microcredential, rather than to rank-order candidates in a population.

Why Assessment Design Matters for Verifiable Credentials

A verifiable credential makes a claim about a person: that they have demonstrated a defined competency to a defined standard. The value of that claim depends entirely on the quality of the evidence behind it. A poorly designed assessment item cannot generate valid evidence, regardless of how well the credential metadata is written.

In traditional certification, poor items are diluted across large item banks and identified through statistical review over many administrations. In microcredential assessment, item banks are typically smaller and each credential claim rests on fewer items. This raises the stakes for individual item quality.

There are two distinct threats to item quality in this context.

Construct-irrelevant difficulty occurs when an item is harder than intended for reasons unrelated to the competency being assessed. A learner who genuinely holds the competency may fail the item because of confusing wording, ambiguous options, or an unnecessarily complex structure. The credential claim becomes inaccurate in the direction of false negatives.

Construct-irrelevant ease occurs when an item can be answered correctly without holding the competency, typically because of structural flaws that give away the answer to a test-wise candidate. The credential claim becomes inaccurate in the direction of false positives.

Good item writing eliminates both threats. The goal is an item that can only be answered correctly by someone who genuinely holds the competency being assessed.

What Should Be Assessed?

Assessment content should be driven by the competency descriptor in the credential specification. In the European Qualifications Framework, each level is defined across three dimensions:

- **Knowledge:** the body of facts, principles, theories, and concepts relevant to the domain
- **Skills:** the cognitive and practical abilities to apply that knowledge
- **Responsibility and autonomy:** the context in which the learner is expected to act

Items should be designed with the credential's EQF level descriptor in mind. An item that only requires knowledge recall, without any application, cannot provide valid evidence for a skills or

responsibility/autonomy claim, regardless of the difficulty of the knowledge being recalled. The table below summarises the cognitive task implied at each EQF level and the corresponding item design approach.

Table 1: EQF Level, Cognitive Demand, and Item Design Approach

EQF Level	Knowledge Descriptor	Skills Descriptor	Responsibility/Autonomy	Appropriate Cognitive Task for Items	Typical Agrifood Roles
4	Factual and theoretical knowledge in broad context	Range of cognitive and practical skills	Work or study under supervision with some autonomy	Identify, select, apply known procedures to defined situations	Farm worker, food production operative, horticultural operative, livestock handler
5	Comprehensive, specialised factual and theoretical knowledge	Comprehensive cognitive and practical skills; creative solutions	Exercise management and supervision in contexts that are unpredictable	Evaluate options, justify selection, adapt procedure to context	Herd manager, farm supervisor, food safety technician, crop technician, plant health officer
6	Advanced knowledge; critical understanding of theories and principles	Advanced skills demonstrating mastery; innovation	Manage complex activities; take responsibility for decision-making	Analyse, synthesise, recommend; integrate multiple information sources	Farm manager, agronomist, veterinary practitioner, food safety manager, quality assurance manager
7	Highly specialised knowledge; original thinking	Specialised problem-solving in research and innovation	Manage and transform complex, unpredictable work contexts requiring new approaches	Critique, design, evaluate systems or strategies	Senior food safety advisor, specialist veterinarian, principal agronomist, supply chain director, plant health specialist
8	Knowledge at the forefront of the field; original research	Most advanced and specialised skills; research and development	Substantial authority, innovation, autonomy, scholarly and professional integrity	Generate, evaluate, and apply original knowledge in novel or contested situations	Research scientist, policy advisor, chief veterinary officer, head of food safety authority

The single-Best-Answer Format

The single-best-answer format is the recommended item type for agrifood skills assessment across all EQF levels. In this format, the candidate reads a scenario (the stem), answers a focused question (the lead-in), and selects the single best answer from a set of options, typically four or five.

A well-constructed one-best-answer item has three components.

- **The stem** contains a scenario that places the candidate in a realistic agrifood context. The scenario should provide enough information to answer the question but should not contain information that is irrelevant to the decision being tested.
- **The lead-in** is a single, closed, clearly worded question. A lead-in is well-formed if a knowledgeable candidate can cover the options and generate the correct answer before seeing them. This is called the cover-the-options rule, and it is the single most useful quality check for a lead-in.
- **The option set** consists of one correct or best answer (the key) and three or four incorrect but plausible alternatives (distractors). Distractors should be homogeneous: they should all be of the same type as the key, so that all options can be evaluated on a single dimension.

The Cover-the-Options Rule

When reviewing an item, the writer should cover the options and ask whether a candidate who genuinely holds the competency could generate the answer from the stem and lead-in alone.

If it could not, the lead-in is too vague or the scenario does not contain enough information; the lead-in should be made more specific, or relevant information added to the scenario.

If it could, the item passes the first quality check.

Direct Interrogatives and Why They Matter

The cover-the-options rule is a test for a deeper principle: a good lead-in asks the candidate to think, not to choose.

There are two families of question phrasing used in multiple-choice assessment.

- **Selective interrogatives** ask the candidate to select from a presented list. The most common forms are “which of the following” and “which of the following is most.” These phrasings invite the candidate to evaluate the options rather than to generate an answer independently. The answer, in effect, lives in the options. A candidate who does not know the answer can still engage in elimination strategies, convergence analysis, or test-wise guessing using structural features of the option set.
- **Direct interrogatives** ask the candidate to retrieve or construct an answer: what, why, when, how, where, who. These phrasings require the candidate to generate an answer from knowledge before encountering the options. The answer lives in the candidate’s knowledge, not in the option set. A candidate who does not hold the competency cannot answer a well-formed direct interrogative by structural means alone.

This distinction matters because the cover-the-options rule is simply a proxy test for whether the lead-in is genuinely direct. A direct interrogative almost always passes the cover-the-options test naturally. A selective interrogative frequently fails it.

Consider the same testing point expressed in both forms:

Selective interrogative (weaker): A food production operative on a poultry processing line notices that the internal temperature reading on the continuous oven monitoring system has dropped below the validated critical limit during production. Which of the following is the most appropriate immediate action? A. Continue production and increase the oven temperature setting B. Stop production, isolate affected product, and notify the supervisor C. Complete the shift and report the finding at the end-of-day debrief D. Apply a time extension to compensate for the reduced temperature A candidate can identify B as the longest and most complete option and guess correctly without knowing anything about food safety critical limits. The answer is findable in the options.	Direct interrogative (stronger): A food production operative on a poultry processing line notices that the internal temperature reading on the continuous oven monitoring system has dropped below the validated critical limit during production. What should the operative do immediately? A. Continue production and increase the oven temperature setting B. Stop production, isolate affected product, and notify the supervisor C. Complete the shift and report the finding at the end-of-day debrief D. Apply a time extension to compensate for the reduced temperature The lead-in now asks directly for an action. A knowledgeable candidate generates “stop production and isolate the product” before reading the options. The options confirm or challenge that answer. A candidate who does not know the correct response cannot arrive at it by structural analysis.
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When Selective Interrogatives Are Appropriate

Direct interrogatives are preferred in the large majority of items. There are, however, a small number of situations where a selective interrogative is genuinely the stronger choice.

Taxonomy and classification questions. Where the question is inherently about identifying a member of a category, “which organism is most likely responsible for this outbreak” is more natural than “what organism is most likely responsible,” and neither form is substantially stronger than the other. In these cases, the selective form is acceptable.

Regulatory reference questions. Where the question asks the candidate to identify the applicable legal instrument or standard, a selective form may be appropriate because the answer is genuinely a selection from a defined set of instruments. However, these items should still be anchored in a scenario and should test application of the regulation, not recall of its name.

Elimination of ambiguity. In a small number of cases, a direct interrogative is grammatically ambiguous or could be read in multiple ways. If rewriting as a direct interrogative introduces ambiguity, the selective form may be retained, but the lead-in should be made as specific as possible.

In all other cases, convert selective interrogatives to direct interrogatives as a default.

The Full Spectrum of Direct Interrogative Forms

The following question words each probe a different cognitive operation. Item writers should select the interrogative form that most directly matches the cognitive task required at the relevant EQF level.

Interrogative	Cognitive operation	Example agrifood lead-in
What	Identification, definition, action	What is the most likely cause of

		these clinical signs?
Why	Causal reasoning, justification	Why is blanket dry cow therapy being phased out across the EU?
When	Timing, sequencing	When should a pre-movement tuberculin test be carried out?
How	Process, mechanism	How does Salmonella contamination most commonly enter a broiler flock?
Where	Location, source	Where in the processing chain does the greatest risk of Listeria amplification occur?
Who	Role, responsibility	Who is legally responsible for ensuring the accuracy of a feed label in this situation?
What action	Procedural decision	What action should the herd manager take first in response to this finding?
What is the most likely	Diagnostic reasoning	What is the most likely explanation for this pattern of soil test results?
What is the most appropriate	Evaluative judgement	What is the most appropriate corrective action in this situation?

The forms “what is the most likely” and “what is the most appropriate” are direct interrogatives that happen to require an evaluative judgement. They are not selective interrogatives. They ask the candidate to generate an answer, even though that answer involves selection among possibilities. The distinction is that the candidate must construct the answer before seeing the options, rather than recognising it within the options.

Worked Example: Converting a Selective to a Direct Interrogative

The following item was written using a selective interrogative. The testing point is the same in both versions.

Original (selective interrogative):

A herd manager on a 400-cow dairy unit records a rolling 12-month culling rate of 32%, with the majority of culls attributed to lameness and reproductive failure. The replacement heifer rearing cost is estimated at €1,800 per animal. **Which of the following is the most appropriate initial step to reduce the culling rate on this farm?**

Revised (direct interrogative):

A herd manager on a 400-cow dairy unit records a rolling 12-month culling rate of 32%, with the majority of culls attributed to lameness and reproductive failure. The replacement heifer rearing cost is estimated at €1,800 per animal. **What should the herd manager prioritise first to reduce the culling rate?**

The revised lead-in asks a direct question. A knowledgeable candidate thinks “I need to understand why animals are being culled before I can intervene, so I should analyse the culling records by cause and age at culling.” The options then present that correct answer alongside plausible but less appropriate alternatives. The candidate who does not hold this competency cannot arrive at the answer by reading the options.

Homogeneous Options

Options must be homogeneous: they must all be the same type of thing, answerable on a single dimension. A lead-in asking for an appropriate management response should have only management responses as options, not a mix of management responses and diagnostic explanations.

Flawed item (heterogeneous options):

What is most important regarding aflatoxin contamination of stored maize?

- A. Aflatoxin is produced by *Aspergillus* species
- B. Grain moisture content should be below 14% at storage
- C. Aflatoxin B1 is the most toxic form
- D. Contamination can be reduced by rapid drying after harvest

Options A and C are factual statements. Options B and D are management recommendations. A candidate cannot rank these on a single dimension.

Revised item (homogeneous options):

A grain store manager is planning to store a maize crop following harvest in a region with a history of aflatoxin contamination. What is the single most important storage practice to reduce the risk of aflatoxin accumulation?

- A. Maintain grain moisture content below 14% throughout storage
- B. Apply a fungicide spray to grain surfaces at intake
- C. Store grain in opaque bags to exclude ultraviolet light
- D. Turn the grain pile weekly to ensure even temperature distribution

Key: A

All options are now management practices, evaluable on the same dimension.

Part 2: EQF-Differentiated Scenario Design

The scenario is the engine of an application-of-knowledge item. Without a scenario, an item tests recall of an isolated fact. With a well-constructed scenario, it tests the candidate's ability to interpret information, apply principles, and make decisions: the cognitive tasks that EQF skills and responsibility descriptors require.

The level of cognitive demand required by the scenario should match the EQF level of the credential being assessed. This is achieved not by making scenarios longer or more obscure, but by varying three things.

The completeness of information provided. At EQF 4, the scenario contains all information needed to identify the correct procedure. At EQF 6 and above, the scenario may contain incomplete, conflicting, or ambiguous information that the candidate must weigh and interpret.

The nature of the decision being tested. At EQF 4, the decision is primarily procedural: what is the correct action in this defined situation? At EQF 6 and above, the decision involves analysis, judgement, or strategic reasoning.

The context of responsibility. At EQF 4, the scenario places the candidate in a supervised role. At EQF 6 and above, the candidate is the responsible decision-maker, often with accountability for others or for systems.

Learner Profiles and Scenario Anchoring

Every scenario must be anchored to a specific learner profile. The profile defines who the candidate is: their role, their level of responsibility, and the system they are operating within. Without this anchor, the scenario floats between EQF levels and the credential evidence it generates is ambiguous.

Learner profiles are drawn from recognisable roles in the European agrifood workforce. They do not need to be exhaustive character descriptions; a single sentence establishing role, setting, and responsibility level is sufficient. The profile should be embedded naturally in the first line of the scenario, not added as a label.

The following are reference profiles for each EQF level across the four domains. These are illustrative: item writers should adapt them to reflect the specific credential and national context.

EQF 4 Profiles

Domain	Example Role	Responsibility Context
Veterinary	Farm animal health worker on a cattle unit	Carries out defined tasks under the direction of the farm manager or attending veterinarian
Farming	Arable farm operative	Operates machinery and carries out fieldwork tasks following written instructions or direct supervision
Food	Food production operative on a processing line	Monitors and records process parameters; escalates deviations to line supervisor
Plant	Horticultural operative in a protected crop nursery	Carries out propagation, planting, and IPM monitoring tasks under the direction of the nursery manager

EQF 5 Profiles

Domain	Example Role	Responsibility Context
Veterinary	Herd health technician on a large dairy unit	Responsible for day-to-day herd health monitoring and first-line intervention within defined protocols; consults veterinarian for diagnosis and prescribing
Farming	Farm supervisor or livestock unit	Manages a production unit within a

	manager	larger farm business; responsible for team supervision and operational decision-making
Food	Food safety technician at a manufacturing site	Carries out HACCP monitoring, sampling, and corrective actions; reports to the food safety manager
Plant	Crop technician or plant health officer	Conducts field assessments, pest and disease monitoring, and advises on control options within defined IPM frameworks

EQF 6 Profiles

Domain	Example Role	Responsibility Context
Veterinary	Veterinary practitioner in mixed or farm animal practice	Responsible for diagnosis, prescribing, and herd health planning; accountable for clinical outcomes
Farming	Farm manager or independent agronomist	Manages the whole farm business or advises on cropping decisions; responsible for compliance, performance, and strategy
Food	Food safety manager or quality assurance manager	Responsible for the site HACCP system, regulatory compliance, and quality management systems
Plant	Agronomist (BASIS-qualified or equivalent)	Provides independent crop protection advice; legally responsible for recommendations under national pesticide regulation

EQF 7 Profiles

Domain	Example Role	Responsibility Context
Veterinary	Specialist veterinary advisor or senior clinical consultant	Provides expert advice to practices, industry bodies, or government; responsible for complex clinical or policy decisions
Farming	Principal agronomist or senior farm business consultant	Advises multiple farm businesses on strategic planning, compliance, and system design
Food	Senior food safety advisor or supply chain director	Responsible for food safety strategy across a multi-site or multi-supplier business
Plant	Plant health specialist or senior regulatory advisor	Designs surveillance programmes, advises on outbreak management, or contributes to regulatory policy development

EQF 8 Profiles

Domain	Example Role	Responsibility Context
Veterinary	Research veterinarian or chief veterinary officer	Generates or evaluates new evidence; responsible for national or international policy

		recommendations
Farming	Agricultural research scientist or senior policy advisor	Conducts original research or advises on evidence-based policy at national or European level
Food	Head of food safety authority or principal research scientist	Responsible for regulatory frameworks, risk assessments, or original research in food safety science
Plant	Plant health scientist or senior policy advisor	Designs regulatory frameworks, conducts original epidemiological research, advises on European plant health policy

Template for an Agrifood Scenario

The following template guides scenario construction. Not all elements are required for every item, but when included they should appear in the order given.

1. **Learner profile** (role, setting, system they are operating in: one sentence, embedded naturally)
2. **Task context** (what is happening, what phase of production or process is underway)
3. **Observed finding, data, or event** (the information that triggers the decision)
4. **Relevant history or background** (prior actions, existing conditions, regulatory context if relevant)
5. **Lead-in** (the direct interrogative question, immediately following the scenario)

The following examples illustrate how the same general topic area can be written at different EQF levels. In each case, the domain and subject matter are held approximately constant; only the cognitive demand, responsibility context, and learner profile change. All lead-ins use direct interrogatives.

Topic: Mastitis and Udder Health (Veterinary Domain)

EQF 4 Item

Learner profile: A farm animal health worker on a 180-cow dairy unit carries out routine milking and basic health monitoring tasks under the direction of the herd manager.

During the afternoon milking, the farm animal health worker notices that one cow is flinching when the cluster is attached to her right rear quarter, and the foremilk from that quarter is visibly clotted. The cow's other three quarters appear normal.

What should the farm animal health worker do immediately?

- A. Remove the cluster from the affected quarter, record the finding, and notify the herd manager before the end of milking
- B. Continue milking the affected quarter and record the finding in the health diary after milking is complete
- C. Administer an intramammary antibiotic tube to the affected quarter from the farm medicine cabinet
- D. Strip the affected quarter fully and reattach the cluster

Key: A

EQF signal: The farm animal health worker is in a supervised role (EQF 4). The correct action is to follow a defined procedure, record the finding, and escalate. Diagnosis and treatment decisions belong to the herd manager or veterinarian.

EQF 5 Item

Learner profile: A herd health technician on a 300-cow dairy unit monitors herd health indicators, carries out defined treatment protocols, and reports to the attending veterinarian.

A herd health technician reviews the monthly milk recording data and notes that the proportion of cows with a somatic cell count above 200,000 cells/mL has risen from 14% to 23% over the past three months. New

infection data from the monthly recording show that the majority of new cases are occurring in first-lactation heifers in the first 30 days of lactation. The farm's dry cow management protocol uses blanket antibiotic dry cow therapy.

What does this pattern of new infection data most likely indicate about the source of the mastitis problem?

- A. Environmental pathogens are infecting heifers around the time of calving
- B. Contagious pathogens are spreading between cows during milking
- C. The dry cow antibiotic treatment is failing to eliminate existing infections
- D. The milking machine is causing teat end damage in older cows

Key: A

EQF signal: The herd health technician must interpret epidemiological data to identify the likely source of infection. This requires applying knowledge of mastitis epidemiology, not simply following a protocol. The answer requires reasoning about the timing and group distribution of new infections.

EQF 6 Item

Learner profile: A veterinary practitioner in farm animal practice is responsible for herd health planning, prescribing, and clinical advice for a 420-cow Holstein dairy unit.

A veterinary practitioner conducting an annual herd health review finds that the 12-month rolling clinical mastitis rate is 48 cases per 100 cows, against a target of below 30. Bacteriological culture results from the past year show that 62% of clinical cases are caused by environmental pathogens (*E. coli*, *Streptococcus uberis*), 21% are caused by *Staphylococcus aureus*, and 17% yield no growth. Milk recording data show that SCC new infection rates are highest in the dry period and in the first two weeks of lactation. The farm uses a once-daily milking system and cubicle housing with rubber mattresses bedded with chopped straw.

What is the most likely explanation for the high mastitis rate on this farm?

- A. Inadequate dry period and transition period management is allowing environmental pathogen challenge to overwhelm the cow's immune defences
- B. A contagious *Staphylococcus aureus* reservoir is being maintained within the herd through milking machine transmission
- C. The once-daily milking system is causing increased teat canal pressure and susceptibility to infection
- D. The bacteriological culture results indicate chronic subclinical infection from the dry period rather than new clinical infection

Key: A

EQF signal: The veterinary practitioner must integrate quantitative data from multiple sources (clinical mastitis rate, bacteriological profile, SCC timing data, housing and management system) to identify the primary driver of the problem. The correct answer synthesises the epidemiological picture; the distractors represent partial or alternative explanations that do not account for the full data set.

Topic: Soil Nitrogen Management (Farming Domain)

EQF 4 Item

Learner profile: An arable farm operative on a mixed arable farm carries out fieldwork tasks following written instructions from the farm manager.

An arable farm operative has been asked to apply a nitrogen fertiliser dressing to a winter wheat crop at growth stage 31. The farm manager has provided a written work instruction specifying the application rate, the field to be treated, and the spreader settings to be used. The operative has collected the spreader from the yard and filled it with the correct fertiliser grade.

What should the operative check before driving onto the field?

- A. That the spreader calibration matches the settings specified in the work instruction
- B. That the nitrogen rate is consistent with the farm's nutrient management plan
- C. That the weather forecast supports application on this date
- D. That the field has been soil sampled in the current season

Key: A

EQF signal: The operative's role is to carry out a defined task correctly (EQF 4). The correct action is to verify that the equipment is set as instructed before beginning the task. Decisions about rates, timing, and agronomy belong to the farm manager.

EQF 6 Item

Learner profile: An independent agronomist provides crop nutrition advice to a portfolio of arable farm businesses across a region designated as a Nitrate Vulnerable Zone.

An agronomist is reviewing a nitrogen management plan for a 120-hectare mixed arable farm. Recent soil analysis shows variable soil organic matter across the farm, ranging from 2.1% to 5.8%. The farm is within a Nitrate Vulnerable Zone. Last season's winter wheat yield averaged 7.2 t/ha against a target of 8.5 t/ha. Post-harvest soil nitrate testing showed elevated residuals in two fields with the highest organic matter values. The farmer currently applies a single farm-wide nitrogen rate for simplicity.

Why is the single farm-wide nitrogen rate likely to be causing both the yield gap and the elevated nitrate residuals simultaneously?

- A. Fields with high organic matter have higher soil nitrogen supply, so the uniform rate over-applies in those fields and under-applies in lower organic matter fields
- B. The nitrogen rate is too high across all fields, causing luxury nitrogen uptake and yield suppression
- C. Fields with low organic matter have poor soil structure, limiting root access to applied nitrogen
- D. The timing of nitrogen applications is mismatched to crop demand regardless of soil organic matter variation

Key: A

EQF signal: The agronomist must explain a causal mechanism that accounts for two apparently contradictory observations simultaneously: a yield gap in some fields and nitrate over-application in others. This requires integrating knowledge of soil organic matter mineralisation, soil nitrogen supply, and crop nitrogen demand. The lead-in asks "why," directly targeting causal reasoning.

Topic: Food Safety Critical Limits (Food Domain)

EQF 4 Item

Learner profile: A food production operative on a cooked poultry processing line monitors oven temperature records and carries out checks at defined intervals as part of the site HACCP plan.

A food production operative on a cooked poultry processing line is carrying out a routine temperature check and notices that the continuous monitoring display shows the oven core temperature has dropped to 71°C, below the validated critical limit of 75°C. This is the first time this has occurred during the current production run.

What should the operative do immediately?

- A. Stop the line, isolate all product that passed through the oven since the last satisfactory check, and notify the line supervisor
- B. Increase the oven temperature setting and continue production, recording the deviation in the end-of-shift log
- C. Retain the affected product at the end of the shift for a supervisor to evaluate
- D. Apply a ten-minute hold time extension to all product in the oven to compensate for the reduced temperature

Key: A

EQF signal: The operative must apply a defined critical control point response procedure (EQF 4). The correct action is immediate line stoppage, product isolation, and escalation. This is a procedural competency; the candidate does not need to evaluate the severity of the deviation, only to follow the established response.

EQF 7 Item

Learner profile: A senior food safety advisor at a large ready-to-eat food manufacturer is responsible for HACCP system design and regulatory compliance across three production sites with different process

technologies.

A senior food safety advisor is reviewing the validation evidence for a heat treatment process used for a new chilled ready-to-eat product. The process was validated against a target 6-log reduction of *Listeria monocytogenes* using a surrogate organism in laboratory conditions. Recent shelf-life studies have shown that *L. monocytogenes* counts in retained samples are approaching the regulatory limit of 100 cfu/g at end of shelf life, despite the product consistently meeting the validated process parameters at manufacture. The product contains modified atmosphere packaging and a lactate-based preservative system.

Why is the validated heat treatment process unlikely to be the primary cause of the end-of-shelf-life contamination findings?

- A. Post-process contamination from the production environment is more likely because the heat treatment is destroying the initial load but the product is being recontaminated after the critical control point
- B. The surrogate organism used in the validation study has a higher heat resistance than *L. monocytogenes*, so the validation overstates the process lethality
- C. The modified atmosphere packaging is suppressing aerobic spoilage organisms, allowing *L. monocytogenes* to grow selectively during chilled storage
- D. The lactate-based preservative is inhibiting the growth of competing microflora, effectively reducing the hurdle effect against *L. monocytogenes*

Key: A

EQF signal: The senior food safety advisor must apply specialist knowledge of HACCP system design to explain why a validated process is not preventing end-of-product-life failures. The lead-in asks “why,” targeting causal reasoning at EQF 7. The correct answer requires understanding of post-process contamination as a HACCP system failure distinct from process lethality. The distractors represent plausible but secondary or incorrect explanations that a non-specialist might reach.

Topic: Integrated Pest Management (Plant Domain)

EQF 4 Item

Learner profile: A horticultural operative in a protected tomato nursery carries out routine crop monitoring and records observations on standardised IPM monitoring sheets.

A horticultural operative in a protected tomato nursery is carrying out a routine scouting walk and notices that several plants in one section of the glasshouse have leaves with silvery streaking on the upper surface and small, pale insects visible on the undersides. The nursery operates an integrated pest management programme. The operative has not seen this symptom before.

What should the operative do with this observation?

- A. Record the location, symptom description, and estimated number of plants affected on the IPM monitoring sheet and report to the nursery manager before leaving the section
- B. Apply the standard aphid control spray from the approved product list before the infestation spreads
- C. Remove all affected leaves and dispose of them outside the glasshouse
- D. Continue the scouting walk and report the finding at the weekly team meeting

Key: A

*EQF signal: The operative’s role is to observe accurately, record, and escalate (EQF 4). The symptoms described are consistent with western flower thrips (*Frankliniella occidentalis*), a notifiable pest in some EU member states, but the operative is not expected to diagnose. Correct and prompt recording and escalation is the competency being assessed.*

EQF 6 Item

Learner profile: A BASIS-qualified agronomist provides independent crop protection advice to a portfolio of protected salad crop producers in a region where resistance to pyrethroid insecticides in western flower thrips (*Frankliniella occidentalis*) has been confirmed.

A crop protection agronomist is reviewing the pest management strategy for a large protected lettuce production unit. Western flower thrips has been confirmed as the primary pest, and resistance to pyrethroid

insecticides has been confirmed in the local thrips population by bioassay. The unit currently uses a scheduled spray programme with alternating pyrethroid and organophosphate active substances. Biological control agents are not currently in use, but the unit has temperature and humidity control that could support their establishment. The crop has a two-week production cycle with a very short pre-harvest interval requirement.

Why is the current scheduled spray programme unlikely to achieve sustainable control of this thrips population?

- A. Resistance to pyrethroids in the thrips population means those treatments have lost efficacy, and the short production cycle limits the ability to rotate to modes of action with longer pre-harvest intervals
- B. Thrips pupate in the growing medium and are protected from contact insecticides during the pupal stage regardless of the active substance used
- C. The humidity levels required for biological control agent establishment are incompatible with the temperature management needed for lettuce quality
- D. Organophosphate insecticides are being phased out under EU regulation and will not be available for continued use

Key: A

EQF signal: The agronomist must explain why the current strategy is structurally inadequate given the confirmed resistance status and production constraints. The lead-in asks “why,” requiring causal reasoning. The correct answer integrates resistance biology, mode of action rotation principles, and the practical constraint of pre-harvest intervals. Distractors represent related but secondary or incorrect factors.

Topic: Traceability and Product Withdrawal (Food Domain)

EQF 5 Item

Learner profile: A food safety technician at a fresh produce packing facility is responsible for HACCP monitoring, batch record management, and first-response to non-conformance events.

A food safety technician at a fresh produce packing facility receives a call from a major retail customer reporting that a consumer has found a piece of blue plastic in a bag of pre-packed salad leaves. The customer has provided the batch code from the packaging. The technician locates the production records for that batch code and confirms the packing date and line.

What should the technician do first?

- A. Place all remaining stock from that batch on hold and notify the food safety manager before taking any further action
- B. Issue an immediate full recall of all salad leaf products packed in the past 30 days
- C. Inform the customer that the investigation is underway and provide an estimated response time
- D. Review the production records to identify the source of the contamination before taking any stock action

Key: A

EQF signal: The technician has a defined first-response role (EQF 5). The correct first action is proportionate: hold the identified batch and escalate to the manager. A full recall (B) is disproportionate at this stage without further investigation. Investigating before holding stock (D) risks allowing potentially contaminated product to leave the site during the investigation window.

EQF 7 Item

Learner profile: A supply chain director at a large fresh produce business is responsible for traceability strategy, supplier management, and crisis response across a multi-country supply base of 47 primary producers.

A supply chain director is reviewing the company’s traceability performance following two voluntary withdrawal exercises in the past 12 months, both of which exceeded the internal target of completing product identification and hold within 24 hours of notification. In both cases, the delay was caused by

incomplete or delayed data transfer from primary producer level to the company's enterprise resource planning system. Primary producers currently record batch and movement data on paper at farm level; this data is manually entered into the ERP system at goods receipt into the company's packing facility. Trade records indicate that in both withdrawal exercises, affected product had already been distributed onward to customers before the withdrawal was initiated.

What is the root cause of the traceability failures in these two withdrawal exercises?

- A. The data discontinuity between farm-level paper records and the company ERP system creates a lag between production events and the company's ability to act on traceability information
- B. The withdrawal target of 24 hours is too short for a supply chain of this complexity
- C. Primary producers lack sufficient training in traceability requirements
- D. The company ERP system does not have sufficient functionality to handle withdrawal exercises at scale

Key: A

EQF signal: The supply chain director must identify a systemic root cause from a description of repeated failure, not just the immediate trigger of delay. The lead-in asks "what is the root cause," targeting analytical reasoning at EQF 7. The correct answer identifies the structural data architecture problem; distractors address symptomatic or secondary factors.

Topic: Notifiable Disease Response (Veterinary Domain)

EQF 5 Item

Learner profile: A herd health technician on a large pig finishing unit is responsible for daily health monitoring of housed finishing pigs and reports to the farm manager and attending veterinarian.

A herd health technician on a 2,000-place pig finishing unit observes that eight pigs in one pen are showing vesicular lesions on the snout and coronary bands, and two of these pigs are lame. The pigs were healthy at the previous morning check 24 hours ago. The farm has had no recent changes in stock or feed sources. The technician is aware that foot-and-mouth disease is notifiable.

What must the herd health technician do before leaving the affected pen?

- A. Avoid moving any animals or equipment from the pen, wash hands and disinfect footwear, and contact the farm manager immediately
- B. Collect swab samples from the lesions and submit them to the regional veterinary laboratory for confirmation
- C. Administer a non-steroidal anti-inflammatory drug to the lame animals and monitor for 24 hours before escalating
- D. Move the affected pigs to the isolation pen to prevent spread to the rest of the finishing group

Key: A

EQF signal: The herd health technician must apply biosecurity principles in a potential notifiable disease situation (EQF 5). The correct response is immediate containment and escalation without taking any action that could spread contamination. The technician is not expected to diagnose or sample; those actions belong to the veterinarian.

EQF 8 Item

Learner profile: A research veterinarian advising a national competent authority on disease control strategy is leading the epidemiological investigation following the index detection of a novel reassortant strain of highly pathogenic avian influenza (HPAI) in a commercial turkey flock.

A research veterinarian advising a national competent authority is reviewing the epidemiological data from the first 72 hours following confirmation of HPAI in a 50,000-bird turkey unit. Whole genome sequencing has identified a novel H5N1 reassortant not previously detected in Europe. Surveillance swabbing of wild bird carcasses within a 10-kilometre radius has yielded two positive detections of a closely related but phylogenetically distinct strain. The index flock was managed under standard biosecurity, with no live poultry vehicle movements in the preceding 21 days. Airborne spread from wild birds is considered the most likely route of introduction based on wind trajectory modelling.

Why does the wild bird phylogenetic data complicate the epidemiological interpretation of this outbreak?

- A. The phylogenetic distance between the wild bird and index flock strains means that a direct spillover from the detected wild bird population cannot be confirmed as the source, requiring consideration of an undetected intermediate host or an independent introduction event
- B. Wild bird HPAI surveillance within 10 kilometres is insufficient to rule out a more distant wild bird source for the index flock infection
- C. The airborne spread hypothesis is inconsistent with what is known about HPAI transmission biology and should be discounted pending further modelling
- D. The novel reassortant status of the index flock strain means that existing serological surveillance tools may not detect further spread accurately

Key: A

EQF signal: The research veterinarian must apply expert-level phylogenetic and epidemiological reasoning to explain an interpretive complication in an outbreak investigation (EQF 8). The lead-in asks “why,” targeting the highest level of causal reasoning. The correct answer requires understanding of phylogenetic inference, source attribution methodology, and the limitations of molecular epidemiology. This level of reasoning is not expected at EQF levels below 7-8.

Part 3: Technical Item Flaws

These flaws make items harder for reasons unrelated to the competency being assessed. They introduce construct-irrelevant variance and reduce the accuracy of the credential evidence.

Long or Complex Options

Options that are substantially longer than the others, or that contain multiple embedded propositions, increase reading load and shift what is being measured from competency to reading speed.

Flawed item:

A pig unit manager has observed an increase in pre-weaning piglet mortality over the past four weeks. What is the most appropriate initial investigation?

- A. Review farrowing house temperature management, sow feeding management in late gestation and lactation, and the colostrum intake monitoring protocol for all litters born in the past four weeks, cross-referencing with records of individual sow body condition score at farrowing and comparing outcomes in gilts versus older sows
- B. Cull all sows with more than two consecutive poor litter outcomes
- C. Increase creep feed provision from day three after birth
- D. Contact the attending veterinarian for clinical investigation

Option A is disproportionately long and detailed, and stands out as the correct answer by length alone.

Principle: Options should be comparable in length. If shared content applies to multiple options, move it into the stem. Remove instructional or explanatory language from the correct option that is not matched in the distractors.

Vague Terms

Frequency qualifiers such as “usually,” “often,” “frequently,” and “sometimes” are interpreted inconsistently across candidates and introduce ambiguity into options that should be clearly evaluable.

Flawed item:

How does *Campylobacter* contamination most commonly behave in broiler processing?

- A. Contamination usually occurs during evisceration
- B. Chilling frequently reduces contamination levels on carcass surfaces
- C. Surface decontamination interventions often achieve a 3-log reduction
- D. Contamination is always introduced at the scalding stage

Options A, B, and C cannot be evaluated with confidence because the frequency terms are subjectively interpreted. Option D uses an absolute term in a context where it is factually incorrect.

Principle: Replace frequency qualifiers with specific, evaluable statements. Reserve absolute terms only where they are factually warranted, and never in incorrect options.

Negatively Phrased Lead-ins

Lead-ins that ask for the exception or the incorrect option require the candidate to hold all options in mind simultaneously and evaluate each one for truth value, rather than generating the best answer. This adds cognitive load that is unrelated to the competency and increases the risk of misreading.

Flawed item:

Each of the following is an appropriate initial response to a positive *Salmonella* environmental swab result in a laying hen unit EXCEPT:

- A. Notify the competent authority
- B. Restrict movement of birds and eggs from the premises
- C. Immediately depopulate the unit
- D. Begin sampling to identify the source and extent of contamination

Principle: Reframe as a positive question wherever possible. Here: “A positive *Salmonella* environmental swab result has been confirmed in a laying hen unit. What is the most appropriate immediate response?” tests the same knowledge without the cognitive overhead of an exception format.

Non-parallel Options

Options should be grammatically and structurally parallel. When some options are full sentences and others are noun phrases, or when verb tense varies, the inconsistency adds irrelevant reading difficulty.

Flawed item:

How should a farmer approaching spray operations for potato blight management reduce the risk of fungicide resistance developing?

- A. Rotation with fungicides of a different mode of action group
- B. The farmer should check the resistance risk rating before applying
- C. Apply at the full label rate to ensure efficacy
- D. Resistance management guidelines published by the fungicide manufacturer

Option A is a noun phrase, B is a full declarative sentence with a subject, C is an imperative clause, and D is a noun phrase. These cannot be compared consistently.

Principle: Choose one grammatical form for all options and apply it consistently. Imperative clauses work well for action-oriented items: “Rotate with fungicides of a different mode of action group,” “Apply at the full label rate to ensure efficacy,” and so on.

Flaws That Cue the Test-wise Candidate

These flaws allow a candidate to arrive at the correct answer through structural analysis of the item rather than by applying the competency. They introduce construct-irrelevant ease.

Grammatical Cues

When a lead-in ends with an incomplete sentence, only options that complete it grammatically are plausible, allowing candidates to eliminate distractors without subject knowledge.

Flawed item:

The primary EU regulation governing the placing of plant protection products on the market is:

- A. Directive 2009/128/EC on the sustainable use of pesticides
- B. Regulation (EC) No 1107/2009
- C. Framework Regulation (EU) 2018/848 on organic production
- D. Directive 91/414/EEC

The incomplete sentence construction means candidates evaluate grammatical fit alongside content.

Principle: Use a complete question as the lead-in. “What is the primary EU regulation governing the placing of plant protection products on the market?” eliminates grammatical cueing entirely.

Correct Option Stands Out

When the correct option is substantially longer than the distractors, or is the only option that contains a qualification or caveat, the length difference itself signals the correct answer to a test-wise candidate.

Flawed item:

What is a critical control point in a HACCP plan?

- A. A step at which the hazard must be eliminated
- B. A step at which control can be applied and is essential to prevent, eliminate, or reduce a food safety hazard to an acceptable level
- C. Any step in a food process
- D. A monitoring step

Option B is substantially longer and more complete than the others, signalling it as the correct answer without any knowledge of HACCP.

Principle: Review all options for length consistency. Remove explanatory language from the correct option that is not matched in the distractors. The option set should look visually balanced.

Word Repetition (Clang Clues)

When a key word from the stem appears only in the correct option, the repetition cues the candidate without

requiring knowledge of the subject.

Flawed item:

A finishing cattle unit is experiencing increased respiratory disease in housed cattle. What is the most likely diagnosis?

- A. Bovine viral diarrhoea
- B. Bovine respiratory disease complex
- C. Leptospirosis
- D. Hypomagnesaemia

The word “respiratory” appears in the stem and in option B, directly cueing the correct answer.

Principle: Scan the stem and all options for repeated words or etymologically related terms. Either replace the repeated word in the correct option with a clinical or technical equivalent, or use the shared term in all options so that it no longer cues.

Collectively Exhaustive Options

When a subset of options covers all possible outcomes on a dimension, the correct answer must lie within that subset. Test-wise candidates can eliminate the remaining options without subject knowledge.

Flawed item:

What happens to the microbial load of a liquid egg product after pasteurisation?

- A. It increases
- B. It decreases
- C. It remains unchanged
- D. The outcome depends on the storage temperature after pasteurisation

Options A, B, and C are collectively exhaustive on the dimension of directional change: the load can only increase, decrease, or stay the same. Option D is not on this dimension, but it does not resolve the problem because the correct answer must lie within A, B, or C.

Principle: Ensure that no subset of options is collectively exhaustive. Restructure the question or replace at least one member of the exhaustive subset with an option that is on a different but relevant dimension of the topic.

Convergence

When the correct answer shares more elements with the other options than any distractor does, a test-wise candidate can identify it by counting shared terms, without any knowledge of the subject. This typically occurs when distractors are written as permutations of the correct answer.

Flawed item:

Under what conditions is grain in store most likely to develop mycotoxin contamination?

- A. High moisture content and low temperature
- B. High moisture content and high temperature
- C. Low moisture content and high temperature
- D. Low moisture content and low temperature

Option B shares “high” with A and “high temperature” with C. It is the option with the most in common with the others, making it identifiable by convergence without knowledge of grain storage science.

Principle: Review options for repeated terms and restructure to distribute shared language more evenly, or rewrite options to avoid the permutation pattern entirely.

Absolute Terms

Options containing “always,” “never,” “all,” and “none” are typically false and are easily eliminated by a test-wise candidate who knows that absolute statements rarely hold in biological or food systems.

Flawed item:

How should antibiotic use in pig production be characterised under current European guidance?

- A. Antibiotics should always be administered via the water supply for herd treatments
- B. Prophylactic use of antibiotics is never appropriate under any circumstances

- C. Metaphylactic treatment may be appropriate when clinical disease is confirmed in part of a group and spread to contact animals is anticipated
- D. All antibiotic use in food-producing animals requires a veterinary prescription

Options A and B contain absolute terms that make them identifiable as incorrect without subject knowledge. Option D contains “all,” which a test-wise candidate will question.

Principle: Avoid absolute terms in distractors. They should only appear in the correct option, and only when the absolute statement is factually accurate.

Summary of Technical Item Flaws

Flaw Type	Category	Solution
Long or complex options	Irrelevant difficulty	Move shared content to stem; ensure options are comparable in length
Vague frequency terms	Irrelevant difficulty	Replace with specific, evaluable statements
Negatively phrased lead-ins	Irrelevant difficulty	Rewrite as a positive direct question wherever possible
Non-parallel options	Irrelevant difficulty	Choose one grammatical form for all options
Grammatical cues	Cues testwise candidate	Use a complete question as the lead-in
Correct option stands out	Cues testwise candidate	Ensure consistent option length and detail level
Word repetition (clang clues)	Cues testwise candidate	Replace repeated word in option or use it in all options
Collectively exhaustive options	Cues testwise candidate	Restructure to eliminate exhaustive subsets
Convergence	Cues testwise candidate	Distribute shared language evenly across options
Absolute terms in distractors	Cues testwise candidate	Remove absolute terms from incorrect options

Part 4: Connecting Items to Credential Specifications

A verifiable credential makes a specific claim. The item that provides evidence for that claim must be deliberately designed to test the competency described, at the level specified. This requires an explicit mapping from credential specification to item design.

The following process is recommended.

Step 1: Identify the competency and EQF level. Start with the competency descriptor as written in the credential specification. Note the EQF level. Identify which of the three EQF dimensions (knowledge, skills, responsibility/autonomy) the credential is primarily claiming to evidence.

Step 2: Determine the cognitive task. Using Table 1, identify the appropriate cognitive task for the EQF level. For EQF 4-5, this is typically identifying, selecting, or applying a procedure. For EQF 6-7, it involves analysis, synthesis, or recommendation. For EQF 8, it involves expert evaluation or generation of knowledge.

Step 3: Select the learner profile. Choose a recognisable workforce role from the EQF level profiles in Chapter 4 that matches the credential's target audience. Embed the profile in the first sentence of the scenario.

Step 4: Design the scenario. Write a scenario that places the candidate in the selected role, in a realistic production or professional context. Include enough information to make the cognitive task possible without providing information that makes it trivial.

Step 5: Write the lead-in as a direct interrogative. Apply the cover-the-options rule. The lead-in should be answerable from the scenario alone. Default to a direct interrogative (what, why, when, how, where, who) unless a selective interrogative is genuinely the stronger choice for this specific testing point. Select a lead-in form from Appendix B that matches the competency domain and cognitive task.

Step 6: Generate options. Write the correct answer first. Then generate distractors that are plausible to a candidate who lacks the specific competency being tested. Distractors should represent common misconceptions, incomplete application of the principle, or contextually inappropriate but superficially reasonable actions.

Step 7: Apply the flaw checklist. Use the summary table in Chapter 8 to review the item for technical flaws before submission.

Complete this template for each item submitted to an item bank. It provides the explicit mapping between the item and the credential it supports.

Item Reference: [unique identifier]

Domain: [Veterinary / Farming / Food / Plant]

Credential Title: [as stated in the credential specification]

EQF Level: [4 / 5 / 6 / 7 / 8]

Competency Descriptor (from credential): [copy from credential specification]

EQF Dimension being tested: [Knowledge / Skills / Responsibility and Autonomy / combination]

Cognitive Task (from Table 1): [e.g., "Analyse and recommend; integrate multiple information sources"]

Learner Profile: [role and responsibility context used in the scenario]

Testing Point: [one sentence: what specifically does this item test?]

Lead-in Type: [Direct interrogative: what / why / when / how / where / who — or Selective interrogative with justification]

Stem:

Lead-in:

Options (mark key with asterisk): - A. - B. - C. - D. - E. (if used)

Rationale for key: [why is the key correct?]

Rationale for each distractor: [why is each distractor plausible but wrong?]

Flaw check completed: [Yes / No]

Subject matter reviewer: [name and affiliation]

Date:

Item Analysis for Small-Scale Credential Assessment

In large-scale assessment programmes, items are refined over time through statistical analysis. In microcredential assessment programmes, candidate numbers are typically smaller and item banks are refreshed more frequently. Statistical analysis remains valuable but should be interpreted with care at small sample sizes.

Track unexpected performance. If an item is consistently answered incorrectly by candidates who perform well overall, this suggests either a flawed item or a genuine gap between what the credential claims and what the item tests. Review the item for technical flaws before concluding that the finding reflects a competency gap in the candidate population.

Review distractor performance. If a distractor is never selected, it is not functioning: either it is too obviously wrong, or it represents a misconception that does not exist in the candidate population. Replace non-functioning distractors with options that are genuinely plausible to a candidate who lacks the competency.

Monitor for item exposure. In small cohorts with shared learning environments, items can become known to candidates before assessment. If a previously difficult item suddenly becomes very easy, consider whether it has been shared. Maintain an item bank large enough to allow rotation.

Use expert review as the primary quality mechanism. At small sample sizes, expert review by a subject matter specialist who was not involved in writing the item is the most reliable quality control mechanism. Expert reviewers should apply the flaw checklist in Chapter 8 and confirm that the key is unambiguously correct and the distractors are genuinely plausible to a non-expert in the specific competency.

Part 5: Constrained short answer questions

The Problem With Open Questions

Short answer questions appear straightforward to write. Questions such as “what causes bovine respiratory disease?” or “how does contamination enter a food processing environment?” may appear reasonable, but they are not well-formed assessment items.

An open short answer question has two structural problems that undermine its validity as assessment evidence.

The brain dump problem. A candidate who knows a great deal about bovine respiratory disease can write everything they know: viruses, bacteria, management factors, housing, stress, transportation, nutritional deficiency, immune suppression, co-infection dynamics. Somewhere in that output will be something the marker accepts. The question has been answered by volume rather than by knowledge of what matters. The question is testing quantity of output, not quality of understanding.

The conceptual frame problem. This is the deeper issue. “What causes bovine respiratory disease?” does not tell the candidate which conceptual framework or category system to apply. One candidate answers with pathogen names. Another answers with management risk factors. Another answers with host susceptibility factors. All are defensible. None is wrong. But the marker had a specific answer in mind, and candidates are left to guess not just what the answer is, but what kind of answer is expected. Marking becomes inconsistent across candidates and across markers.

Consider a question asking for the categories of mastitis pathogens. A candidate could organise their answer by transmission route (contagious, environmental, opportunistic), by bacterial classification (Gram-positive, Gram-negative, other), by clinical significance (major pathogens, minor pathogens), or by genus (Staphylococci, Streptococci, coliforms). Each candidate has made a reasonable choice of framework. None can be marked wrong. But the marker expected one framework, and others will be penalised inconsistently depending on the marker’s interpretation.

The constraint solves both problems simultaneously. It prevents the brain dump by limiting the scope of the expected answer. More importantly, it specifies which mental model or category system the candidate must apply, so that two competent candidates working independently converge on the same answer, and two markers working independently converge on the same mark.

What a Constraint Does

A constraint is a specification embedded in the question that tells the candidate:

- **How many** answers are expected
- **Which conceptual framework** to apply
- **In what context** the answer applies
- **To what level of specificity** the answer should be pitched
- **In what order or by what criterion** answers should be ranked or prioritised

A well-constrained short answer question is one where two markers working independently, and two candidates working competently, would converge on the same answer set. If that convergence is not possible, the constraint is insufficient.

Types of Constraint

1. Quantity Constraint

Specifying the number of expected answers is the most basic form of constraint. It prevents the brain dump but does not by itself specify the conceptual frame. It should almost always be combined with at least one other constraint type.

Unconstrained: What are the causes of lameness in dairy cattle?

Quantity-constrained only: What are three causes of lameness in dairy cattle?

This is better but still permits candidates to answer from entirely different frameworks: one lists specific conditions (white line disease, sole ulcer, digital dermatitis), another lists management factors (concrete surfaces, overcrowding, wet conditions), another lists nutritional factors. All are defensible.

Quantity and framework constrained: Name the three most common infectious causes of lameness in housed dairy cattle during the winter housing period.

Now the candidate knows: infectious causes are required, not structural or nutritional causes; the three most common are required, not an exhaustive list; the context is housed cattle in winter. Two competent candidates will converge on digital dermatitis, foul of the foot, and heel horn erosion.

2. Framework Constraint

The framework constraint specifies the conceptual category system the candidate must apply. This is the most important form of constraint and the most frequently omitted.

Unconstrained: What are the categories of food safety hazards?

A candidate could list specific pathogens, hazard types, hazard sources, or process stages at which hazards arise. All are defensible.

Framework constrained: Name the four main categories of food safety hazard as classified by hazard type under the Codex Alimentarius framework.

The phrase “classified by hazard type under the Codex Alimentarius framework” specifies both the organising principle and the reference framework. The expected answer is: biological, chemical, physical, and allergenic. There is now only one correct organisational system.

3. Context Constraint

The context constraint anchors the question to a specific production system, regulatory context, species, or stage of a process. It prevents the candidate from answering in generalities and requires them to apply knowledge to a defined situation.

Unconstrained: What are the risks associated with pesticide application?

A candidate could discuss operator safety, environmental contamination, residues in food, resistance development, non-target organism impacts, and much more. The question has no defined scope.

Context constrained: What are the two most important risks to operator safety when applying an organophosphate insecticide by knapsack sprayer in warm weather?

The context is now specific: organophosphate chemistry, knapsack application, and warm weather conditions that affect both product volatility and operator physiology. The answer converges on dermal absorption and heat stress or inhalation exposure, with correct understanding of the acute toxicity risk profile of organophosphates.

4. Specificity Constraint

The specificity constraint tells the candidate at what level of detail the answer should be pitched: the pathogen genus, the specific species, the management category, the regulatory instrument name, the mechanism.

Insufficiently specific: What organism is most commonly associated with food poisoning from cooked rice?

A candidate could answer “a bacterium,” “a spore-forming bacterium,” “*Bacillus*,” or “*Bacillus cereus*.” These are at four different levels of specificity, all technically answering the question.

Specificity constrained: Name the bacterial species most commonly responsible for the emetic form of food poisoning associated with cooked rice held at ambient temperature after cooking.

“Bacterial species” specifies that a genus-species binomial is required. “Emetic form” distinguishes the toxin type and the specific clinical syndrome. There is now one correct answer at one level of specificity: *Bacillus cereus*.

5. Prioritisation Constraint

Prioritisation is the most cognitively demanding form of constraint. It does not simply ask the candidate to list; it asks them to demonstrate understanding of relative weight, significance, urgency, or risk, and to apply that understanding to rank their answers by a defined criterion.

A quantity constraint asks how many. A framework constraint asks what kind. A prioritisation constraint asks which matters most, and requires the candidate to understand why that ordering holds in the specific context given.

This is what separates knowledge of content from understanding of significance. Two candidates might both know that digital dermatitis, foul of the foot, and heel horn erosion are common infectious causes of lameness in dairy cattle. Only the candidate who understands the comparative prevalence data, welfare implications, and economic impact of each condition can correctly prioritise them in a specific production context and justify that ordering.

The prioritisation criterion must be stated explicitly in the question. This is critical. “Prioritise by clinical urgency” and “prioritise by ease of prevention” will produce different correct answers for the same list of conditions. The criterion is part of the constraint, not an implicit assumption.

Prioritisation criteria in agrifood contexts include:

- **Clinical or welfare urgency:** which requires the most immediate intervention to prevent animal suffering or death
- **Frequency or prevalence:** which occurs most commonly in this production system or context
- **Economic impact:** which has the greatest effect on production performance or profitability
- **Regulatory consequence:** which carries the greatest legal or compliance risk
- **Preventability:** which can most effectively be reduced by feasible management changes
- **Public health significance:** which poses the greatest risk to the food chain or consumer safety
- **Reversibility:** which causes irreversible harm if not addressed promptly

A well-written prioritisation question states the criterion, specifies the number of items to be ranked or the ranking position required, and provides enough contextual information for the prioritisation to be unambiguous.

Unconstrained: What are the most important food safety hazards in a fresh poultry processing operation?

A candidate can list any hazard and cannot be marked wrong. The question provides no basis for prioritisation and no framework for deciding which hazards are more important than others.

Prioritisation constrained: In a fresh poultry processing operation producing whole birds for retail sale, identify the two microbiological hazards that pose the greatest public health risk at the point of consumption, and for each one state the primary process step at which contamination most commonly occurs.

The candidate must now: identify the relevant hazard category (microbiological, not physical or chemical), prioritise by public health risk at the point of consumption (not at the point of processing), and demonstrate knowledge of the process steps at which each hazard is introduced. The expected answer prioritises *Campylobacter* and *Salmonella* by public health significance in this product category, and correctly identifies scalding and evisceration as the primary contamination points respectively. A candidate who lists “foreign bodies” or “chemical contamination” has not applied the prioritisation criterion correctly.

Worked Examples Across the Four Domains

The following pairs show unconstrained questions revised into well-constrained short answer questions. Each example identifies the constraint types applied and explains why the prioritisation criterion, where used, produces a specific and defensible answer set.

Veterinary Domain

Unconstrained: What are the causes of neonatal calf diarrhoea?

Problem: Pathogens, management factors, nutritional factors, and environmental factors are all valid causes. The conceptual frame is entirely open. A candidate can write a page and hit every marking criterion by accident.

Constrained: A beef suckler herd has experienced an outbreak of diarrhoea in calves aged between five and fourteen days. List the four infectious agents most likely to be responsible, ranked in order of the age at

which each typically causes peak disease in the first three weeks of life.

Constraints applied: - Quantity: four agents - Framework: infectious agents only, not management or nutritional causes - Context: beef suckler calves, five to fourteen days of age, outbreak situation - Specificity: agent level, not category level - Prioritisation criterion: ranked by age at peak disease

Why the prioritisation criterion matters here: The age-at-onset ranking requires the candidate to know the epidemiology of each agent, not just its name. A candidate who knows that *Cryptosporidium parvum*, rotavirus, coronavirus, and *E. coli* K99 are all relevant but cannot rank them by age at onset has demonstrated recall without the epidemiological understanding the credential is assessing.

Expected answer: *E. coli* K99 (days 1 to 4, septicaemic and enteric forms), rotavirus (days 5 to 10), *Cryptosporidium parvum* (days 5 to 14), coronavirus (days 7 to 21, often as a secondary pathogen after rotavirus). The ranking separates this answer from a simple list: a candidate who places *Cryptosporidium* first or coronavirus second cannot be awarded full marks even if all four agents are named.

Farming Domain

Unconstrained: What are the problems with compacted soil?

Problem: Physical, chemical, biological, agronomic, and environmental effects are all valid. The question provides no basis for selection or prioritisation and no specified conceptual frame.

Constrained: A farm manager on a heavy clay arable farm is reviewing soil management following two consecutive seasons with poor establishment of winter oilseed rape. Identify the two most significant effects of surface soil compaction on crop establishment in this system, ranked by the immediacy of their impact on seedling survival in the first four weeks after drilling.

Constraints applied: - Quantity: two effects - Framework: effects on crop establishment specifically, not on long-term soil health, water quality, or machinery performance - Context: heavy clay soil, winter oilseed rape establishment, two poor seasons providing a diagnostic anchor - Specificity: effects must be linked to seedling survival, not described in abstract soil science terms - Prioritisation criterion: ranked by immediacy of impact on seedling survival in the first four weeks

Why the prioritisation criterion matters here: Ranking by immediacy requires the candidate to understand the temporal sequence of compaction effects on a germinating crop. Capping (surface crusting preventing emergence) acts within the first week. Poor root penetration into compacted layers acts over a longer period. A candidate who reverses the ranking or conflates them demonstrates incomplete understanding of establishment biology.

Expected answer: First in immediacy: surface capping and reduced soil-seed contact, preventing or delaying emergence (impact within days one to seven after drilling). Second: restricted root development into the compacted layer, limiting access to moisture and nutrients during the critical establishment phase (impact from weeks two to four). Effects on drainage, earthworm activity, and nitrogen mineralisation are real but operate over longer timescales and do not directly threaten seedling survival in the first four weeks.

Food Domain

Unconstrained: What are the principles of HACCP?

Problem: This is a recall question dressed as a short answer question. A candidate who has memorised the seven Codex principles lists them. A candidate who has not writes whatever they know about food safety. Neither version produces useful evidence of competency in applying HACCP.

Constrained: A small ready-to-eat cooked meat manufacturer is implementing a HACCP plan for the first time. From the seven Codex Alimentarius HACCP principles, identify the two that are most critical to implement correctly in the first instance to ensure the plan provides genuine food safety control, rather than being a paper exercise. For each one, state why it takes priority over the remaining principles at this stage of implementation.

Constraints applied: - Quantity: two from seven - Framework: Codex Alimentarius seven principles specifically (not general food safety management concepts) - Context: small manufacturer, ready-to-eat cooked meat, first implementation (a context in which resource constraints make prioritisation genuinely necessary) - Specificity: must name the principle and explain the prioritisation rationale, not just list - Prioritisation criterion: most critical to genuine food safety control at first implementation, explicitly

distinguished from administrative compliance

Why the prioritisation criterion matters here: The distinction between “genuine food safety control” and “paper exercise” is doing important work. A candidate who prioritises documentation principles (record keeping, verification) over hazard analysis and critical limit establishment is demonstrating a compliance mindset rather than a food safety understanding. The expected prioritisation reflects the logical dependency of the whole system on hazard analysis and CCP identification: without these being correct, the rest of the plan is built on the wrong foundations.

Expected answer: Principle 1 (conduct a hazard analysis) takes priority because every subsequent element of the plan depends on correct identification of significant hazards; a flawed hazard analysis cannot be corrected by good monitoring or documentation. Principle 3 (establish critical limits for each CCP) takes second priority because a CCP without a validated critical limit provides no actual control of the hazard; it creates the appearance of control without the substance. A candidate who prioritises record keeping (Principle 7) or verification (Principle 6) has not applied the prioritisation criterion correctly, even if those principles are correctly named.

Plant Domain

Unconstrained: What are the factors that affect pesticide efficacy?

Problem: Weather, application technique, formulation, canopy characteristics, target organism biology, and resistance status are all valid. A candidate can write extensively without demonstrating depth of understanding in any area.

Constrained: A BASIS-qualified agronomist is investigating why a foliar fungicide application failed to control septoria leaf blotch (*Zymoseptoria tritici*) in a winter wheat crop despite being applied at the full label rate. Identify the three most likely reasons for the control failure, ranked in order of the likelihood that each was the primary cause in this specific situation, and for each reason identify the stage of the application process at which the failure would have occurred.

Constraints applied: - Quantity: three reasons - Framework: application-related failure reasons, not product resistance or pathogen biology (those are excluded by the phrase “applied at the full label rate,” which removes dose as a variable) - Context: foliar fungicide, septoria in winter wheat, full label rate applied (which constrains the candidate to application quality and timing factors, not dose) - Specificity: ranked by likelihood as primary cause in this situation, and located by stage of application process - Prioritisation criterion: likelihood of being the primary cause in this specific context

Why the prioritisation criterion matters here: Ranking by likelihood as primary cause requires the candidate to understand the relative frequency with which each failure mode occurs in practice. Timing errors (missing the T2 or T3 application window) are statistically the most common cause of septoria control failure in commercial practice. Rain-fastness failures are real but require rainfall within a specific post-application window. Nozzle and water volume errors are possible but less common when applications are made by experienced contractors. A candidate who ranks rain-fastness first is not wrong in principle, but is not applying the likelihood criterion correctly given what is known about septoria control failure patterns.

Expected answer, ranked by likelihood as primary cause:

First: Timing failure at the application stage. The T2 and T3 timing windows for septoria control are narrow and linked to specific growth stages and infection periods. Applications made outside these windows, even at full label rate, will not control established infection. This is the most common cause of control failure in commercial practice.

Second: Insufficient canopy penetration at the application stage. Inadequate water volume or an inappropriate nozzle type for the crop canopy density at the target growth stage will result in poor coverage of the lower and middle canopy, where septoria infection typically initiates on lower leaves and progresses upward.

Third: Post-application rainfall before the product achieved rain-fastness. If significant rainfall occurred within the product-specific rain-fast period (typically one to two hours for triazole fungicides), active ingredient may have been washed from leaf surfaces before uptake was complete.

Marking Short Answer Questions With Constraints

A well-constrained short answer question should have a marking key that specifies:

1. **The required number of acceptable answers** and whether partial marks are available
2. **The conceptual framework** within which answers will be accepted, with explicit statements about which frameworks are not acceptable for this question
3. **The required level of specificity**, distinguishing between answers that are technically correct but insufficiently specific and those that meet the constraint
4. **The prioritisation criterion and expected ordering**, with clear statements about which answers demonstrate correct prioritisation and which demonstrate recall without understanding of significance
5. **The basis for awarding marks for explanation or justification** where the question requires the candidate to explain why, not just what

A marking key that requires individual marker judgement to decide whether an answer is acceptable indicates that the question was insufficiently constrained. If experienced markers working independently cannot converge on the same mark for a given response, the question must be revised before use.

Constraints and EQF Level

The type of constraint appropriate for a question should reflect the EQF level of the credential being assessed.

At **EQF 4**, quantity and context constraints are most appropriate. The question specifies what to do in a defined situation and asks for a defined number of steps or actions. The cognitive task is procedural and the answer is largely non-negotiable.

At **EQF 5**, framework constraints become important alongside quantity and context. The candidate is expected to apply a specific analytical or professional framework to a problem, not just identify correct actions. Prioritisation by frequency or by standard procedure is appropriate at this level.

At **EQF 6**, prioritisation constraints become central. The candidate must demonstrate understanding of relative significance, not just knowledge of content. Prioritisation criteria at this level typically involve clinical judgement, risk assessment, or evidence-based reasoning. Explanation of the prioritisation rationale is required, not just the ranked list.

At **EQF 7**, prioritisation is by systems-level significance, policy consequence, or public health impact. The candidate must account for competing considerations and justify a prioritisation that acknowledges trade-offs. The question may ask the candidate to prioritise under conditions of incomplete evidence or conflicting data.

At **EQF 8**, the candidate may be asked to critique a prioritisation made by others, identify the assumptions on which it depends, and propose an alternative prioritisation with justification from primary evidence. The constraint at this level specifies the analytical perspective and the standard of evidence required.

Quick Reference: Constraint Checklist for Short Answer Questions

Before submitting a short answer question, confirm the following.

Scope and frame: - The question specifies how many answers are required, or makes clear that a single best answer is expected - The question specifies the conceptual framework or category system the candidate must apply - The question specifies the context within which the answer applies

Specificity: - The question specifies the level of specificity required - A candidate who gives a correct but insufficiently specific answer (for example, naming a genus when a species is required) can be distinguished from one who gives a fully correct answer

Prioritisation: - If ranking or prioritisation is required, the criterion by which items should be ranked is explicitly stated in the question - The prioritisation criterion is specific enough that two competent candidates would rank the items in the same order - The question makes clear whether justification of the prioritisation is required or whether the ranked list alone is sufficient

Marking convergence: - Two competent markers working independently would produce the same marking key from the question as written - Two competent candidates working independently would produce answers that converge on the same content and ordering - A candidate who knows the subject but applies the wrong conceptual frame can be clearly distinguished from one who applies the correct frame

If any of these conditions is not met, the question requires further constraint before use in credential assessment.

Quick Reference Checklist for Item Writers

The scenario: - Embeds a recognisable learner profile in the opening sentence - Places the candidate in a role consistent with the EQF level responsibility descriptor - Contains enough information to answer the question - Does not contain irrelevant information (window dressing or red herrings) - Presents information in logical order: role and setting, task context, observed finding, relevant history

The lead-in: - Is a complete, direct interrogative question (what, why, when, how, where, who) - Passes the cover-the-options test: a knowledgeable candidate can answer it without seeing the options - Is positively phrased unless a negative form is genuinely unavoidable - Uses a selective interrogative only where explicitly justified

The options: - Are homogeneous: all of the same type, evaluable on a single dimension - Are comparable in length and grammatical structure - Are free of absolute terms in distractors - Contain no repeated key words from the stem (clang clues) - Do not form a collectively exhaustive subset - Are free of convergence patterns - The key is unambiguously correct - Each distractor is plausible to a candidate who lacks the specific competency

The EQF alignment: - The cognitive task matches the EQF level (see Table 1) - The responsibility context in the scenario matches the EQF level descriptor - The lead-in type (direct interrogative form) matches the cognitive task - The item specification template has been completed

Lead-in Bank by Domain, Competency Area, and EQF Level

The following lead-ins are provided as starting points, written in direct interrogative form. They should be adapted to fit specific scenarios. The EQF range in brackets indicates the level range for which each lead-in type is most appropriate based on the cognitive task implied.

Direct interrogative forms are listed first in each section. Selective interrogative alternatives are noted where they remain appropriate.

Veterinary Domain (Farm Animals)

Diagnosis and Clinical Reasoning - What is the most likely cause of the clinical signs described? [EQF 5-6] - What is the most likely diagnosis in this animal? [EQF 5-6] - What does this pattern of clinical findings most likely indicate? [EQF 5-6] - What additional diagnostic test is most appropriate at this time? [EQF 5-7] - Why are these laboratory findings inconsistent with the initial working diagnosis? [EQF 6-7] - How did this pathogen most likely enter this herd? [EQF 5-6]

Management and Treatment - What should the farm animal health worker do immediately? [EQF 4] - What is the most appropriate treatment for this animal at this stage? [EQF 5-6] - What is the most appropriate next step in managing this disease outbreak? [EQF 5-6] - Why is the current treatment protocol unlikely to resolve this problem at herd level? [EQF 6-7] - What change to herd management is most likely to reduce the incidence of this condition? [EQF 6-7]

Regulation and Reporting - What must the herd health technician do before leaving this pen? [EQF 5] - What action is required under statutory disease control regulations in this situation? [EQF 5-6] - When must the competent authority be notified in this situation? [EQF 5-6]

Medicines and Prescribing - What is the most appropriate pharmacological treatment for this condition? [EQF 5-6] - Why is this antimicrobial choice inappropriate for this condition in this species? [EQF 6-7] - How should the prescribing cascade be applied in this situation? [EQF 6-7]

Farming Domain

Husbandry and Production - What should the farm operative do before beginning this task? [EQF 4] - What is the most appropriate response to this observation? [EQF 4-5] - Why is this management approach likely to be causing the performance problem described? [EQF 5-6] - What change to the production system is most appropriate to address this performance gap? [EQF 6]

Nutrition and Feed Management - What is the most appropriate adjustment to the ration at this stage of production? [EQF 5-6] - Why is this nutritional factor the most likely explanation for the clinical signs described? [EQF 5-6] - How should the ration be modified to prevent recurrence of this condition? [EQF 6]

Soil, Water, and Environmental Management - What should the operative check before beginning fertiliser application? [EQF 4] - What does this soil analysis result most likely indicate about the nutrient status of this field? [EQF 5-6] - Why is the single farm-wide nitrogen rate likely to be causing both the yield gap and the elevated residuals? [EQF 6] - What is the most appropriate strategy to improve nitrogen use efficiency across this farm system? [EQF 6-7]

Business and Systems Management - Why is this production system unlikely to remain viable under current market and regulatory conditions? [EQF 6-7] - What is the root cause of the performance gap described across this farm business? [EQF 6-7]

Food Domain

Food Safety and HACCP - What should the operative do immediately when this critical limit is exceeded? [EQF 4] - What is the most appropriate corrective action in this situation? [EQF 5] - Why is this process step classified as a critical control point rather than a control measure? [EQF 5-6] - What is the most likely hazard associated with this process step? [EQF 5-6] - Why is the validated heat treatment process unlikely to be the primary cause of this microbiological failure? [EQF 6-7]

Traceability and Withdrawal - What should the food safety technician do first? [EQF 5] - What is the most appropriate scope for the initial product hold in this situation? [EQF 5-6] - What is the root cause of the traceability failures identified in these two exercises? [EQF 6-7] - Why is the current traceability system design structurally inadequate for the supply chain described? [EQF 7]

Regulation and Compliance - What action is required under food safety legislation in this situation? [EQF 5-6] - What is the most appropriate response to the findings of this official control inspection? [EQF 5-6] - When must the food business operator notify the competent authority in this situation? [EQF 5-6]

Quality Systems and Continuous Improvement - What is the most likely root cause of the non-conformance described? [EQF 6] - Why is this corrective action unlikely to prevent recurrence? [EQF 6] - What system change is most likely to address the underlying cause of this repeated failure? [EQF 6-7]

Plant Domain

Crop Health and Pest Management - What should the horticultural operative do with this observation? [EQF 4] - What is the most likely cause of the symptoms described? [EQF 5-6] - What should be done before selecting a control intervention for this pest? [EQF 5] - Why is the current spray programme unlikely to achieve sustainable control of this pest population? [EQF 6] - How should resistance risk be managed in this spray programme? [EQF 6]

Crop Nutrition and Agronomy - What do these soil analysis results most likely indicate about this field's nutrient status? [EQF 5-6] - What is the most appropriate adjustment to the nutrient management plan in this situation? [EQF 6] - Why is the current nitrogen management strategy likely to be causing both of the problems identified? [EQF 6]

Plant Health Regulation and Surveillance - What must the grower do immediately following this observation? [EQF 4-5] - What action is required following confirmation of a notifiable plant pest in this situation? [EQF 5-6] - Why does the epidemiological data complicate the source attribution for this outbreak? [EQF 7-8] - What surveillance design feature is most critical to address the specific epidemiological challenge presented by this pathogen? [EQF 7-8]

Environmental and Sustainability - What action is most appropriate to reduce the environmental risk associated with this application? [EQF 5-6] - Why is the current cropping system likely to be contributing to the soil health decline described? [EQF 6-7]

Glossary of Key Terms

Cognitive task: The mental process required to answer an item correctly, ranging from recall of a fact to synthesis and expert evaluation.

Competency descriptor: The formal statement of what a learner is expected to know, be able to do, and the level of autonomy at which they can act, as specified in a credential.

Construct-irrelevant difficulty: Item difficulty arising from factors unrelated to the competency being assessed, such as confusing wording or unnecessarily complex structure.

Construct-irrelevant ease: The ability of a candidate to answer an item correctly without holding the competency, typically through exploitation of structural flaws.

Cover-the-options rule: The principle that a well-formed lead-in can be answered by a knowledgeable candidate without seeing the option set.

Direct interrogative: A lead-in phrased using a question word (what, why, when, how, where, who) that asks the candidate to generate an answer from knowledge. Preferred over selective interrogatives in the large majority of items.

Discrimination index: A statistical measure of whether an item differentiates between candidates who perform well overall and those who perform poorly.

Distractor: An incorrect option in a one-best-answer item. Distractors should be plausible to a candidate who lacks the specific competency being tested.

EQF: The European Qualifications Framework, an eight-level framework for describing qualifications in terms of learning outcomes (knowledge, skills, responsibility and autonomy).

Item: A single question in a multiple-choice assessment, consisting of a stem, a lead-in, and an option set.

Key: The correct or best answer in a one-best-answer item.

Lead-in: The question component of an item, following the stem.

Learner profile: A description of the candidate's role, level of responsibility, and working context, embedded in the opening of the scenario to anchor the item to a specific EQF level.

Microcredential: A credential that certifies learning outcomes from a short learning experience, typically linked to a specific competency or set of competencies.

One-best-answer format: An item format in which the candidate selects the single best answer from a set of options, where remaining options may be partially correct but are less correct than the key.

Option set: The full set of answer choices presented to the candidate, including the key and all distractors.

Selective interrogative: A lead-in phrased as "which of the following." Appropriate in a small number of situations where the question is inherently about selecting from a defined set, but should not be used as the default phrasing.

Stem: The scenario component of an item, providing the context within which the lead-in question is situated.

Testing point: A one-sentence statement of what a specific item is designed to test.

Verifiable credential: A digital credential that can be cryptographically verified, typically linked to a defined competency and issued following an assessment that provides evidence of that competency.

This guide was developed to support agrifood skills assessment linked to microcredentials and verifiable credentials across EQF levels 4 to 8. It is intended for use by subject matter experts, item writers, and credential developers across the veterinary, farming, food, and plant sectors. Examples are illustrative and should be localised to reflect national regulatory frameworks and production systems.

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Appendix D. From occupational profile to curriculum: building the 130-hour learning product

Appendix A explains how a competency statement is written. Appendix B explains how lesson plans are stacked into pathways. The present appendix sits between them and addresses the question that trainers and curriculum designers ask most often: given a complete ESCO occupational profile, with its essential and optional knowledge, skills and competences, how is it turned into the actual training product, namely sixty hours of contact learning, sixty hours of work-based learning and ten hours of assessment? The appendix sets out a six-step method and then works the method through a complete example for the Precision Crop and Irrigation Management Specialist profile.

D.1 The target product

Each AGRIFOODSKILLS occupational profile is delivered as a 130-hour learning product, in line with the structure agreed for Task 5.2: sixty hours of contact learning, sixty hours of work-based learning, and ten hours of assessment. Across the eight occupational profiles within scope of Task 5.4 this amounts to approximately 1,040 hours of designed training effort. The 130-hour product is the final deliverable that the curriculum design process produces. The method below shows how the abstract architecture in the main report and in Appendices A and B becomes that concrete product.

Component	Hours	Principal content	Principal evidence
Contact learning	60	Essential knowledge and cognitive skills, delivered through the methodologies in Section 6 (online guided learning, micro-learning, peer demonstration, flipped sessions, in-person workshops).	Engagement and assessment assertions.
Work-based learning	60	Performance competences, applied in or reflected on within the real working context. Undertaken flexibly: doing, producing work product, project-based work, observation and analysis, or structured reflection (Section 6.2 and Appendix D.6).	Performance attestations; the strongest evidence for the third assertion.
Assessment	10	Formative and summative assessment across the product, using the instruments in Section 7.	Assessment assertions and the metadata that supports credential composition.
Total	130	One occupational profile, decomposed into stackable microcredentials.	All three assertion categories.

D.2 The starting point: the ESCO occupational profile

The starting point is the validated occupational profile produced in WP3 and structured in the ESCO format. An AGRIFOODSKILLS occupational profile has four parts: essential knowledge (the factual, theoretical and conceptual material the role requires); essential skills and competences (the cognitive and practical operations the role performs); optional skills and competences; and optional knowledge. The essential parts define the core of the 130-hour product. The optional parts define the branches that different learner profiles may add, in line with the pathway mapping matrix in Appendix B.2.

D.3 Step 1: Scope the profile into a microcredential set

The first step clusters the essential knowledge, skills and competences into a small set of microcredentials, each of which is independently meaningful and independently earnable (Appendix A.5). A 130-hour profile typically decomposes into three to five microcredentials. Each microcredential is assigned a share of the 60 contact hours, the 60 work-based learning hours and the 10 assessment hours. Optional skills and competences are assigned to optional microcredentials or to optional episodes within the core microcredentials; they enrich the pathway for some learner profiles without being required of all. The EQF level of the profile is confirmed at this step against the responsibility-and-autonomy strand (Section 3) and Webb's Depth of Knowledge (Appendix A.4.1).

D.4 Step 2: From competence to atomic learning episodes

The atomic learning episode is the smallest meaningful unit of learning that produces evidence toward one competency. An episode has a single objective, a single delivery mode, a defined duration and a defined output. Episodes are the building blocks that fill the hour budget. Each essential competence in the profile is decomposed into a sequence of episodes: contact episodes that build the knowledge and cognitive skill; work-based episodes that demonstrate the performance; and assessment episodes that capture the evidence. The decomposition is the practical counterpart to the stacking architecture in Appendix B: episodes declare prerequisite and output competencies and chain together in the same way that lessons do.

Episode type	Fills which budget	Typical duration	Typical output
Contact episode	The 60 contact hours	10 to 90 minutes	A completed online unit, a micro-learning unit, attendance and participation at a peer demonstration, or a flipped synchronous session.
Work-based episode	The 60 WBL hours	Distributed over weeks	Evidence of applied or reflected practice: a workplace artefact, a project deliverable (implemented or fully planned), an observation-and-analysis record, or a structured reflection against a framework, authenticated by a mentor.
Assessment episode	The 10 assessment hours	5 to 60 minutes	A scored assessment result, a rubric judgement, a performance attestation, with the metadata in Section 7.8.

D.5 Step 3: Allocate the 60 contact hours

Essential knowledge and cognitive skills are mapped to contact episodes. The delivery mode for each episode is chosen from the methodology catalogue in Section 6 on the basis of fit to the learner's time pattern and EQF level (Section 6.13), not on the basis of habit. For a time-poor EQF 3-5 workforce, the contact hours are weighted toward micro-learning and short online guided units consumed on mobile devices, with peer demonstration for embodied skills and a small number of short flipped or in-person sessions reserved for content that genuinely benefits from synchronous co-presence. The 60 contact hours are allocated across the microcredentials in proportion to the knowledge and cognitive-skill load each carries.

D.6 Step 4: Allocate the 60 work-based learning hours

Performance competences are mapped to work-based episodes. Work-based learning is understood broadly here, as the application of learning within, or structured reflection on, the real working context. It is distinct from passive observation, but it does not require that every hour be spent performing a task: the sixty hours may be ascribed flexibly across several forms, and the project deliberately allows providers latitude in how each episode is undertaken and evidenced.

- **Doing:** performing a real task in the workplace, such as delivering a session, conducting a procedure or running a process.
- **Producing work product:** creating a real workplace artefact, such as a protocol, a map, a plan, an audit or a teaching resource.
- **Project-based work:** undertaking a defined improvement project over time, with the flexibility to implement it fully or, where implementation is not within the learner's control, to plan it fully and document the intended process and outcomes.
- **Observation and analysis:** recording and analysing a real workplace episode, such as a consultation, a teaching session or an operation, against a defined framework. The observation is active and analytical, not passive watching.
- **Structured reflection:** reflecting on workplace experience against a competency or professional-standards framework, where the structured reflection on genuine experience is itself the evidenced learning.

These forms are not mutually exclusive; a single work-based episode commonly combines doing, producing and reflecting. The Developing Mastery in Clinical Education programme illustrates the range within one microcredential: one episode asks the learner to plan and deliver a teaching session and to submit the session plan, an account of what happened and the feedback received; another asks the learner to design a teaching improvement and either to implement it or to plan it fully, documenting process and outcomes; a third asks the learner to record and evaluate their own teaching against the Zones of Engagement and Depth of Knowledge frameworks through structured self-reflection. Each is a legitimate work-based episode, each produces evidence, and the form of the evidence differs by episode. The balance of forms also differs by occupational profile: a precision-agriculture profile weights its work-based hours toward doing and producing; a clinical-educator profile weights them toward project work, observation and structured reflection.

The sixty work-based hours are distributed over the working weeks of the programme rather than delivered in a block, and are absorbed mostly within the learner's normal working time. The episodes carry the principal share of the performance-attestation evidence for the profile, and each is signed off against a published rubric by a mentor or supervisor who can attest to the authenticity of the work, the observation or the reflection. The hours are allocated across the microcredentials in proportion to the performance competence each carries.

D.7 Step 5: Allocate the 10 assessment hours

The 10 assessment hours are allocated across the product using the assessment instruments in Section 7, chosen by what they can evidence (Section 7.5). Formative assessment is embedded throughout the contact and work-based episodes and feeds the Learning Record System (Section 7.4); summative assessment is concentrated at microcredential completion. The 10 hours include the EQAVET-required assessment effort (Section 8). Each assessment episode produces an assertion carrying the metadata in Section 7.8.

D.8 Step 6: Sequence and stack

The episodes are ordered using the prerequisite and output competency declarations (Appendix A) and the stacking metadata (Appendix B.1.7). The order is exposed to the learner through the pathway mapping matrix (Appendix B.2), so that the learner can navigate the product as a self-directed pathway rather than a fixed sequence. Optional episodes are presented as optional cells in the matrix, available to the learner profiles that need them. The result is the complete 130-hour product: a set of stackable microcredentials, each decomposed into contact, work-based and assessment episodes, that compose into the full occupational profile and can also be earned independently.

D.9 Full worked example: Precision Crop and Irrigation Management Specialist (EQF 5)

The example below works the six-step method through the Precision Crop and Irrigation Management Specialist occupational profile, which was introduced in Appendix A.5.1 through its drone-survey competency stack and developed further in Appendix B. The profile is delivered as a single 130-hour product, decomposed into four microcredentials.

D.9.1 Step 1: the profile and its microcredential set

The essential skills and competences in the profile (apply precision agriculture technologies; plan and manage efficient irrigation; interpret sensor, satellite and drone data; produce site-specific maps and plans; calibrate and maintain precision equipment; use GIS and remote sensing; analyse weather, soil and crop data; implement water-use efficiency strategies; provide technical guidance) cluster into four microcredentials. The optional skills (machine learning models; IoT network management; cost-benefit analysis; renewable energy integration) become optional episodes within the relevant microcredentials.

Microcredential	Essential competences covered	Contact h	WBL h	Assess h
MC1 Precision data acquisition	Apply precision agriculture technologies; interpret sensor, satellite and drone data; calibrate and maintain precision equipment. Includes the drone-survey stack from Appendix A.5.1.	15	15	2.5
MC2 Spatial analysis and mapping	Use GIS and remote sensing; produce site-specific nutrient and irrigation management maps; analyse weather, soil and crop data.	15	15	2.5
MC3 Irrigation design and water-use efficiency	Plan and manage efficient irrigation systems; implement water-use efficiency strategies.	15	15	2.5
MC4 Advisory practice	Provide technical guidance and advisory services; collaborate with farmers, engineers and researchers; communicate technical findings.	15	15	2.5
Total	Full occupational profile	60	60	10

D.9.2 Step 2: MC1 decomposed into atomic episodes

MC1 (Precision data acquisition) is decomposed below into its atomic learning episodes, showing how the 15 contact hours, 15 work-based hours and 2.5 assessment hours are filled. The drone-survey lesson PCS-L-03 from Appendix B.1.8 appears as one of the contact-plus-WBL episodes.

Episode	Type	Mode (Section 6)	Hours	Output
E1 Principles of precision data acquisition	Contact	Online guided learning, 4 units	4 contact	Completed units; embedded checks.
E2 Read and interpret an existing land survey (PCS-L-01)	Contact	Micro-learning pathway	2 contact	Pathway completion; check.
E3 UAV and GDPR framework (PCS-L-02)	Contact	Online guided + secure SBA	3 contact	SBA result (assessment).
E4 Plan and conduct a drone flight (PCS-L-03)	Contact + WBL	Flipped: planning + workshop + supervised flight	3 contact + 6 WBL	Signed flight log (performance attestation).
E5 Calibrate and maintain precision equipment	Contact + WBL	Peer demonstration + workplace task	3 contact + 4 WBL	Calibration record signed by mentor.
E6 Sensor and satellite data acquisition on own holding	WBL	Work-based project	5 WBL	Acquired dataset with quality report.
MC1 assessment	Assessment	SBA + observation + portfolio entry	2.5 assessment	Composed microcredential assertion.
MC1 total			15 contact + 15 WBL + 2.5 assess	MC1 awarded.

The other three microcredentials are decomposed in the same way. MC2 weights its contact hours toward GIS and remote-sensing online units and its WBL toward the production of a real prescription map for the learner's holding (the imagery-processing competency from Appendix A.5.1). MC3 weights its contact hours toward irrigation engineering and its WBL toward the design and measurement of a water-use efficiency improvement on the holding. MC4 weights its contact hours toward communication and advisory practice and its WBL toward the production and presentation of an advisory note to a real client.

D.9.3 The complete product

Assembled, the four microcredentials constitute the complete 130-hour Precision Crop and Irrigation Management Specialist product: 60 hours of contact learning delivered through mobile-first online units, micro-learning, peer demonstration and a small number of flipped sessions; 60 hours of work-based learning producing real workplace artefacts (a flight log, a calibration record, a prescription map, a water-use improvement, an advisory note) signed off by mentors; and 10 hours of assessment producing the assertions that compose the credential. A learner who completes all four

microcredentials holds the full occupational profile; a learner who completes only MC1 holds a recognised, independently meaningful microcredential in precision data acquisition that they can present in its own right or stack toward the full profile later, in line with the pathway mapping matrix in Appendix B.2.

Design summary: from ESCO profile to the 130-hour product

- Start from the validated ESCO occupational profile; separate essential from optional knowledge, skills and competences.
- Cluster the essential competences into three to five microcredentials; assign optional competences to optional episodes or branches.
- Decompose each competence into atomic learning episodes: contact, work-based and assessment.
- Allocate the 60 contact hours to knowledge and cognitive skills, choosing delivery modes from Section 6 by learner-constraint fit.
- Allocate the 60 work-based hours to performance competences, each producing a mentor-signed workplace artefact.
- Allocate the 10 assessment hours across the product using the instruments in Section 7, formative throughout and summative at completion.
- Sequence the episodes using the prerequisite and output competency declarations; expose the order to the learner through the pathway mapping matrix in Appendix B.2.
- Confirm the EQF level against the responsibility-and-autonomy strand and Webb's Depth of Knowledge before sign-off.

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